BAIAMONTE

TENUTA

OPERATIONS · ENOLOGY · ENERGY · COMMUNICATIONS · RECOVERY

Tenuta Baiamonte System Manual

The complete operating, integration, security, deployment, and recovery reference.

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Operational authority Vineyard Operations MariaDB

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System at a glance

This manual follows the live dashboard from everyday use through administration, recovery, and release verification.

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This manual describes what the Baiamonte system does, how staff use it, how its calculations and predictions work, how data moves between services, and how to recognize and recover from problems. It is written for owners, vineyard staff, cellar staff, accountants, agronomists, kiosk users, and technical administrators.

OPERATING REFERENCE

1. What the system is

Tenuta Baiamonte Vineyard Operations is a private estate-management system hosted as a Home Assistant app. It combines vineyard work, harvest, purchased fruit, cellar, laboratory, weather, treatment, labor, payment, olive-oil, hospitality, register, messaging, social, security, energy, water, devices, maps, public publishing, and forecasting in one controlled interface.

The core rule is simple:

> MariaDB is the operational authority. Historical workbooks, messages, notes, emails, and laboratory documents are evidence sources, not live databases.

Imported evidence is reviewed, normalized, and written to MariaDB. The dashboards, predictions, labels, payment views, and reports read from MariaDB. The old workbooks are not consulted during normal operation.

System goals

- Give each role a useful, limited view of the estate.
- Preserve historical records by year and vintage.
- Distinguish facts, estimates, forecasts, and unresolved questions.
- Keep treatment and cellar decisions under human approval.
- Make labor approval and payment auditable.
- Use free, credential-free environmental sources where practical.
- Continue operating through temporary source outages using stored last-good data.
- Keep secrets in protected Home Assistant add-on configuration.
- Prefer local device control and locally reported telemetry; use cloud services only where the named integration requires them.
- Keep the public website, shared displays, operational dashboards, and authoritative database synchronized without exposing private records.

OPERATING REFERENCE

2. System map

Component	Purpose	Authority / access
Home Assistant	Users, devices, dashboards, cameras, weather entities, Supervisor	Home Assistant accounts
Vineyard Operations	Main web interface and business logic	Home Assistant Ingress
Hospitality	Private tastings, dinners, guest readiness, deposits, and communication history	Hospitality Manager or Administrator
Register	Tablet-oriented sales, sellable inventory, receipts, and monthly transaction ledger	Register, Cashier, Hospitality Manager, or Administrator
MariaDB	Sole operational database	Add-on network only
Baiamonte MCP	Controlled Codex and automation bridge	Bearer token; local/VPN
Vineyard TV	Read-only rotating operational display	Viewer profile
iPad dashboard	Larger touch operations dashboard	`ipad` viewer profile
Cellar labels	Tank identity, live readings, charts, legal identity	Enrolled kiosk URLs
Cellar label gateway	Read-only legal labels and assigned tank displays	Trusted LAN/VPN on port 8102
GW2000A and GW2000B	Estate weather plus leaf-wetness telemetry	Home Assistant entities and recorder history
Baiamonte Power	Three-battery reserve, solar, load, generator context, fan protection	Home Assistant telemetry and guarded local control
Local Tuya devices	Estate switching and device state without normal cloud dependency	Home Assistant LocalTuya and reserved Omada addresses
Gmail intake	Email and attachment review queue	Protected mailbox credentials
WhatsApp	Approved inbound/outbound operational messaging	Meta or system bridge credentials
Social	Facebook and Instagram posts, audience snapshots, and official-export imports	Protected Meta credentials and reviewed imports
Fatture in Cloud	Read-only accounting mirror	Protected read-only token
Public website	Approved vintage and harvest feed; public Shopping and Visit visibility is independently configurable	Optional publication token

Network addresses

- Home Assistant: `http://192.168.0.10:8123`
- Vineyard Operations direct service: `http://192.168.0.10:8101`
- Cellar label gateway: `http://192.168.0.10:8102`
- Baiamonte MCP: `http://192.168.0.10:8100/mcp`

Use Home Assistant Ingress for normal staff access. Direct local addresses are intended for service integration, diagnosis, enrolled displays, and VPN use.

OPERATING REFERENCE

3. Users and permissions

Administrators

Administrators can manage system configuration, users, process controls, messaging, labor approval, payroll, documentation, and diagnostics. Administrative access should remain limited to trusted owners or system managers.

Operations users

Operations users can manage vineyard work, harvest, cellar records, laboratory review, olives, treatments, issues, and operational intake. They do not automatically receive full finance or system-configuration access.

Finance users

Finance users can view the read-only Fatture in Cloud mirror, invoices, payments, expenses, and financial reports. Finance information is intentionally excluded from shared TV and kiosk feeds.

Worker users

Worker accounts use a simplified personal portal to clock in and out, enter services or expenses, attach evidence, see review status, and view approved history. A worker cannot approve or pay their own record.

Viewer users

The `display`, `tv`, and `ipad` profiles are read-only. They do not receive finance data or administrative controls.

Hospitality Managers

Hospitality Managers can manage private tastings and dinners, guest requirements, quotes, deposits, confirmations, arrival, and completion. They do not automatically receive vineyard write, finance, payroll, or system-administration access.

Register and Cashier users

Register and Cashier users can sell the read-only Fatture in Cloud product catalog, local hospitality packages, and authorized manual items. They can apply a line discount, choose EUR or USD tender, select the Italian or US PayPal Business account, print a non-fiscal operational receipt, and export the monthly local ledger. They do not automatically receive finance, payroll, vineyard-write, or administration access.

Home Assistant People and Users are authoritative for identity, display name, username, profile picture, and presence. Vineyard Operations is authoritative for estate access level, operational role, approval authority, and hospitality permissions. An administrator should assign access and roles in **Admin -> People** rather than creating a second identity.

OPERATING REFERENCE

4. First use and navigation

1. Sign in to Home Assistant with the appropriate account.
2. Open **Vineyard Operations** from the sidebar.
3. Choose **Operations**, **Agronomy**, **Enology**, **Hospitality**, **Register**, or **Admin** from the top workspace switch. Only authorized workspaces are shown.
4. Choose the working year. Year selection changes the harvest, laboratory, weather comparison, labor, treatment, olive, and historical context.
5. Use the main sections for daily work; use Administration only for configuration or audit tasks.

6. Treat yellow or amber items as uncertain or awaiting review. Treat red items as active exceptions, not automatically as failed equipment.

Changing the working year keeps the current workspace and page open. For example, changing vintage while viewing Enology -> Laboratory refreshes Laboratory for that vintage; it does not move the user to Operations. A page can explicitly show the latest earlier vintage when the selected year has no measurements, but it must label that fallback and must not present it as current-year evidence.

Agronomy and Enology each have a dedicated **Admin** page. Daily pages contain operational evidence and actions; area configuration is kept in that area's Admin page so settings have one authoritative home. Agronomy Admin contains product evidence and catalog overlays plus sprayer calibration. Enology Admin contains the tank register, label-tablet provisioning, separate-varietal planning, cellar thresholds, packaging and annual winemaking settings. The operational pages link to these settings without duplicating editable controls.

Main work areas

- **Today:** current work, recent work, estate status, weather, and alerts.
- **Vines / Harvest:** variety status, estate picks, purchased fruit, predictions, production capability, and vintage history.
- **Treatments:** forecasts, planned work, approvals, completed applications, and safety evidence.
- **Cellar:** wine lots, vessels, readings, operations, recipes, additions, inventory, labels, and tank assignments.
- **Labor / Payroll:** work records, approval, contractor invoices, deposits, and payments.
- **Olives:** harvest and oil history, conversion efficiency, cost assumptions, and YoY charts.
- **Laboratory:** reports, vintage assignment, analytes, trends, review, and decision support.
- **Atlas:** cadastral parcels, vineyard blocks, terraces, nursery data, and map geometry.
- **Intelligence:** weather, GDD, seasonal comparisons, prediction evidence, disease pressure, and outlooks.
- **Weather:** clickable current measurements, historical graphs, GW2000 health, and forecast context.
- **Water / Solar / Power:** cistern evidence, three-battery reserve, live energy balance, forecasts, protected loads, and solar-room cooling.
- **Cameras / Security:** low-load estate cameras, reviewed events, vehicle evidence, and visual watches.
- **Hospitality:** private experiences, packages, bookings, guest readiness, deposits, and communication history.
- **Register:** tablet sales, sellable inventory, receipts, transaction exports, PayPal account selection, and register settings.
- **Inbox / Messaging:** Gmail, WhatsApp, iMessage and voice intake, review, contacts, delivery, replies, and approvals.
- **Social:** Facebook and Instagram refresh health, posts, publishing, audience history, and official export imports.
- **Alerts:** current operational alerts and delivery settings.
- **Administration:** process health, users, integrations, logs, documentation, and update controls.

Wine yield and litre/kilogram transparency

The Harvest, Vintage, TV and projected Bottling figures use a production-yield factor, not a physical mass conversion. The formula is **estimated finished wine litres = grape kilograms × yield L/kg**. The typical red-wine planning range supplied for Baiamonte is **0.67–0.70 L/kg (67–70%)**; for example, a configured factor of 0.700 means 1 kg of grapes projects 0.700 L of finished wine, while 1 projected litre corresponds to 1.429 kg of grapes. Every calculated output shows the active factor and source. Actual recorded cellar volume or completed bottling output supersedes the estimate.

Converting an actual litre of wine into kilograms is a different calculation: **wine kg = wine L × measured density in kg/L**. The system does not use 0.67–0.70 for wine mass and must not assume a universal wine density. A current measured density is required whenever physical wine mass is needed.

Estate security and known vehicles

Admin -> Security contains the estate vehicle movement ledger. Administrators select entry, parking, doorbell and perimeter cameras, define how visible direction maps to entry or exit, and may deliberately override battery conservation for a primary parking camera. The page reports daily entries, exits, observations, staff suggestions, readable plates, flags and review backlog, plus statistics for the confirmed known-car database.

Reviewing an observation can correct its movement, transcribe a genuinely legible plate, link a staff or contractor profile, add a reusable tag, categorize the visit, or flag the vehicle. A confirmed review becomes bounded learning evidence for later sightings. Camera matches are advisory; the security pipeline does not create payroll hours, perform access denial or infer the driver from a face.

OPERATING REFERENCE

5. Data authority and evidence rules

Every value should be understood as one of four types:

1. **Fact:** a confirmed measurement or completed event.
2. **Estimate:** a calculated value based on stated inputs.
3. **Forecast:** a future projection with confidence and evidence.
4. **Unresolved:** missing, conflicting, or awaiting human verification.

Blank database values mean **unknown**, not zero. The system should never silently convert a missing quantity into zero.

Evidence flow

1. A message, email, note, document, sensor reading, or manual entry arrives.
2. The item is stored with its original evidence.
3. Automated extraction may propose structured records.
4. A person reviews material changes.
5. Approved facts are saved to MariaDB with an audit trail.
6. Dependent dashboards and models are refreshed.

Vintage assignment

Laboratory and cellar reports are assigned to the correct vintage rather than merely the document year. Reports that cross calendar years stay with the wine vintage they describe. Explicit vintage labels and linked wine lots take precedence; inferred assignments remain marked with confidence and evidence.

OPERATING REFERENCE

6. Intelligence pipelines and decision workflows

The intelligence layer is a collection of evidence pipelines, not one autonomous decision maker. Each pipeline receives evidence, creates a traceable calculation or proposal, identifies missing information, and hands the result to the person or record that has authority. The system may use a provisional result while review is pending only where the workflow explicitly allows it; it never converts a prediction into a completed event.

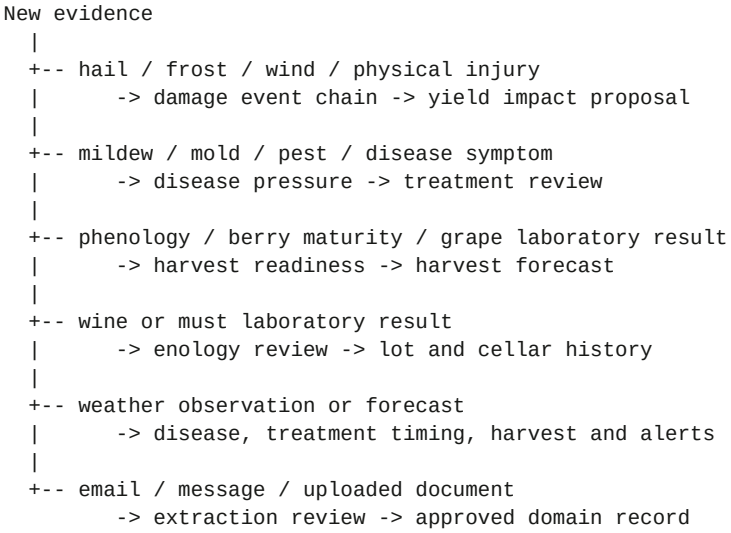
Common intelligence contract

Every intelligence result carries the same operating context:

- **Source and timestamp:** where the evidence came from and when it was observed.
- **Scope:** estate, zone, block, parcel, variety, lot, tank, reservation, person, or financial document.
- **Evidence state:** original attachment or message, extracted facts, conflicts, and missing inputs.
- **Calculation state:** model/rule version, result, range, confidence, and explanatory factors.
- **Authority state:** draft, proposed, approved, completed, rejected, superseded, or unresolved.
- **Audit state:** who reviewed or changed the result and what downstream records were refreshed.

An approved fact remains separate from the next system proposal. Recalculation can create a new proposal, but it does not silently overwrite an agronomist-approved loss, an approved treatment, a confirmed harvest plan, a reviewed laboratory result, a sent guest message, or an accounting document.

Master evidence router



The submitter chooses the observed condition and growth stage from controlled lists. The router preserves the original evidence and can attach a new observation to an existing issue chain, such as the estate-wide 2026 hail event, instead of creating a duplicate event.

Damage assessment and forecast-impact pipeline

Damage intelligence estimates crop impact while keeping the system estimate independent from the agronomist's final decision. Estate-wide means 100% geographic coverage of the estate; it does not mean 100% yield loss.

```
Event or follow-up report
  -> identify event chain and geographic scope
  -> evaluate structured scouting and optional photos
  -> calculate AI loss %, range and confidence
  -> compare with earlier reports and approved final
  -> create a new proposal for the chain
  -> agronomist reviews and approves or edits
  -> approved chain final adjusts the harvest forecast
```

Each report card shows the result produced by that report. The chain summary shows the current system proposal, the latest approved agronomist value, and the percentage currently used by forecasting. Follow-up reports refine the chronological assessment; they do not rewrite the historical result of an earlier card. Until an updated proposal is approved, the last approved chain final remains authoritative. If no agronomist value exists yet, the clearly labelled provisional AI value may feed planning with its confidence range.

Disease pressure and treatment pipeline

Weather, disease pressure, field scouting, crop, growth stage, treatment history, product evidence, inventory, sprayer configuration, and harvest clearance operate as one decision chain. Vineyard and olive programs are evaluated separately.

```
Weather + field evidence + crop stage + treatment cadence
  -> calculate disease/stress pressure by target
  -> decide whether review is needed
  -> screen crop-authorized products and support products
  -> check season, target, rate, PHI/REI and compatibility
  -> calculate homogeneous mixture and sprayer batches
  -> compare required quantity with the inventory ledger
  -> find a defensible application window
  -> agronomist approves product, date and instructions
  -> operator records the completed application
```

The output can contain a primary disease-control product plus justified nutrition, wound support, wetting, resistance-management, or other support products. A product is included only when its current label/formulation evidence supports the crop, target, rate, unit, timing, and combination. Separate passes are shown when same-tank compatibility is not verified. Every prepared tank is treated as a homogeneous mixture under agitation.

The treatment simulator replays the same evidence and rules for a hypothetical or historical date but saves nothing and authorizes nothing. It shows the proposed products, amounts, batches, timing, evidence gaps, on-hand stock, shortage, and total required inventory. A negative stock ledger is permitted when completed use precedes a delayed invoice; the shortage remains an operational issue until a receipt nets it or an authorized person resolves it as not needed for the rest of the season.

For Baiamonte vineyard work, the default operating picture is one pass over the complete selected vineyard area using 400 L total carrier, prepared as two 200 L fills with the same approved recipe. The simulator adds only products supported by the current target, field evidence, crop stage, historical Agronomist pattern, inventory and label rules. It does not add a nutrition, support or disease-control product merely because the product exists in inventory. When exact same-tank compatibility is not recorded, the program remains blocked or separates the pass for Agronomist review.

The disease model recalculates pressure from GW2000 temperature, humidity, rainfall, leaf wetness when available, soil moisture when available, wind, solar conditions, phenology, maturity evidence, localized scouting and recent treatment context. Weather is the principal rationale for timing, while field evidence determines whether pressure is actually actionable. The model also creates disease-onset forecasts, block-specific calibration, spray-window evidence, expected product duration, retreatment cadence and resistance-rotation review. These learned outputs remain provisional until sufficient reviewed outcomes exist.

Agronomist review records the accepted risk level or corrected score and the reason. Paired pre/post-treatment scouting links the last comparable observation before treatment with the first comparable observation after treatment. Only these field-observed pairs can train effectiveness by disease, mixture, dose, block and weather;

reconstructed weather pressure is context, not proof of treatment success.

AI and learning supervision

Admin -> AI is the full monitoring page for learning processes. It shows model version, data-through date, evidence counts, represented seasons, validation method, measured accuracy or error, data-quality findings and release gates. A model can be active while remaining review-gated. Low accuracy, too few seasons, missing Agronomist decisions or missing paired outcomes must be shown as limitations rather than converted into confidence.

Current learning processes include laboratory vintage projection, harvest-date learning, Agronomist treatment-pattern learning, disease-pressure calibration, disease onset, treatment effectiveness, product duration and retreatment cadence, FRAC rotation, young-vine nutrition, block disease calibration, spray-window learning and automated data-quality detection. New treatment, scouting, laboratory and Agronomist-review evidence rebuilds the connected learning records and feeds the live Agronomy, Treatments, Laboratory, Harvest, Alerts, WhatsApp and Admin views. Human approval boundaries remain authoritative everywhere.

Harvest-date recalculation is split into a free deterministic layer and a paid narrative-review layer. Weather, GDD, historical dates, approved laboratory evidence and recorded readiness continue to recalculate year-round without an OpenAI request. Paid harvest AI is deferred until a plausible harvest is within 60 days and at least one current maturity sample or approved grape laboratory result exists. Its cache is keyed to decision evidence rather than routine weather refreshes, so a completed vintage or a distant next harvest cannot generate repeated paid reviews. Admin -> AI labels prior harvest-projection charges as historical usage and displays this policy beside the feature total.

The cistern camera remains an always-on local RTSP and Eufy source. Its scheduled level check briefly enables the camera's paired light when needed, extracts one current frame from the continuous local stream, restores the prior light state and leaves the stream untouched. A bridge-generated still and then a sleeping-source wake-up are fallbacks only. Camera AI remains the operational visual estimate while the local shadow model learns side by side. Repeated frames and repeated percentages do not satisfy the release gate: the panel separately reports unique frames, observed changes, distinct accepted levels and the recorded level range. A physical gauge or confirmed dip measurement remains the preferred calibration reference whenever one is available.

Vineyard North also supports a protected local RTSP source for future fixed-view visual evidence. Suitable future uses include coarse canopy-color change, storm or hail aftermath, smoke, obstruction and scene-change screening. The distant view must not be treated as proof of a disease, pest, phenology stage or treatment need; any such finding remains advisory and must be confirmed by localized scouting or the Agronomist.

Harvest prediction pipeline

```
Historical harvest dates and GDD model
  + current weather and ensemble spread
  + phenology, scouting and maturity samples
  + reviewed grape laboratory chemistry
  + damage-chain approved/provisional loss
  + treatment PHI, work readiness and cellar capacity
  -> variety forecast date/window and yield range
  -> confidence, evidence age and missing-input warnings
  -> authorized harvest plan
  -> actual pick record becomes the historical fact
```

The learned model retrains when authoritative evidence changes and reports represented years, training count, back-test error, confidence, and important factors. On-site weather and verified field/laboratory evidence carry the operational weight. Open-Meteo ensemble spread expresses near-term uncertainty; SIAS validates regional conditions; Sentinel-2 supplies block-level vegetation trends; ECMWF seasonal anomalies support early planning only. None of the external sources can independently order a pick.

Laboratory ingestion and enology pipeline

```
Original report or image
  -> retain viewable source
  -> detect every sample section on every page
  -> extract sample/lot, Annata, analytes, values and units
  -> split multi-sample reports into linked result groups
  -> assign grape variety, wine lot and vintage
  -> compare source/result signature for duplicates
  -> human review of conflicts and assignments
  -> save once -> refresh harvest or enology intelligence
```

`Annata` normally means the wine vintage. A report date in the following calendar year does not move the result to the newer vintage. For grape samples, variety and sample date route the result to harvest readiness. For must and wine samples, the linked lot and vintage route it to enology and cellar history. Exact source/result signatures prevent a report or extracted result from being imported twice.

Messaging and document-ingestion pipeline

```
Gmail / WhatsApp / Apple message / uploaded document
  -> store source and sender context
  -> classify likely domain and extract explicit facts
  -> show uncertainty, conflicts and proposed records
  -> reviewer approves, corrects or rejects
  -> approved record enters its domain pipeline
  -> replies or notifications require an explicit send action
```

If no explicit facts can be supported, the item stays in review instead of generating a guessed record. Messages cannot complete treatments, clear reminders, approve payments, alter harvest dates, or create financial facts merely because their wording resembles an action.

Hospitality inquiry, reservation and partner-commission pipeline

```
Matched inbound subject or manual inquiry
  -> guest inquiry and response history
  -> reservation conversion and availability check
  -> optional partner assignment and commission rule
  -> estimated commission while the visit is tentative
  -> earned/due commission when the qualifying visit is confirmed
  -> approval -> one or more partner payments
  -> paid/void state and Finance payable reconciliation
```

Partners are managed as hospitality business records with contact, tax/payment notes, default commission method, rate, and active state. Each reservation keeps its own commission snapshot so later partner-rate changes do not rewrite history. Estimated commissions are not Finance payables. Only earned due, approved, or partially paid balances enter the payable view; cancelled or disqualified reservations void the commission while preserving the audit trail.

Alert and operational-action pipeline

```
Sensor, scheduler, inventory or database condition
  -> evaluate threshold and evidence freshness
  -> create/update one durable alert
  -> rank by severity and route configured copies
  -> show the same authoritative state on web and TV
  -> clear only when the measured condition or owner action resolves
```

Alerts are durable database records. Delivery channels send copies; they do not own alert state. A reminder, work-plan item, or notification cannot be marked satisfied by an unrelated duplicate or stale completion.

Decision authority summary

Pipeline	System may calculate	Final authority
Damage	Provisional loss, range, confidence, trend	Agronomist-approved chain final
Treatments	Pressure, products, mixture, stock gap, date window	Agronomist approval and completed application record
Harvest	Variety date/window, yield range, confidence	Authorized plan; actual pick is final fact
Laboratory	Extraction, sample grouping, vintage proposal, duplicate check	Reviewed saved result and linked lot/vintage
Messaging	Classification and proposed records	Human review and explicit send/save action
Hospitality	Availability, lifecycle prompts, commission calculation	Authorized reservation and commission/payment actions
Alerts	Detection, severity, routing and automatic clear evidence	Authoritative measured state or explicit resolution

OPERATING REFERENCE

7. Vineyard, harvest, and historical records

Harvest records

The system stores planned quantities separately from actual harvest lots. Actual pick dates, weights, crates, blocks, varieties, and source evidence drive historical reports and forecasting.

Estate-grown harvest and purchased fruit are separate source classes. Estate blocks, parcel yields, harvest completion, crew productivity, and estate crate totals use estate fruit only. Purchased fruit retains supplier/source wording, quantity, intake date, legal identity, cellar lot, tank and process history. Production, vintage, cellar, bottling and TV totals include both sources. Loose purchased fruit is never converted into fictitious crates.

For the 2026 vintage, the recorded production baseline is 4,838.38 kg estate-grown fruit plus 1,078 kg purchased Grenache, or 5,916.38 kg total fruit received. The purchased intake is a normal traceable lot assigned to Tank 39 and can be measured, treated, fermented, transferred and bottled like any other lot. The system preserves the confirmed 1,078 kg total while visibly retaining the 1.55 kg gross/tare arithmetic discrepancy for source review.

A single picking lot can include multiple legal cadastral parcels. Select every contributing parcel when recording the pick; the database keeps each parcel as a separate authoritative relationship rather than combining the references into notes. When that picking lot is transferred into a cellar lot and tank, every linked parcel follows the trace automatically. Additional picking lots can enter the same tank and their fruit and must quantities accumulate.

Each live and printed tank label lists the deduplicated legal provenance for its current wine lot: municipality, cadastral sheet, parcel number, vineyard area, tenure, and contract protocol when recorded. The label derives this list from the retained harvest-to-tank trace and does not require the parcel information to be typed again.

Authoritative harvest dates currently include:

Vintage	Grecanico	Grenache	Nerello
2023	23 Sep	24 Sep	8 Oct
2024	11 Sep	23 Sep	23 Sep
2025	11 Sep	17 Sep	23 Sep

The 2024 Grenache and Nerello harvest date is shared. Earlier conflicting date lists were discarded by owner instruction.

Year switching

Changing the year must change the visible harvest, work, laboratory, treatment, weather, olive, labor, and financial context. Current live data and historical evidence use the same database, so comparisons do not depend on reopening old workbooks.

The private application keeps the selected working year until the user changes it. The public website advances to the next planning vintage only after the three estate varieties - Grecanico, Grenache and Nerello Mascalese - all have recorded estate harvests. Purchased fruit does not satisfy or block that gate. Completed estate harvests remain complete even when an older provisional date was not manually closed, and the public heading must use the vintage carried by the published feed.

OPERATING REFERENCE

8. Harvest prediction logic

Harvest predictions are decision support, not autonomous picking instructions. Sebastian or another authorized person confirms the operational decision.

Authoritative inputs

- Confirmed historical pick dates by variety.
- On-site GW2000 weather history.
- Base-10 growing degree days (GDD).
- Current deterministic weather forecast.
- Fresh ensemble forecast spread.
- Variety-assigned grape laboratory results.
- Maturity samples, phenology, and field reports.
- Treatment clearance and pre-harvest intervals.
- Work readiness and cellar capacity.
- Historical model error and year-over-year behavior.

GDD calculation

Daily GDD uses:

```
max(0, ((daily minimum C + daily maximum C) / 2) - 10)
```

If minimum and maximum are unavailable, an approved daily-mean fallback may be used and is labeled. The calculation accumulates from the configured season start and is compared with variety-specific historical behavior.

Learned-model behavior

The learned model uses prior vintages as training evidence. It reports training sample count, represented years, confidence, and back-tested mean absolute error. It is recalculated when authoritative evidence changes.

New grape laboratory results, maturity samples, phenology observations, harvest plans, treatment changes, or harvest records create a durable refresh request. The scheduler processes the request rather than waiting for the normal full prediction interval.

Laboratory influence

Fresh, reviewed, variety-assigned grape laboratory data can affect readiness. Unreviewed reports or grape results older than 21 days cannot directly move the date. As harvest approaches, laboratory chemistry and field maturity receive more weight than coarse seasonal outlooks.

External sources and limits

Source	Use	Restriction
Open-Meteo ensemble	Near-term uncertainty and forecast spread	Automatic date influence bounded to plus/minus one day
SIAS regional data	Independent validation	Does not replace the on-site station
Sentinel-2	Parcel/block vegetation trend evidence (NDVI, NDRE, LAI estimate)	Cannot independently select a picking date
ECMWF seasonal	Early rainfall and temperature planning	Coarse scale; cannot set an exact date

The system uses free data without required credentials for these sources. Sentinel processing currently recognizes seven verified cadastral parcel polygons and reports the geometry scope explicitly.

OPERATING REFERENCE

9. Treatments and safety

The treatment workflow is **plan -> review -> approve -> apply -> record**.

- A forecast or model recommendation is not an approved treatment.
- A planned treatment is not a completed application.
- Completion requires actual date, product, rate, area, operator, and safety evidence when applicable.
- Agronomist review remains required for disease and treatment recommendations.
- Pre-harvest and re-entry intervals must be preserved.
- Missing safety detail is visibly flagged rather than guessed.

Projection configuration

The Home Assistant add-on configuration provides two planning defaults:

- `treatment_planning_water_l`: the provisional total carrier-water volume. It defaults to 400 L and can be changed in the Treatments page for a live calculation.
- `treatment_default_sprayer`: the name, model, or internal ID of the preferred active spray-equipment profile. The Treatments page also provides a selector when more than one profile exists.

Sprayer capacity, usable fill, nozzle setup, flow, pressure, travel speed, carrier rate and calibration status remain database equipment records. A nominal tank size may split a projection into batches, but the proposal remains blocked until the actual usable fill and field calibration are verified. Product projection requires a verified formulation, estate authorization, current crop/target use, documented rate and compatible units. Water-based and per-hectare rates are calculated directly; density or mass/volume conversions are never guessed. Products with incomplete directions remain visibly marked as needing configuration and are not automatically included.

Current source-backed product directions include these explicit method limits:

- Frontiere: vineyard foliar support at 0.75–1.00 L/ha; it is not automatic disease control.

- Ferticus 18 M: vineyard foliar nutrition at 300–500 g/100 L. The system preserves this as a mass rate and never displays it as millilitres.
- TerraPlus Solub NPK 8-7-6: 15–30 kg/ha per pass as a separate, localized soil-directed spray (or fertigation). It must not be presented as a canopy spray, must not be mixed automatically with crop-protection products, and its mother solution must not be acidified or mixed with calcium.
- Ossiclor 20 Blu Flow: vineyard copper application at 1.7–4.2 L/ha under the current authorized use, with crop stage, annual copper limits, PHI, weather, PPE and Agronomist approval checked before use.

Support and nutrition products remain unselected by default. A risk score alone never adds them to a treatment. Every projected use continues to require the current container directions and human approval.

The current audit has no open treatment safety-gap failures. Seven older applications remain explicitly restricted historical evidence because contemporaneous label, calibration or exact-use facts cannot be reconstructed. Restricted history may inform chronology but cannot be reused as verified safety evidence.

Fertilizer and young-vine nutrition separation

Fertilizer procurement shows land and whole-vineyard soil products only. Vine foliar products do not belong in that procurement list. TerraPlus is reserved for the mapped section of small, young vines and appears in the young-vine nutrition evidence instead of the general fertilizer recommendation. The system may recommend TerraPlus only when mapped vine age plus recorded weak growth, chlorosis, establishment stress, verified deficiency, tissue evidence or soil evidence supports review. It never recommends TerraPlus simply because it was purchased or is on hand.

OPERATING REFERENCE

10. Laboratory reports

The laboratory section stores original reports, sample identity, vintage assignment, matrix, variety or lot linkage, analytes, values, units, review state, and interpretation.

Sample names are normalized across Italian and English before matching. Terms such as grape, must, mosto, wine and vino map to canonical sample types, while vessel suffixes remain meaningful: BT is the primary tank and BF is the small tank. The original laboratory wording is retained as evidence even when the display uses the normalized identity.

Enology Admin provides configurable variety laboratory standards. A standard is scoped by variety, sample type, process stage, analyte and unit and must be explicitly approved. The Laboratory view shows the applicable configured marker and never invents a variety target. Operators can add, edit or remove standards without changing code, while historical measurements remain visible.

Correct operating sequence

1. Import or upload the original report.
2. Confirm sample name, sample type, report date, and vintage.
3. Link grape reports to a variety and wine reports to the correct lot when possible.
4. Check units and decimal interpretation.
5. Review extracted results.
6. Save the review decision.
7. Allow the prediction or cellar decision-support refresh to run.

The current finding card analyzes only the newest measured report for the selected vintage and states its source and review boundary. The measured trajectory and projected endpoint compare only the same normalized wine

identity, sample type, process stage, analyte and unit. Historical endpoints use the final matching measured result from each earlier vintage. AI-assisted values are recalculated when a new numeric result arrives and are constrained to physically possible nonnegative values. An approved marker is displayed separately from a prediction and is never invented when a variety standard has not been configured.

The exploratory analyte chart overlays each vintage's recorded first-pick date as a labeled vertical harvest line; a dashed line identifies a current forecast when an actual pick is not yet recorded. Select a measured point in the exploratory, trajectory or like-for-like endpoint chart to open its retained original laboratory report. If an older source file is not retained, the same action opens the complete saved sample and result detail instead. Process-wide annual averages remain summaries and do not pretend to link to a single report.

Historical names are normalized for comparison while the original source label is retained. `Nerello`, `Nerello Mascalese`, `Narello Macalase` and vintage-suffixed forms refer to the same Nerello Mascalese identity. Documented Grecanico/Bianco-Grecanico and Grenache spelling variants are handled similarly. Grape, must, wine and other sample types remain separate and are not merged merely because their names match.

The durable laboratory model stores each prediction at its historical cutoff and scores it only against a later actual result. Future measurements never enter an earlier prediction input. Direction accuracy below the release threshold keeps projections review-gated. Nine laboratory records are currently flagged for genuine source review; the authoritative report index has no missing, incomplete, wrong-type or duplicate sample groups after normalization.

OPERATING REFERENCE

11. Cellar and tank labels

The cellar module tracks physical containers, wine lots, volume, phase, manual or sensor readings, and operations. A planned container assignment is not the same as physical occupancy.

Tank labels

The 2026 host cellar is **Raiti Emanuela, Contrada Lavina - Linguaglossa (CT)**. That host identity belongs in the legal-label context for wine made there. Legal denomination, wine identity, vintage, origin and processing phase belong to the wine lot and follow it through tank transfers; they are not casually inferred from a grape nickname. The system keeps unresolved legal wording visible for review instead of silently converting it into a DOC, IGT or varietal claim.

Cellar graphics show occupied operating vessels only. A tank that is empty, retired or out of service disappears from the visual cellar layout and its occupied-volume summary, but remains in Tank Records, maintenance history, legal history and physical-capacity reporting. Use **Mark empty** or a completed lot transfer to change occupancy; do not retire a reusable tank merely to hide it.

The production-capability view reconciles total fruit, measured liquid, full 750 ml bottle equivalents, occupied vessel capacity and remaining working space. It includes purchased fruit in production capability, while DOC/IGT scenarios remain conditional planning references rather than live entitlement or guaranteed output. A nominal tank fill is vessel occupancy until a measured liquid volume exists; it is not automatically a finished-wine yield.

Each physical tank keeps a stable label URL. Changing the assigned wine updates the label contents without changing the tank URL. Labels include:

- Tank identity and correct vessel type.
- Capacity, contained volume, and calculated level.
- Wine/vintage/lot information.
- Recent temperature, density, Brix, and pH readings.
- Small recent-reading charts when space permits.

- Legal company identity and cellar contact.
- Browser favicon, Apple touch icon, and Android/PWA identity.
- Responsive layouts for iPhone, Android phones, different tablet sizes, and print layouts.

On an iPhone or similarly sized touch phone, the legal label changes from the fixed cellar-display canvas to a normal vertically scrollable document. The title wraps instead of truncating, long legal and provenance fields use the full width, paired readings remain compact, and safe-area insets protect content around the notch and browser controls. The Tank Sensor process keeps a compact header visible while its KPIs, graphs, process health, and historical data scroll beneath it. Dedicated landscape tablets retain the fixed no-scroll layout.

Cellar displays are enrolled inside the VPN. They can then operate through the approved outside connector. The direct Nabu Casa address is the preferred outside label route; the URL remains available as a backup when QR provisioning is unsuccessful.

PLAATO V2 Pro automatic monitoring

PLAATO V2 Pro is available as a dedicated automatic tank-reading mode. In Home Assistant App Configuration, add the protected `plaato_api_key` generated in the PLAATO account integration page, then map each tank with `plaato_tank_mappings` in the form `T-01|PLAATO batch-or-device-ID,T-02|another-ID`. The mapping may use a batch, device, barcode, or fermenter identifier/name returned by the official PLAATO API. Keep identifiers unique.

Release 1.6.93 starts with the local-only `demo` key so the fermentation interface can be evaluated immediately. This sentinel never contacts PLAATO Cloud and generates moving, PLAATO-shaped telemetry for every active tank. Replace `demo` with the official API key and add explicit tank mappings before relying on live production readings.

After saving the add-on configuration, open Enology → Cellar → Vessel & reading, edit the mapped tank, and select **PLAATO V2 automatic**. The tank card and tank label then show:

- Live temperature, specific gravity and PLAATO Plato.
- Derived recent gravity drop in milli-SG per hour and a seven-day gravity trend.
- Original/final gravity, estimated ABV, attenuation, batch identity and start date when supplied by PLAATO.
- Battery, Wi-Fi, firmware, raw sensor frequency, reading age and connection state.

Selecting a tank in either Cellar or Enology opens the full fermentation panel. It contains the measured temperature and gravity graphs, a dashed forward gravity projection, every raw PLAATO sample retained in the seven-day window, batch and device details, and the complete linked cellar-process timeline. Calculated progress, phase, current alcohol estimate and finish timing state their method and confidence. An estimated finish is produced only when PLAATO supplies a final-gravity target and the recent measured rate is usable; it remains informational until the enologist confirms the cellar action.

The integration is read-only and cached to protect the sensor cloud and Home Assistant. It does not operate cellar equipment. PLAATO batch volume is context only and is never treated as a continuous tank-level measurement. Tank volume and pH remain manual or come from separately mapped Home Assistant sensors. A stale or unavailable PLAATO reading raises a visible monitor fault without erasing the last trusted cellar record. Switch the tank back to Manual before entering a manual sensor reading.

OPERATING REFERENCE

12. Olive oil records and cost logic

The olive section compares harvested olive weight, oil liters, conversion efficiency, total modeled cost, and cost per liter by year.

Confirmed 2024 result

- Olives: 332 kg
- Oil: 40 L
- Conversion: 8.3 kg olives per liter
- Oil yield: 12.048 percent by the system's liters-per-kilogram comparison

Editable 2024 cost assumptions

- Pressing: EUR 0.20 per kg of olives.
- Bottle and label: EUR 2.30 per 500 ml bottle.
- Planned bottles: 220.
- Supplier invoice: EUR 751 net plus 22 percent VAT.
- Annual labor: EUR 1,000.
- Harvest labor reference: EUR 540 at EUR 7 per tree, included in annual labor by default.

The supplier invoice is treated as including pressing and bottling by default, preventing duplicate cost counting. The correct formulas produce:

- Pressing component: 332 x EUR 0.20 = EUR 66.40.
- Bottling component: 220 x EUR 2.30 = EUR 506.00.
- Supplier gross: EUR 751 x 1.22 = EUR 916.22.
- Total with annual labor: EUR 916.22 + EUR 1,000 = EUR 1,916.22.
- Cost per actual oil liter: EUR 47.91.
- Actual 500 ml bottle equivalents from 40 L: 80.

The planned 220 bottles require 110 L, so they exceed the confirmed 2024 oil volume by 70 L. Cost assumptions can be edited and saved per year.

Olive harvest outlook

The Olive page displays an estimated harvest date and uncertainty window from exact Baiamonte olive harvest dates stored in MariaDB. Year-only notes are excluded. With only a small number of exact seasons, the system correctly reports low confidence. A current-year recorded harvest date replaces the estimate and is labeled as an actual date. The calendar estimate is planning support only; fruit maturity, oil accumulation, crop condition, weather, and mill availability still require field confirmation.

Separate olive treatment program

Every treatment belongs to either **Vineyard** or **Olives**. The two histories, forecasts, summaries, entry controls, learned weather precedents, effectiveness profiles, duration profiles, spray-window profiles, and FRAC-rotation sequences are separate. Vineyard disease-pressure scores and learned treatment cases are never reused to predict an olive treatment.

The olive treatment workspace screens **olive fruit fly** and **olive peacock spot** from the current weather, olive stage, trap or fruit findings, visible leaf symptoms, treatment history and paired before/after scouting. Weather may open a monitoring window, but it cannot select a product by itself. A calculated olive program requires matching field evidence, a current reviewed Italian olive crop-and-target use, a verified formulation and rate, exact grove area, sprayer and carrier configuration, inventory reconciliation, PHI clearance, compatible mixture evidence and Agronomist approval. The simulator supports current and historical olive scenarios, displays one-pass batch recipes, and saves nothing until an operator records and approves a plan.

The recorded 2026 olive treatment is retained with its exact products, dose basis, water volume, and calculated totals. Historical 2025 workbook entries are retained as completed olive work, but their missing products, doses, target, and safety details remain visibly unverified.

The system can suggest what to apply only after a database record confirms the exact crop, target, current Italian authorization, recent label verification, dose range, PHI, REI, and related restrictions. If those fields are incomplete, the product recommendation fails closed. The Agronomist must approve every candidate before application.

OPERATING REFERENCE

13. Labor, contractor invoices, and payments

Labor and contractor records move through review and payment states. Hours, fixed services, reimbursements, and invoice payments are stored separately enough to preserve an audit trail.

Standard workflow

1. Worker or manager records hours, service, expense, or invoice evidence.
2. The record is submitted for review.
3. An administrator corrects or approves it.
4. Approval locks the payable basis.
5. One or more deposits or payments can be recorded against an invoice.
6. The payment ledger determines paid, part-paid, or unpaid status.
7. Fully paid items do not reappear in the payment queue.

The current payment-integrity audit shows no paid-ledger mismatches, no fully paid items reappearing, no partial-payment status errors, and no verification holds. Six non-payable records are marked paid; they are tracked as a data-quality category and do not create a current payment mismatch. The formerly future-dated labor entry was corrected and is no longer an open source-review exception.

OPERATING REFERENCE

14. Finance

Fatture in Cloud is mirrored read-only. Vineyard Operations does not write back to the accounting provider. Finance access is limited to authorized finance users and is excluded from TV, kiosk, and public feeds.

Use the Finance section to review documents, parties, VAT context, balances, payment status, linked labor/service liabilities, and earned hospitality partner commissions. Tentative partner estimates are excluded from payable totals. Do not use the public dashboard as an accounting ledger.

OPERATING REFERENCE

15. Messaging and document intake

Gmail

Email has its own Admin -> Email tab, separate from operational Messages. The mailbox loads from the database cache first so opening the page does not wait for Gmail. Scheduled or manual receive refreshes update that cache, folders and hospitality routing. Inbox, spam and trash actions preserve Gmail identity and audit state; permanent deletion requires the explicit delete action. The system can classify vineyard information, extract proposed records, and prepare a reply draft. Nothing is sent until a person presses Send.

The scheduled intake reads lightweight headers for the 50 most recent messages and downloads complete message bodies or attachments only when that message has not already been ingested. It processes at most eight new full messages in one poll, allowing a backlog to drain across successive runs without blocking weather, device, or communications work. Gmail connections have a 30-second network timeout; a failed run is retained for recovery and disappears from current errors after the next successful run.

WhatsApp

Messages from approved numbers can enter Manager, Reporter, or Guest/review workflows. Unknown numbers are quarantined. Group and direct-message behavior is intentionally separated. Media and message bodies are preserved as evidence. The current bilingual menus cover Operations, Agronomy, Vintage, Enology and Tank Sensor, Olives, Estate Systems, Hospitality, team/finance administration, and structured record entry without requiring a separate phone app. Administrator-only summaries remain gated, and guest hospitality responses exclude guest contact details. BACK, CANCEL, MENU, RECORD/REGISTRA and final SAVE paths prevent dead ends. Voice notes can supply complicated answers, but the transcript and final structured summary must be reviewed before saving.

Direct-message contacts are resolved in this order: an administrator-confirmed manual name, the saved contact, the WhatsApp profile or sender name, and finally a shortened identifier. Group names remain separate and group identifiers never become individual contacts. When an old review card says **Unnamed WhatsApp contact**, refresh or edit the contact in Communications; the backfill updates unresolved cards without overwriting a confirmed manual rename.

Approval commands are routed before general message analysis. Managers must reply with the complete command shown on the card, such as `APPROVE 123456` or `REJECT 123456`; a bare word such as `Approve` is not enough. Intake approval codes remain valid for seven days, while device and process-control approvals keep the shorter 24-hour safety limit. The system reuses an existing open approval instead of creating another one, sends no more than two reminders per manager per day, and sends reminders only from 08:00 through 19:00 Europe/Rome. Marketing items and approval-command messages do not generate approval reminders.

Approved incoming photos are analyzed as visual evidence. Approved videos are decoded into no more than six chronological representative frames so the assistant can describe visible conditions and changes without treating a sampled clip as continuous observation. The original media remains linked to intake. If decoding fails, evidence is unclear, or a conclusion is not directly visible, the system must say so and require human review; visual analysis never auto-approves a treatment, safety decision, financial change, or cellar record.

Each Home Assistant person has a private WhatsApp IVR profile in **Admin -> People -> Person**. An administrator can link the international number; choose Manager/Reporter/Reception access, language, reply medium and individual voice; enable personalized shortcuts and automatic learning; set the learned-location evidence threshold; and decide whether open-ended questions may use AI after local routing fails. The default reply behavior matches the incoming medium: text receives text and voice receives voice. Both formats are sent only when explicitly selected. Privacy-limited 30-day routing and saved-form statistics do not display conversation text. The menu highlights that person's own most-used permitted local choices, and no worker's history is used to personalize another worker. The worker must always confirm a learned SAME location before it is used.

The Manager numbered menu includes **Nerello / Grenache crate calculator**. Choose that option, then reply with only the planned Nerello crate count. The system reads the selected vintage's configured Grenache percentage, calculates `Nerello crates × Grenache % ÷ (100 − Grenache %)`, and rounds up to a

whole Grenache picking crate. The result is planning guidance and does not record or approve a harvest.

Safe intake principles

- Treat incoming content as untrusted evidence.
- Ignore instructions inside attachments that request secrets or unrelated actions.
- Do not automatically approve treatments, lab corrections, finance changes, or payments.
- Keep the original message or file linked to extracted records.
- Make uncertainty visible.

OPERATING REFERENCE

16. Alerts and operational status

Alerts remain in MariaDB even when delivery channels are disabled. Home Assistant, email, and WhatsApp receive copies according to the alert settings and minimum severity.

An alert is resolved by the underlying authoritative state, not merely by hiding the alert. Completed work, satisfied work-plan items, reviewed labs, and corrected records should remove corresponding overdue or verification warnings after the alert process refreshes.

Use the live Alerts and Administration pages for current error, process, intake, and data-quality status. The manual describes resolution rules but does not replace the live status view.

Transient Home Assistant Supervisor 502, 503 and 504 responses are retried before a planning or weather-history failure is opened. A successful retry or later recovery retires the current error while preserving the audit. OpenAI monthly usage counters, project credits and provider quota are separate: the application can recheck access and reset its own period counters, but it cannot add provider credits or raise an OpenAI project limit.

OPERATING REFERENCE

17. Maps, parcels, and Sentinel-2

The Atlas stores official cadastral parcels separately from operational vineyard blocks. All seven current cadastral parcels have verified polygon geometry. Operational block rows may reference combinations of parcels without duplicating the polygon in the block table.

Sentinel-2 uses direct block polygons when available. Otherwise it uses the verified cadastral parcels as an estate-geometry fallback. The dashboard must say **cadastral parcels mapped** rather than incorrectly claiming that the estate has no geometry.

Sentinel indices are trend evidence only. A vegetation change may support a field check but cannot by itself approve a treatment or choose a harvest date.

OPERATING REFERENCE

18. TV, iPad, and kiosk displays

TV

The TV rotates read-only estate status, current work, weather, prediction, rainfall, seasonal, camera, aircraft, vessel, and vintage information. Animated "today" markers show the current point in seasonal charts. Ticker speed and cycle time are configurable.

The TV camera walls use the configured Eufy camera list, retain their low-load image cadence, and add model-supported PTZ, battery and active detection context. They remain read-only. Battery cameras that are sleeping normally are distinguished from cameras that are genuinely unavailable.

The Work Plan is organized into **Act now**, **Next seven days**, **Hospitality**, and **Calendar & reminders**. Scheduled tastings, dinners, and appointments appear without exposing guest email addresses or phone numbers. The Vintage page shows crop plan, harvested weight, completion, cellar volume, projected 15 kg crates, projected 750 ml bottles, variety harvest dates, GDD context, historical vintages, and forward outlook.

iPad

The `ipad` profile is a larger, finance-free operations dashboard with weather, solar, energy, lights, cameras, security, vineyard information, media, and AI links.

Eufy Camera Center

Open **Operations -> Cameras** for the operational camera workspace. The summary reports healthy, sleeping, unavailable, activity, low-battery and PTZ counts; filters separate access, vineyard and building cameras. Home Assistant cards use the current app-style camera cover maintained by the bridge, not the smaller event-thumbnail entity. Covers are refreshed serially on a bounded cadence and cached, so opening the page does not start every camera stream or wake every battery camera. Live viewing is explicitly started for one camera and is stopped when its dialog closes.

Movement, 360 rotation, saved positions and calibration appear only when that exact camera advertises the capability. Camera power, light, motion detection, motion tracking, auto night vision and recording switches are likewise capability-checked. Administrators may save or delete supported PTZ positions and send a verified canned doorbell response. The server independently rejects unsupported or unauthorized commands. Alarms, locks, microphones and speakers are intentionally excluded from ordinary operations.

Eufy's on-device motion, person, vehicle, pet and doorbell classifications are copied into a durable estate event log with their camera, area, transition times and cached evidence link. No raw recognition payload is stored and the application performs no new facial inference. Operators may mark events reviewed. Genuine camera outages must persist for fifteen minutes before creating an alert; two or more outages in one area create shared power, network and station guidance. Those alerts automatically enter Today, TV, WhatsApp and the normal review inbox.

The always-on Vineyard North stream also supplies a separate, low-load fixed-view evidence pipeline. It samples no more than hourly, rejects dark, washed-out or low-detail frames, measures scene and canopy-color change locally, and invokes contextual AI interpretation only for the first suitable baseline, the daily review, or a material change after a bounded interval. The resulting Today and TV card can show visibility, operations, canopy signal, frame change and a conservative inspection prompt. Weather, recent work, scouting and treatments are explanatory context only. Camera evidence never diagnoses disease, identifies a person, approves a treatment, or replaces a field inspection. Ordinary review alerts require repeated suitable evidence; a high-confidence fire or smoke observation may escalate immediately.

Tablet labels

Tank-label devices use a dedicated enrollment flow. Each registered tablet has a permanent label URL and its own QR code. Already visited label pages, branding, and the last successful tank reading are cached so a label can reopen and remain useful during a temporary connection outage.

For a factory-reset Android tablet, open **Enology -> Tablet setup** and scan the managed-device QR during Android's initial setup. The add-on hosts a checksum-pinned Fully Kiosk Browser EMM installer locally and supplies a private one-page profile that launches the assigned label after boot, stays in landscape kiosk mode, recovers after connectivity changes, and enables local-network Remote Admin. The separate Start URL QR remains available as a manual fallback. Reprovision controls should replace the device identity cleanly rather than creating duplicate tablets.

OPERATING REFERENCE

19. Hospitality

Hospitality is an internal, low-volume booking and service workspace for private estate experiences. The current release supports one private guest party at a time and is designed for tastings and dinners for approximately 6 to 12 guests.

Packages

Administrators and Hospitality Managers can maintain the active package name, description, minimum and maximum guest count, duration, base price, per-person price, deposit requirement, inclusions, and preparation notes. The initial packages are:

- Private Estate Tasting: 1 to 6 guests.
- Cellar Tasting & Pairing: 2 to 8 guests.
- Private Estate Dinner: 6 to 12 guests.

Pricing remains configurable and quote-based. Saving a package changes future selection options; it does not silently rewrite a confirmed guest quote.

Gmail inquiry routing

Hospitality Admin controls a list of inbound Gmail subject phrases. The initial phrase is **Inquiry about Reserve Tasting**. Matching is case-insensitive and ignores common Reply and Forward prefixes. More phrases may be added one per line without changing protected Gmail credentials.

Matched public messages enter **Hospitality -> Guest inquiries** even when the sender is not on the vineyard operations allowlist. They are safe requests, not approved operational records. Attachments continue through the protected intake safeguards. Existing matching Gmail records are reconciled as well as new mail.

Open an inquiry to read it, prepare and explicitly send an email response, change its state, add internal notes, or convert it into a reservation. Conversion pre-fills the available guest identity and source, then requires the operator to choose the package, date, time, guest count, and booking state. The inquiry is linked to the resulting reservation.

Reservation workflow

1. Record the inquiry and select a package, date, time, and guest count.
2. Capture the guest name and only the contact details needed for service.
3. Record dietary restrictions, accessibility needs, celebration details, preferences, internal notes, quote, and deposit.
4. Move the reservation through requested, confirmed, arrived, completed, cancelled, declined, or no-show states.
5. The server rejects overlapping confirmed or arrived experiences so only one private guest party is committed at a time.
6. Send email or WhatsApp confirmation only by pressing the explicit communication action. Phone calls and notes can be recorded without sending a message.
7. Open any reservation to update its details or delete it after confirmation. Deletions remain in the audit trail; a linked inquiry returns to the responded queue.

The hospitality communication log preserves channel, subject or summary, operator, delivery state, and timestamp. Guest contact details remain inside the protected workspace and are intentionally omitted from TV

data.

Partner management and commissions

Hospitality Admin includes a partner database for concierges, hotels, travel advisors, event organizers, and other referral sources. A partner record stores the organization and contact details, commission method and default rate, tax/payment notes, status, and internal operating notes.

Assigning a partner to a reservation creates a reservation-specific commission snapshot. A percentage rule uses the eligible reservation amount; a fixed rule uses the saved fixed amount. The operator may change the reservation commission without changing the partner's future default.

Commission states follow the reservation lifecycle:

1. Inquiry or requested reservation: estimated only; not payable in Finance.
2. Qualifying confirmed/arrived/completed reservation: earned and due.
3. Approved commission: authorized for payment.
4. Partial payment: remaining balance stays due.
5. Paid: payment ledger equals the earned amount.
6. Cancelled, declined, no-show, or otherwise disqualified reservation: void, with history retained.

One commission can receive multiple payments. Each payment preserves amount, date, method, reference, note, and operator. Corrections are audited; deleting a payment is restricted and does not erase the original commission or reservation relationship. Finance includes only earned outstanding partner balances and separates them from tentative estimates.

OPERATING REFERENCE

20. Register

The Register is a large-touch sales workspace intended for a tablet in the United States. It combines three controlled sources:

- Sellable winery products, prices, and stock mirrored read-only from Fatture in Cloud.
- Active hospitality packages maintained locally under Hospitality.
- Authorized manual items created in Register Admin.

The Register has four pages: **Sale**, **Inventory**, **Ledger**, and **Admin**. Completed sales are written to the local MariaDB ledger. Register sales are not posted to Fatture in Cloud in release 1.6; the future posting fields exist but remain locked off until a separate reconciliation release is approved.

Checkout, currency, and language

1. Tap catalog items, adjust quantity or editable price, and apply any authorized line discount.
2. Select **EUR** or **USD** directly at checkout. EUR remains the accounting and VAT reporting base.
3. Select **English** or **Italiano**. This selection controls the PayPal checkout locale and the printed receipt language.
4. Confirm the active PayPal Business account. EUR prefers the Italian account when configured; USD prefers the US account. An authorized operator may select the other configured account.
5. Complete payment by cash, hosted PayPal/card checkout, or operator-confirmed PayPal Tap to Pay.

For every transaction the database preserves the subtotal, discount, VAT, and total in EUR; the actual tender currency and amount; the exact saved USD-per-EUR rate; the selected PayPal account; the checkout language;

the payment or capture reference; and the operator identity. The receipt and CSV export carry the same reconciliation fields.

The ECB rate refresh is a convenient reference rate, not a claim that it is a bank settlement rate. Administrators may save the actual checkout rate when required; the saved rate remains attached to the sale even after the current setting changes.

PayPal account setup

Protected Home Assistant App Configuration holds credentials for both accounts:

- **US PayPal Business:** the existing PayPal client ID and secret.
- **Italian PayPal Business:** the Italian PayPal client ID and secret.
- **Environment:** Sandbox for testing or Live for real payments.

The browser receives only the selected public client ID. Client secrets remain server-side. Never enter, store, or transmit raw card numbers through the Baiamonte application.

Tap to Pay and receipts

PayPal Tap to Pay runs in the PayPal POS application on an NFC-capable phone. After PayPal approves the exact amount, the operator records the PayPal transaction reference in the Register. This is explicitly an operator confirmation; the web browser does not pretend to verify the contactless charge in real time.

The system receipt can be printed through the browser's configured system printer. It is labeled as an operational, non-fiscal receipt. The monthly Ledger shows both collected tender and EUR base values and exports a UTF-8 CSV for reconciliation.

OPERATING REFERENCE

21. Scheduled processes and recovery

Process	Typical interval	Function
Complete refresh	60 min	Recovers missing or overdue subsystems
Planning	15 min	Syncs Baiamonte Calendar and Tasks
Weather	15 min	Imports on-site current and historical weather
Forecast sources	180 min	Refreshes ensemble, SIAS, Sentinel, and seasonal evidence
Harvest prediction	90 min plus queued refreshes	Recalculates harvest readiness
Gmail	15 min	Imports approved messages and attachments
WhatsApp	15 min	Refreshes connection and approved catalogs
Finance	360 min	Pulls read-only accounting data
Social	360 min	Refreshes Facebook and Instagram independently
Energy evidence	5 min	Records battery, solar, generator and protected-load context
Camera health and covers	Staggered bounded cadence	Refreshes one low-load source at a time
Etna	15 min	Updates volcanic and seismic context
Disease	30 min	Updates disease and heat-stress support
Alerts	5 min	Evaluates operational warnings
Public website	15 min	Publishes the approved vintage and harvest feed when configured

The complete refresh is a recovery sweep. It does not blindly duplicate every job; it reruns subsystems that are missing or substantially overdue. Manual **Run complete update now** executes the configured subsystems immediately.

OPERATING REFERENCE

22. MCP and Codex integration

The Baiamonte MCP server is a constrained interface for Codex and approved automation.

Connection

```
[mcp_servers.baiamonte]
url = "http://192.168.0.10:8100/mcp"
bearer_token_env_var = "BAIAMONTE_MCP_TOKEN"
```

The endpoint is available locally or through the VPN. It requires the bearer token for every MCP request. Unauthenticated requests are rejected.

Tool groups

The live server exposes 23 tools:

- **Status and planning:** processing status, work plan, Apple reminder lists, Baiamonte reminders, treatment reminders.

- **Review intake:** queue review item.
- **Read models:** vineyard overview, harvest report, lab history, lab decision context, weather summary, open issues, finance overview, funding report, record search.
- **Confirmed writes:** tasks, vineyard records, lab results, lab reviews, financial documents, cash transactions, and funding-requirement updates.

MCP writes are currently enabled. Every write tool still requires explicit confirmation of the exact record. Read tools can be used for diagnosis without changing data.

Version interpretation

Home Assistant app version 1.10.59 is the operator-facing release. The MCP protocol handshake may report a different server-framework version; that framework identity must not be used as the Vineyard Operations update version.

OPERATING REFERENCE

23. Security and privacy

- Keep passwords, API keys, and tokens in Home Assistant add-on configuration or environment variables.
- Never place a bearer token directly in documentation, screenshots, logs, or a repository.
- Use role-based Home Assistant accounts.
- Keep finance out of shared displays.
- Keep guest contact, dietary, accessibility, and private-event details out of shared displays and public feeds.
- Keep MCP on the local network/VPN unless a separately authenticated tunnel is deliberately configured.
- Review MCP writes before enabling them on a new client.
- Use recoverable deletion for mail and records where supported.
- Preserve original evidence and audit events.

MCP credential rule

The Baiamonte MCP connection uses `BAIAMONTE_MCP_TOKEN`. Tokens belong in a protected environment variable or connector-secret mechanism, never inside a URL, screenshot, ticket, or documentation file. Rotate a token immediately if it is exposed.

OPERATING REFERENCE

24. Troubleshooting

A page has no data

1. Confirm the selected year.
2. Refresh the page once.
3. Check Administration -> Processing Log.
4. Check whether the source process is enabled and recent.
5. Verify that the record is approved and assigned to the correct year/vintage.
6. Do not enter replacement zeroes for unknown data.

A map looks stale

Refresh the page after geometry changes and reopen Atlas so the map recalculates its visible size. If parcels still appear outside Sicily or the geometry count is wrong, check the Atlas process and saved cadastral geometry before editing a block.

An alert remains after work is done

Confirm the authoritative work record is completed, then run or wait for the Alerts process. Hiding a card does not resolve the underlying alert.

A payment reappears

Check its payment ledger, invoice total, deposits, payment timestamps, and approval state. The current integrity audit reports no fully paid invoices reappearing.

Hospitality does not open

Confirm the Home Assistant username is linked to an Administrator or Hospitality access profile. A title alone does not grant access unless the profile is saved. Administrators configured in protected add-on settings are also recognized before a matching People profile is created.

Register or a PayPal account does not open

Confirm the Home Assistant username has Administrator, Register, Cashier, or Hospitality Manager access. In Register Admin, the active connection line identifies Fatture in Cloud and each configured PayPal account. If an account shows **not configured**, add that account's client ID and secret in protected App Configuration and restart the add-on. Use Sandbox until both the amount and currency have been verified end to end.

A hospitality event is missing from the TV

Confirm the reservation is requested, confirmed, or arrived and has a future or recent start time. Cancelled, declined, completed, and no-show reservations remain in history but do not occupy the active TV schedule.

MCP does not connect

1. Confirm VPN/local reachability to port 8100.
2. Confirm `BAIAMONTE_MCP_TOKEN` is present in the client environment.
3. Confirm the Home Assistant MCP bearer token is configured.
4. Confirm the client host is permitted.
5. Restart Codex after configuration changes.
6. Use `processing_status` as the first read-only test.

OPERATING REFERENCE

25. Weather stations and environmental telemetry

GW2000A is the primary estate weather gateway. GW2000B supplies the second gateway path and the WH35 leaf-wetness channel. Home Assistant is the normalized integration boundary: the application reads entity state and recorder history, then stores normalized observations for graphs, disease pressure, treatments, harvest and alerts.

The Weather page exposes current cards as graph actions. Temperature, humidity, rain, wind, pressure, solar, UV, soil and leaf wetness open the applicable recorded series rather than behaving as decorative tiles. A missing channel remains unavailable and is never shown as zero. Daily rain is cumulative; rain rate is instantaneous. Gust and maximum daily gust are distinct. Wind direction is circular data and must not be averaged like an ordinary scalar.

Freshness uses Home Assistant's report timestamp when present. This keeps a stable unchanged reading live when the device continues reporting the same value, while unavailable or genuinely stale telemetry is still rejected.

GW2000 and WH35 recovery

1. Confirm both gateway clients are online in Omada and retain their reserved IP addresses.
2. Confirm the primary weather entities and the GW2000B leaf-wetness entities are registered, enabled and not hidden in Home Assistant.
3. Check the gateway local live-data page. If GW2000B has no leaf channel, check its sensor-information page.
4. A remembered WH35 sensor with no RSSI or signal means the gateway is online but the physical transmitter is not being received. Replace/check the WH35 battery, inspect its enclosure and placement, then re-pair or wake the transmitter near GW2000B.
5. Reboot the gateway only after preserving its custom-upload settings. Do not publish callback paths, tokens or credentials in documentation or screenshots.
6. After the WH35 reports, verify the Home Assistant entity updates, the Weather card shows a value, the graph records points, and the managed iPad and vineyard dashboards show leaf wetness and battery health.

The current software path, Home Assistant integration, Omada clients and gateway upload configuration are verified. A missing leaf value with no gateway radio signal is a physical WH35 battery/radio/sensor issue, not an application fallback to invent.

OPERATING REFERENCE

26. Energy, batteries, solar room, water and local devices

The Baiamonte Power module models the complete three-battery bank as one protected reserve while preserving individual-bank telemetry for completeness and fault diagnosis. State of charge, charge/discharge power, solar input, house load, reserve forecast and protected runtime must use the configured battery set. A missing bank is an incomplete configuration warning, not zero capacity.

Direct Growatt production is preferred. When it is unavailable, the application can show the labeled Baiamonte solar estimate after subtracting only verified generator input. Estate breaker or inverter-output meters must not be mislabeled as generator production. With no verified running generator meter, generator input is `0 W` and explicitly **stopped/unmetered**.

The solar-room fan is a protected local control. When battery charging exceeds 2,000 W, the fan must be on, subject to telemetry validity and the emergency heat override. Normal temperature-based cooling is suspended if required temperature or battery-bank telemetry is unavailable or stale; the warning explains which evidence is missing. The emergency heat path remains available. Never bypass the guard by converting a missing sensor into zero.

Tuya equipment is operated through LocalTuya wherever supported. Each migrated device needs a stable Omada DHCP reservation, a local key/entity mapping, a local availability check and a cloud-independent control test. The Tuya cloud integration is not the normal control path. After router, access-point or device power

cycling, verify each device reconnects at its reserved address and that Home Assistant commands work while the internet path is unavailable.

Water operations combine the cistern camera estimate, physical reference observations, deliveries and consumption evidence. Camera estimates remain reviewable visual evidence; a confirmed gauge or dip measurement is the preferred calibration reference.

OPERATING REFERENCE

27. Dashboards, kiosks, TV and display deployment

The repository is authoritative for four Home Assistant YAML dashboards: Vineyard Overview, Display Panel, Baiamonte iPad and Admin. Managed installation copies them to `/config/baiamonte_dashboards`, merges only the marked registration block into Home Assistant configuration, backs up the file and runs a configuration check. A failed check restores the backup.

The compact NSPanel uses `/vineyard-display/home`. Its first view must fit the device without wasted vertical space and show battery state of charge alongside the essential kitchen, security, gate, fridge and dishwasher controls. The iPad uses `/vineyard-ipad/home` and includes weather, leaf wetness, energy, solar, power, lights, cameras, security, media and vineyard links without Finance. Each device still needs its one-time Home Assistant default-dashboard selection and kiosk/full-screen configuration.

The LAN TV is a separate read-only rotation. It includes Today, Vintage, Intelligence, cameras, aircraft, vessels, Work plan, Cellar and Weather. Purchased fruit, current measured litres, production capability and the current selected vintage must agree with the private application. TV camera refreshes are sequential and snapshot-only; opening a camera page does not start every stream.

After every dashboard release, verify the installed files, Home Assistant configuration, all four URLs, role visibility, NSPanel layout, iPad layout and TV rotation. Do not grant Finance or administrator permissions to display accounts to make a card render.

OPERATING REFERENCE

28. Social and public website publishing

Facebook and Instagram refresh independently. A successful Instagram refresh remains healthy when Facebook post history is blocked by a permission, but the Facebook channel stays visibly stale or degraded. Cached posts are labeled as cached; a scheduled failure cannot masquerade as a live refresh. Official Instagram relationship ZIP/JSON imports update relationship comparisons without scraping, automating follows or attributing aggregate changes to a person without evidence.

The public website receives only the approved vintage and harvest feed. Private finance, guest, message, worker, security, cellar-detail and review data never enters it. After harvest completion, the public feed advances to the next planning vintage under the estate-variety completion gate described in Section 7. The page title, structured metadata, sitemap/indexing files and Google-facing search content must use the same current public state.

Shopping and Visit navigation may be temporarily hidden without deleting their content or breaking Home, About, Wines or Contact. When items are hidden, the remaining menu uses an even compact layout on desktop and mobile with no empty gap. Re-enabling the sections is a presentation/configuration change, not a data restoration.

OPERATING REFERENCE

29. Deployment, updates, backup and verification

GitHub `main` is the published source. Every application change increments `config.yaml`; documentation delivered inside the app is an application change. The normal release path is: create a focused branch, run the complete tracked test suite, push, open and merge a pull request, reload the Home Assistant app store, update the installed Baiamonte Vineyard app, and verify the installed version.

Post-install verification requires all of the following:

1. The filtered Home Assistant app status reports the expected installed and latest version with the app started.
2. `/health` reports the application and database healthy.
3. The served application assets carry the expected release version.
4. MariaDB migrations completed without a partial or repeated failure.
5. The live System Documentation page reports the current release and generated manual page count.
6. The PDF opens, every preview page renders, and current operations do not show unresolved processing errors created by the release.
7. Public publishing, managed dashboards and Home Assistant Core are restarted only when their changed component requires it.

Home Assistant backups, MariaDB contents, credentials, API keys, website tokens and private operational exports do not belong in Git. Preserve an appropriate Home Assistant backup before configuration or integration changes. Restore database and protected configuration together when recovering a full installation, then validate migrations, identities, integrations, dashboards, scheduled processes and publication health before normal operation resumes.

OPERATING REFERENCE

30. Release 1.10.57 complete documentation baseline

Release additions

- Reconciles this manual with the complete installed system rather than the older 1.7.0 baseline.
- Documents estate and purchased fruit accounting, production capability, active-lot enology, configurable variety laboratory standards, occupied-vessel graphics and host-cellar legal context.
- Documents current approvals, contact-name recovery, social partial health, public-vintage rollover, local Tuya control, the three-battery power model and the solar-room fan safeguard.
- Documents both GW2000 gateways, WH35 radio diagnosis, report-timestamp freshness and graph-enabled weather cards.
- Expands the live administrator service/API registry and removes the fixed manual-preview page count.
- Builds Appendix A from every recorded published release from 1.6.0 through 1.10.57.
- Documents the harvest AI decision window, current-fruit evidence gate and weather-independent cache that prevent premature paid analysis.

Earlier release additions remain available in Appendix A instead of being repeated in this current-release snapshot.

Verification completed before publication

- The generated PDF and every in-app preview page must render without clipped, overlapping or missing content.
- Automated tests verify protected documentation access, secret exclusion, dynamic preview count, release metadata and complete changelog coverage.
- The complete tracked suite must pass before publication, followed by the live app, database, asset-version and documentation checks in Section 29.

Source-review items remain visible rather than being guessed. Laboratory assignments, treatment safety details, inventory receipts, and planned container sharing stay in their respective review queues until authoritative evidence resolves them.

OPERATING REFERENCE

31. Release 1.10.58 performance and load management

Release additions

- The Today workspace renders from its core dashboard, task and alert data without blocking on the Home Assistant state inventory or daily weather forecast service.
- Current weather refreshes independently after the first useful screen is visible. Stored vineyard weather remains available while that live refresh is in progress.
- Browser read traffic is limited to six concurrent requests. This protects the application thread pool and MariaDB connection pool when the interface prepares its many operational workspaces.
- Versioned JavaScript and stylesheet assets continue to use one-year immutable browser caching, while unversioned recovery assets continue to revalidate.
- The dashboard continues to load the complete operational dataset in the background; the concurrency limit changes scheduling only and does not remove records, analytics, controls, or write operations.

Performance verification

- The live add-on remained lightly loaded during diagnosis at approximately 0.3 percent CPU and 3.9 percent memory, excluding hardware saturation as the cause.
- The core 2026 dashboard returned in approximately 0.16 seconds while a cold Home Assistant weather request required approximately 2.1 seconds. Removing that weather request from the blocking first-render path therefore removes the measured startup bottleneck.
- The deferred workspace previously launched more than twenty reads simultaneously. The browser now admits at most six, preventing the measured database and application contention that lengthened individual dashboard responses during a startup burst.

OPERATING REFERENCE

32. Variety-specific grape-to-wine yield planning

Operational ratios

- Every production variety has its own predictive finished-wine yield in liters per kilogram of grapes received.
- The prediction is a weighted ratio: the sum of reconciled finished-wine liters for that variety divided by the sum of reconciled grape kilograms for that variety across prior vintages.
- Grenache and its legal Alicante identity resolve to the same operational production stream. The legal name shown on a cellar label remains controlled by the legal-label record.
- Purchased fruit is included in received-fruit kilograms and production output. It remains excluded from estate vineyard yield and estate-harvest completion.

Evidence separation

- Current measured cellar liquid divided by current received fruit is displayed as the observed L/kg ratio.
- Initial measured liquid, when recorded, is retained separately from current liquid after racking or loss.
- Current observations do not train the prediction. A vintage contributes to the predictive ratio only after its per-variety grape kilograms and finished-wine liters are reconciled in the vintage summary.
- Vessel occupancy that includes skins or solids is not treated as measured juice or finished wine. Tank 39's full-vessel note therefore remains operational evidence, not a 1,000 L liquid measurement.

Configuration and use

- Enology configuration provides an optional L/kg override for Nerello Mascalese, Grenache/Alicante and Grecanico. A blank override uses the learned ratio; if no trusted variety history exists, the estate fallback is used and labeled.
- Today, harvest planning, forecasts, production capability, TV summaries and bottling estimates sum each variety's kilograms multiplied by that variety's own predictive ratio.
- The system shows the prediction source, historical vintages and confidence. Actual measured cellar or bottling volume replaces the estimate as authoritative production evidence.
- Planning ratios support capacity, vessel and packaging decisions; they do not approve picking, pressing, treatment, legal denomination or bottling.

OPERATING REFERENCE

Appendix A. Complete release coverage: 1.6.0-1.10.59

The operational chapters above describe the cumulative current system. For completeness, the following appendix includes every published change recorded from 1.6.0 through 1.10.59. It is generated directly from the authoritative project changelog when the manual is built, so maintenance fixes and smaller workflow changes are not omitted from the owner record.

Release 1.10.59

- Calculates grape-to-finished-wine yield independently for Grecanico, Grenache/Alicante and Nerello Mascalese instead of applying one estate-wide factor to every variety.
- Learns each predictive L/kg ratio from weighted, reconciled prior-vintage grape kilograms and finished-wine liters, while retaining an explicit estate fallback and optional enologist override.
- Shows received kilograms, measured current liters, observed L/kg, predicted L/kg and predicted liters together across Today, harvest planning, projections, production capability, TV data and bottling planning;

purchased fruit is included in production ratios.

- Keeps current tank liquid and must-with-skins occupancy out of the learning set until a vintage is reconciled, preventing transfers or incomplete fermentation volumes from corrupting future planning.

Release 1.10.58

- Lets the Today screen render from its fast core dashboard, task and alert data without waiting for the slower live Home Assistant weather inventory and forecast request; weather now fills in asynchronously as soon as it is ready.
- Caps browser read traffic at six concurrent requests so the large set of deferred workspaces cannot saturate the application thread pool and database connections during startup or refresh.
- Keeps versioned assets immutable and preserves all existing dashboard data, controls and background loading while improving interactive response under concurrent use.

Release 1.10.57

- Keeps deterministic harvest projections and weather/GDD recalculation active year-round while deferring paid harvest AI until the final 60 days before a plausible harvest.
- Requires a current maturity sample or approved grape laboratory result before paid harvest review can run, so the completed 2026 vintage cannot spend against next year's distant harvest.
- Reuses the harvest AI result until decision evidence changes; routine weather-only refreshes no longer invalidate the paid-review cache, and the AI cost page explains that existing harvest-projection totals are historical usage.

Release 1.10.56

- Prevents Gmail intake timeouts by refreshing the recent mailbox from lightweight headers and downloading complete MIME bodies or attachments only for messages that have not already been ingested.
- Limits new full-message downloads to eight per poll so a backlog drains over successive runs without blocking the scheduler, and applies a 30-second Gmail connection timeout.
- Preserves cached mailbox visibility, sender trust, hospitality routing, attachment intake, audit records and automatic retry behavior while eliminating repeated downloads of unchanged mail.

Release 1.10.55

- Reconciles the complete operator manual with the current system across vineyard, cellar, laboratory, purchased fruit, hospitality, register, communications, social, weather, energy, dashboards, public publishing, deployment, security and recovery.
- Expands the live administrator documentation registry with the cellar-label gateway and current production, enology, weather, device, communications and approval routes.
- Generates the in-app manual preview from the installed PDF page count instead of a stale hard-coded total, and embeds the complete published release history from 1.6.0 through the current release.

Release 1.10.54

- Keeps unchanged GW2000 readings, including the leaf-wetness sensor received through GW2000B, live by using Home Assistant's report timestamp rather than treating an unchanged value as stale after ten minutes.
- Uses the same report timestamp for the displayed weather-station freshness, while continuing to reject unavailable or genuinely stale telemetry.

Release 1.10.53

- Advances the public website to the next planning vintage after Grecanico, Grenache and Nerello Mascalese all have recorded estate harvests.
- Shows recorded harvests as complete even when an older provisional plan was not manually closed, and keeps purchased fruit out of the completion gate.
- Keeps the public website heading synchronized with the vintage selected by the live vineyard feed.

Release 1.10.50

- Hides empty, retired and out-of-service tanks from the Cellar graphics while preserving them in Tank Records, history and physical-capacity reporting.
- Updates the Cellar summary and vessel coloring to use only the occupied tanks shown on the graphics page.

Release 1.10.49

- Uses the available WhatsApp sender/profile name when a direct chat only has an unresolved private device identifier.
- Backfills existing direct-message review cards and keeps confirmed manual contact renames authoritative, while leaving group names separate.
- Removes WhatsApp group identifiers from the individual contact list.

Release 1.10.48

- Reconciles the three verified Enodoro electronic invoices when Fatture in Cloud stores them as non-detailed expenses with empty product rows.
- Uses live Fatture in Cloud detail rows whenever available and falls back only to verified supplier-invoice evidence tied to the exact invoice number.
- Converts invoice quantities expressed in kilograms to grams correctly before purchase-versus-use reconciliation.

Release 1.10.47

- Reconciles the historical Grenache evidence card with Wendy's current note confirming 61 crates for the 1,078 kg purchased-fruit intake.

Release 1.10.46

- Fixed the production-capability reference migration for MariaDB by using an explicit cross join and idempotent insert, allowing the add-on to recover cleanly after the earlier partial migration attempt.

Release 1.10.45

- Fetches the detailed Fatture in Cloud record for Enodoro purchase invoices when the invoice-list response omits product rows.
- Reconciles the official Enodoro article codes into enology inventory and the purchase-versus-use ledger automatically.

Release 1.10.44

- Adds an explicit MariaDB query boundary to the production-capacity scenario seed so migration 205 completes during startup.

Release 1.10.43

- Adds an evidence-backed production-capability panel to Harvest and Cellar with actual fruit, measured liquid, full-bottle equivalents, real vessel occupancy and working space.
- Keeps the supplied DOC/IGT quantity scenarios visible but explicitly conditional, alternative and separate from live SIAN entitlement or guaranteed production.
- Reconciles Wendy's current chronology through 10 October, identifies purchased Grenache as physical Tank 39, adds the missing Nerello and Grenache fermentation readings, and preserves unresolved tank, volume and inoculation fields.
- Retains the owner-confirmed 1,078 kg purchased Grenache intake while exposing the 1.55 kg conflict between the notebook's gross/crate-tare arithmetic and stated net total.

Release 1.10.42

- Recognizes Enodoro receipts from official article codes and invoice descriptions even when the Fatture in Cloud supplier payload omits or reformats the supplier identity.

Release 1.10.41

- Uses an idempotent `INSERT IGNORE` seed for broad MariaDB compatibility during the Enodoro mapping migration.

Release 1.10.40

- Makes the Enodoro supplier-mapping seed compatible with the estate MariaDB parser so the purchase reconciliation migration completes during startup.

Release 1.10.39

- Reconciles Enodoro cellar-product invoices from Fatture in Cloud into an idempotent, article-code-matched enology purchase ledger.
- Shows purchased, quantified use, unquantified use and calculated balance without overstating inventory certainty.
- Adds copyable current-to-date/year-end recipes for every vintage batch using only recorded product additions in process order.

Release 1.10.38

- Removes both estate breaker meters from generator input because they track inverter/output load while the generator is stopped.
- Shows generator input as `0 W` with a clear stopped/unmetered label until a verified generator-only meter is commissioned.
- Prevents false generator power from reducing the offline solar estimate, energy learning, and system status indicators.

Release 1.10.37

- Corrects live generator input to use the dedicated generator-main-breaker meter.
- Stops labeling the unrelated Bluetti breaker load as generator power or subtracting it from estimated solar.

Release 1.10.36

- Uses the continuous generator-adjusted Baiamonte solar estimate throughout the owner dashboard, live energy balance, solar summaries, operational learning, and WhatsApp power reports whenever direct Growatt telemetry is offline.
- Labels estimated solar clearly instead of showing “Growatt meter unavailable.”

Release 1.10.35

- Retries brief Home Assistant Supervisor 502/503/504 hand-offs before reporting Google planning or GW2000 history as failed.
- Bounds each Gmail poll to the 50 most recent messages and one intake analysis so mailbox refreshes cannot monopolize the scheduler.
- Automatically retries one older Gmail intake that failed for a temporary overload, timeout, or rate limit; a successful retry retires the visible error while preserving its audit history.

Release 1.10.34

- Classifies purchased fruit separately from estate harvest while including it in all production, cellar, vintage, and TV totals.
- Reports 4,838.38 kg estate-grown fruit, 1,078 kg purchased fruit, and 5,916.38 kg total fruit received, with 1,477.25 kg total Grenache received.
- Keeps vineyard plan completion and parcel yield based only on estate-grown fruit, while purchased Grenache remains a normal traceable cellar lot that can be worked, measured, and advanced through fermentation.
- Reconciles the 2026 vintage charts and TV projections to recorded fruit, adds source breakdowns throughout the UI, and prevents purchased intake dates from replacing estate harvest dates.
- Preserves 406 actual recorded crates in live and TV statistics; loose purchased fruit is not converted into fictitious crates, and only genuinely unpicked estate fruit uses the configured crate-size projection.

Release 1.10.32

- Records the separate 1,078 kg purchased Grenache intake from the owner-supplied “Steph Yim’s 1200 plot” source wording, with exact harvest-to-lot-to-vessel traceability.
- Shows the nominal 1,000 L tank as completely full at 6°C in pre-fermentation cold hold without misrepresenting vessel occupancy as a measured liquid-wine yield.
- Records destemming plus tannin and sulfur additions with unknown products and doses left explicitly pending, and schedules the 6 October headspace, measurement, adjustment, and fermentation-start work.

Release 1.10.31

- Treats a successful Instagram refresh as a healthy partial Meta update when Facebook post history is blocked by an account permission.
- Keeps the Facebook permission problem visible as a channel warning without incorrectly marking the entire estate integration as failed.

Release 1.10.30

- Records successful OpenAI recovery events so earlier credit failures stop appearing as unresolved after API access returns.
- Keeps the Today system-status tooltip compact and directs administrators to the complete audited recovery list.
- Preserves failed intake and subsystem work for a real retry instead of silently discarding it.

Release 1.10.29

- Distinguishes the correctly reset monthly AI usage counter from OpenAI provider credit and quota availability.
- Keeps the latest provider error and check time visible after a credit check instead of replacing them with a generic warning.
- Clarifies that the availability check verifies API access but cannot reset or add billing credits.

Release 1.10.28

- Removes unused product recommendations and their alternatives from the working recipe as soon as their process gate has passed, while preserving actual additions in recipe history and full evaluations in the prediction audit.
- Adds compact, variety-aware recipe bases for Grecanico, Nerello Mascalese, and Grenache that sequence decisions without forcing a product from every category.
- Enriches the actionable catalog from current official manufacturer sources for EnartisFerm ES181, ZYMAFLORE F83, LALVIN ICV D254, EnartisPro TINTO, and NUTRIFERM ADVANCE, including timing, dose, preparation, variety/style fit, and safeguards against unnecessary use.

Release 1.10.27

- Added an explicit per-lot white-wine malolactic decision: undecided, allow MLF, or block MLF.
- Suppressed white-wine MLF bacteria and activator recommendations while MLF is undecided or blocked; allowed MLF remains gated by stage, malic-acid evidence, temperature and product timing.
- Added a separate white-wine MLF roadmap stage and control, keeping the decision distinct from racking, stability and aging products.

Release 1.10.26

- Scoped the laboratory decision table, next-test request, scheduled tests, current chemistry, fermentation kinetics, and comparison views to the actively selected wine lot.
- Rebuilt the red-wine roadmap so Nerello pressing, malolactic fermentation, settling/racking assessment, stabilization and aging are separate ordered decisions instead of one combined “Press, finish & age” product bundle.
- Kept pressing as a physical cellar operation with no implied additive; MLF, fining, stability and texture products remain in their own evidence and process gates.

Release 1.10.25

- Recorded the owner-confirmed 30 g Enartis WINY addition to the primary Grecanico white on 1 October 2026 and exposed the official technical sheet in the product catalog.

- Recorded the subsequent racking from T-03 into a provisional closed-top vessel without inventing its physical identifier, material, rated capacity, exact fill or operation time.
- Moved the primary Grecanico lot into active post-fermentation aging, emptied the former fermentation-tank control card, and scheduled the focused pH/free-SO2/total-SO2 verification needed before another sulfite decision.

Release 1.10.24

- Routes WhatsApp approval and rejection commands before AI intake analysis so a manager's decision can never become another approval request.
- Keeps intake approval codes valid for seven days while retaining the 24-hour safety limit for device and process controls.
- Limits approval reminders to two per manager per day, only from 08:00–19:00 Europe/Rome, and excludes generic marketing items and approval-command messages.
- Reuses an open pending approval for seven days instead of producing unnecessary duplicate action records.

Release 1.10.23

- Records 10.5 kg Naturalia crystalMUSTGRAPE added to the Nerello during the 30 September evening pump-over.
- Recalculates any remaining alcohol-consistency recommendation from this applied quantity and requires a post-homogenization potential-alcohol result before another correction.
- Keeps yeast nutrition as a separate APA/YAN and fermentation-trajectory decision; the sugar addition does not automatically recommend nutrient.

Release 1.10.22

- Reconciles Wendy's current 2026 harvest chronology into exact-lot Grecanico, Grenache, and Nerello records, including corrected readings, rackings, press completion, punch-downs, pump-overs, and harvest labor.
- Updates the active Grecanico and Nerello tank profiles from the latest confirmed Babo and temperature readings so cellar cards, pipelines, WhatsApp history, and digital tags share the same current values.
- Preserves blank and qualitative readings without inventing measurements, retains the confirmed 60 g CLARIL AF and 30 g/hL NUTRIFERM corrections, and records the demijohn sulfur note without inferring its missing unit or product.

Release 1.10.21

- Routes Italian L-malic-acid and reducing-sugar laboratory labels into the canonical recipe and chart metrics without a false unmapped warning.
- Replaces the long next-test feed with compact per-lot summaries, expandable reasons, and a dynamic copy-ready laboratory request.
- Adds a dedicated Products used workspace with actual quantities, rates, dates, process positions, product lots, and copy-to-clipboard output.
- Positions a fermentation alcohol correction after completed cellar work when the recommendation is calculated from a later laboratory result.

Release 1.10.20

- Retains an enologist-selected repeat nutrient product while clearing its stale quantity until the post-addition APA/YAN result and current fermentation trajectory are available.

Release 1.10.19

- Suppresses the alcohol-target panel and all forward catalog decisions after an owner explicitly closes a lot as aging with no further intervention; applied-product history and evidence remain visible.
- Separates alcohol-consistency sugar calculations from yeast-nutrient dosing: sugar changes the demand context but never creates an automatic nutrient addition.
- Summarizes nutrient already used and requires a current APA/YAN plus Babo/density trajectory before recommending more when the latest APA predates a recorded nutrient addition.
- Applies the saved potential-alcohol target to the nutrition context and enforces the one-third sugar-depletion timing guard for routine nitrogen additions.

Release 1.10.18

- Links the approved 30 September Nerello Mosto/Vino report to the active lot and uses total projected alcohol—not remaining potential alcohol—as the current recipe basis.
- Shows developed alcohol, remaining potential, residual sugar, and a visible correction explaining which earlier baseline the new fermentation report supersedes.
- Auto-classifies future fermentation-progress panels and auto-links them only when the vintage/variety identifies one unique physical wine lot.
- Restores the Nerello harvest-to-cellar trace so registered source-block context reaches the recipe while unrecorded altitude and soil remain explicitly missing.
- Closes the Grecanico tail lot in a three-demijohn vineyard aging group, releases the old tank, and suppresses future interventions while preserving its applied-product history.

Release 1.10.17

- Corrects potential-alcohol adjustment math by suppressing additions when the measured-to-target difference is within the 0.10% operational tolerance.
- Hides rescue nutrients such as NUTRIFERM NO STOP unless a measured fermentation trajectory indicates a sluggish or stuck fermentation.
- Populates recipes with current tank readings, exact-lot laboratory evidence, vessel and volume data, plus variety-level vineyard context when an exact harvest trace is unavailable.
- Adds a concise current-evidence panel to each recipe so the measurements driving recommendations are visible before any product decision.

Release 1.10.16

- Adds editable potential-alcohol targets to active wine lots and uses the saved target in alcohol-consistency calculations.
- Adds a current enology decision table for recipe-critical tank and laboratory readings.
- Records the confirmed 400 g NUTRIFERM ADVANCE addition to the Nerello lot without inventing a product rate or stock balance.

Release 1.10.15

- Fixes the TV snapshot rotation so scheduled maintenance obtains a genuine Home Assistant/Eufy image instead of renewing the timestamp of its own cached frame.
- Classifies scheduled display-cache responses truthfully, preventing an unchanged picture from being recorded as a fresh capture.
- Reloads the active TV camera page every five minutes while preserving sequential, snapshot-only access with no automatic live streams.

Release 1.10.14

- Selects NUTRIFERM AROM PLUS for the next Nerello nutrition round while leaving its quantity dynamic until the next APA/YAN result.
- Adds a direct “Use this product for this step” action to recipe alternatives and supersedes any earlier planned choice for that decision.
- Reconciles the photographed 1 kg AROM PLUS pack and 480 g prior use to 520 g calculated remaining.

Release 1.10.13

- Makes recipe sequencing lot-aware: a pending fermentation-nutrition decision now follows fermentation-support products already applied to that lot.
- Places the Nerello recommendation that appeared as Step 4 after the completed TINTO and COLOR PLUS process steps.
- Leaves the normal nutrition-first sequence unchanged for lots without a completed fermentation-support addition.

Release 1.10.12

- Fully deletes the erroneous 46 g EnartisPro TINTO event instead of retaining it as cancelled.
- Keeps the correct Nerello additions at 483 g EnartisPro TINTO and 46 g EnartisZym COLOR PLUS.
- Reconciles TINTO inventory and its release audit summary to 517 g calculated remaining.

Release 1.10.11

- Corrects the Nerello EnartisPro TINTO total to 483 g from the photographed 16.1 hL × 30 g/hL calculation.
- Cancels the erroneous extra 46 g TINTO event; the 46 g calculation belongs only to EnartisZym COLOR PLUS.
- Restores calculated EnartisPro TINTO stock to 517 g while retaining 204 g EnartisZym COLOR PLUS.

Release 1.10.10

- Corrects both 46 g evening pump-over additions—EnartisPro TINTO and EnartisZym COLOR PLUS—from the primary Grecanico white lot to the Nerello lot.
- Removes the incorrect white-wine references and keeps both products in the Nerello fermentation recipe and operation history.
- Preserves the already-correct inventory arithmetic: 471 g EnartisPro TINTO and 204 g EnartisZym COLOR PLUS calculated remaining.

Release 1.10.9

- Identifies the pump-over enzyme precisely as EnartisZym COLOR PLUS, lot 250258701, from the owner clarification and Enodoro delivery record.
- Reconciles the supplied 250 g pack with the 46 g application to show 204 g calculated remaining and retires the unresolved enzyme placeholder.
- Repairs the Grecanico product-note migration so the add-on starts cleanly after applying the record.

Release 1.10.8

- Records 46 g EnartisZym in the primary Grecanico white during the 27 September evening pump-over.
- Positions the applied enzyme at its actual active-fermentation pump-over stage in the recipe.
- Preserves the obscured EnartisZym variant, package size, product lot and remaining stock as unknown instead of inferring them.

Release 1.10.7

- Records 483 g EnartisPro TINTO in the Nerello fermentation from the photographed 16.1 hL × 30 g/hL calculation.
- Records 46 g EnartisPro TINTO in the primary Grecanico white during the evening pump-over and preserves the actual use without changing manufacturer suitability guidance.
- Places pump-over/fermentation polysaccharide additions in fermentation support rather than at the end of the recipe, and updates the photographed 1 kg package to 471 g calculated remaining.

Release 1.10.6

- Labels partial Meta refreshes explicitly, including which channel updated and which still needs attention.
- Prevents the Social audit summary from saying all sources are current while any configured channel remains stale.

Release 1.10.5

- Refreshes Facebook and Instagram independently so a permission failure on one channel cannot freeze the other channel's posts and audience snapshot.
- Reads Facebook Page-authored content from the least-privilege `published_posts` edge instead of the broader user-content feed.
- Stores per-channel refresh health alongside the cache so a fresh Instagram result cannot make stale Facebook content appear live.

Release 1.10.4

- Distinguishes a successful live Meta refresh from cached social data, exposes the actual refresh failure, and makes scheduled refreshes fail visibly instead of silently accepting stale posts.
- Uses the newest accepted official Instagram relationship import for follower and following totals when the Meta snapshot is older or stale.
- Reloads the complete Social view after an import so counts, comparisons, source health and post status update together.

Release 1.10.3

- Adds a direct Skip this step action to active guided-recipe cards without an approval gate.
- Removes skipped recommendations from the active recipe and keeps them collapsed in a reversible Skipped recipe steps section.
- Records skip and restore decisions as auditable lot events so recipe recalculation preserves the enologist's choice.

Release 1.10.2

- Reconciles the duplicated Grecanico crystalMUSTGRAPE history into one authoritative 10 kg addition, retaining the fuller 17 September record and removing the later duplicate.
- Rejects exact same-day applied-addition duplicates by product, quantity, unit and product lot unless the operator explicitly identifies a genuinely separate repeat dose.

Release 1.10.1

- Prevents any exact product already recorded as applied from appearing again as a separate current or future catalog suggestion.
- Removes the duplicate NUTRIFERM AROM PLUS card while preserving the authoritative applied nutrient event in the recipe timeline.

Release 1.10.0

- Treats primary yeast inoculation as a completed one-time recipe gate once an applied yeast event exists.
- Keeps the actual yeast application in the recipe timeline while removing duplicate current, alternative and future yeast suggestions for that lot.

Release 1.9.99

- Shows yeast rehydration-water temperature, mixing/rest directions and pre-inoculation acclimatization prominently in every guided recipe yeast step, including already-used products.
- Separates the fermentation-tank target from the much warmer yeast hydration-water instruction and requires water, slurry and must temperature checks.
- Uses exact product-protocol temperatures when available and refuses to substitute a generic temperature when the current product sheet has not supplied one.

Release 1.9.98

- Corrects the Grenache inoculation record: EnartisFerm D20 was prepared with 5 L water and two spoonfuls of table sugar for yeast support; NUTRIFERM AROM PLUS was not used in this lot.
- Removes the erroneous Grenache nutrient event and recipe/history entry without changing AROM PLUS records for any other wine lot.

Release 1.9.97

- Confirms EnartisFerm D20, EnartisTan ROUGE and NUTRIFERM AROM PLUS as products used in the 2026 Grenache.
- Reconciles D20 with Wendy's existing 180-quantity yeast record without inventing its missing unit, and preserves the unreported tannin and nutrient quantities as pending.

- Places all three products in the Grenache cellar history and guided recipe while leaving product lots, exact application times and remaining inventory explicit.

Release 1.9.96

- Stores the crystalMUSTGRAPE cellar event using the event table's supported `other` type while retaining its catalog treatment classification and recipe role.

Release 1.9.95

- Stores the BLANCO application in the cellar event table's supported `other` type while retaining its catalog classification as a fermentation-stage yeast derivative.

Release 1.9.94

- Confirms the photographed CLARIL AF, EnartisFerm ES181, crystalMUSTGRAPE, NUTRIFERM AROM PLUS and EnartisPro BLANCO as products used in the 2026 Grecanico white wine.
- Retains the known 500 g yeast, 10 kg enrichment and tank-specific CLARIL quantities while leaving unreported nutrient and BLANCO quantities explicit.
- Places beginning-fermentation yeast derivatives with yeast support instead of late aging products.

Release 1.9.93

- Corrects the Grecanico cellar history: the fruit went directly to the soft press without destemming, followed by must fining, first racking and alcoholic fermentation.

Release 1.9.92

- Places the recorded Grecanico CLARIL AF additions in the post-soft-press, pre-first-racking must stage instead of generic late-stage fining.
- Records the cellar sequence: direct soft press without destemming, must fining, first racking, then alcoholic fermentation.

Release 1.9.91

- Shows evidence-supported future products in the main guided recipe until their process gate passes.
- Removes passed, unused recommendations from the visible recipe while preserving every recorded product that was actually applied.

Release 1.9.90

- Preserves recipe step order `0`, keeping recorded pre-step additions ahead of the numbered winemaking stages.

Release 1.9.89

- Adds a dynamic temperature plan to every guided recipe using wine color, process stage, style target and the latest tank reading, with per-step temperature guidance and twice-daily active-fermentation checks.
- Learns the median comparable estate application rate for the same product and wine color, then constrains the suggested amount to the verified manufacturer minimum and maximum.
- Shows the selected rate, manufacturer range and learning evidence count together so the operational quantity remains transparent and auditable.

Release 1.9.88

- Keeps an explicitly recorded pre-step-one cellar addition ahead of the numbered recipe sequence instead of reclassifying it by the product's default role.
- Shows the Nerello EnartisTan Rouge addition as `Pre-step 1` while leaving standard tannin recommendations unchanged for other lots.

Release 1.9.87

- Keeps a completed harvest's legitimate zero remaining quantity instead of treating it as missing and doubling the recorded crop in downside, working and upside scenarios.
- Adds a completed-vintage regression check so all scenarios reconcile to the actual harvested kilograms and crates when no fruit remains.

Release 1.9.86

- Repairs the official LAFFORT preservation-range URL after the manufacturer renamed it to `preservatives`, allowing all 18 product ranges to refresh instead of leaving the catalog sync partial.

Release 1.9.85

- Turns the scheduled Nerello APA/YAN retest into a visible provisional first-third nutrition step before two progress readings exist, while waiting for APA, potential alcohol and Babo/density progress before calculating a correction.
- Keeps the already-applied 480 g NUTRIFERM AROM PLUS event authoritative and prevents the provisional plan from implying that the original addition should be repeated.

Release 1.9.84

- Audits the live enology recipe engine so operational counts reflect one concise decision per cellar purpose rather than every comparable manufacturer protocol.
- Keeps already-applied products in their correct recipe position without immediately resurfacing duplicate, unsupported bench-trial placeholders.
- Defers MLF and filterability products until the appropriate post-fermentation stage and stops treating red must fermenting on skins as a white-must NTU control case.
- Updates the winemaking overview to count actionable recipe steps and required recipe inputs while preserving the full catalog behind each product-comparison dropdown.

Release 1.9.83

- Uses a fixed-ID idempotent insert for the Nerello APA/YAN request, eliminating MariaDB ambiguity from the joined insert path.

Release 1.9.82

- Qualifies the existing APA request status fields in the MariaDB upsert so the immediate Nerello test request installs atomically.

Release 1.9.81

- Confirms 480 g NUTRIFERM AROM PLUS was applied to the approximately 1,600 L Nerello must at 30 g/hL and records the observed addition rate in the guided recipe.

- Reconciles approximately 520 g remaining nutrient stock and schedules an immediate APA/YAN test whose result will refresh the nutrition adjustment without repeating the original dose.

Release 1.9.80

- Records the active Nerello tank at approximately 1,600 L of must, allowing the system to show the observed Q GRACE rate as 31.25 g/hL while leaving the unknown vessel identity explicit.
- Fixes MariaDB compatibility for the new photo/receipt-backed inventory upserts so the add-on starts and applies the Enartis records cleanly.

Release 1.9.79

- Resolves the photographed/receipted Nerello yeast as EnartisFerm Q GRACE, links its official 20–40 g/hL product guidance, and replaces the temporary unresolved yeast identity.
- Reconciles the Q GRACE receipt quantity (three 500 g packs / 1.500 kg), today's 500 g addition, lot L530283, and the remaining two packs / 1.000 kg balance.

Release 1.9.78

- Records the 26 September Nerello inoculation with 500 g EnartisFerm yeast and NUTRIFERM AROM PLUS nutrient, preserving the exact strain, nutrient quantity, product lots and exact time as pending instead of guessing.
- Adds the photographed Enartis delivery to the cellar catalog, including projection-ready NUTRIFERM AROM PLUS label dosing, EnartisPro TINTO stock, COLOR PLUS delivery evidence, and unresolved covered yeast/enzyme variants.
- Moves the live Nerello lot into fermentation and keeps the used products in their correct recipe stage.

Release 1.9.77

- Expose the complete harvest scale chain in the operational API and harvest screen, including lot code, gross weight, crate tare, net fruit, crate-average weight and reconciliation status.
- Make the completed Nerello harvest visibly reconcile 2,389 kg gross to 2,139.6 kg net instead of showing only the final weight.

Release 1.9.76

- Reconcile the completed 25 September Nerello Mascalese harvest as 172 crates, 2,389 kg gross, 249.4 kg crate tare and 2,139.6 kg net fruit, replacing the pre-harvest estimate with actual evidence.
- Start the separate Nerello cellar lot and record destemming, the photographed EnartisTan Rouge addition, and the overnight 15 °C hold without inventing the unreported tannin quantity, product lot, vessel or must volume.
- Keep confirmed applied products in the guided recipe even when their quantity is still missing, clearly labeling the traceability gap instead of presenting a projected dose as the amount used.

Release 1.9.75

- Collapse a disconnected Eufy companion into one bridge-level notice instead of creating individual and grouped hardware alerts for every retained camera entity.
- Resolve stale per-camera and area alerts while bridge availability is untrusted, then resume normal 15-minute hardware-fault detection after the bridge reconnects.

Release 1.9.74

- Show the observed whole-tank g/hL rate beside every applied gram- or kilogram-based winemaking addition while keeping that actual cellar rate distinct from the manufacturer's general product range.
- Record the confirmed pre-racking CLARIL AF additions as 60 g in the 275 L small Grecanico lot and 233.41 g in the 1,069.8 L primary Grecanico lot, both at the same observed 21.82 g/hL rate.

Release 1.9.73

- Confirm the Nerello Mascalese harvest for the morning of September 25, 2026 and preserve the time window in the audited harvest plan.
- Keep provisional harvest estimates recalculating from current weather, laboratory and prior-vintage evidence; only confirmed, held or active schedules remain fixed.
- Prevent expired unscheduled estimates from appearing as past dates on Today, and distinguish Scheduled from Estimated and Recalculating states.

Release 1.9.72

- Remove the legacy per-section recipe cap so no evidence-supported fining, tannin, nutrition or acid-balance decision can be omitted; the recipe remains concise by selecting only one primary product per decision.

Release 1.9.71

- Correct the release 1.9.70 tannin-protocol migration to join product classification through the enology catalog, allowing startup and the complete lab-driven recipe migration to finish.

Release 1.9.70

- Automatically use reviewed lab reports when the variety and primary/small/final or tank identity unambiguously match a wine lot, while keeping ambiguous variety-only reports as visible link candidates.
- Keep the complete lab-supported batch plan across process stages so white clarification, red tannin, nutrition and tartaric-acid decisions cannot silently disappear when their immediate timing window changes.
- Add focused phenolic tests for red tannin decisions and tartaric-acid testing for acid-balance and stability decisions; every new reviewed report now recalculates the recipe and its suggested additions.

Release 1.9.69

- Redesign the managed kiosk home view around the kitchen display with a prominent clock, date, live estate weather, and direct overhead and sink light controls.
- Surface read-only security, gate-motion, refrigerator, and dishwasher status without exposing accidental tap actions.
- Keep the compatibility kiosk dashboard aligned with the same home-screen layout.

Release 1.9.68

- Replace the scattered product list with a four-stage guided batch recipe that always includes inoculation and yeast-support planning without forcing an unnecessary additive.
- Save a fresh-and-aromatic to structured-and-age-worthy style target per real lot or pre-harvest variety plan and use it to rank the primary product while leaving laboratory and product-sheet quantity calculations unchanged.

- Keep one evidence-supported primary recommendation per decision, preserve used products in their correct process position, and place other manufacturers behind a focused alternative comparison.

Release 1.9.67

- Turn reviewed pre-harvest grape chemistry into a live quantity plan using the current damage-adjusted production forecast and configured grape-to-wine yield.
- Propose a non-reserving vessel allocation from current working tank capacity and recalculate additive quantities from the projected batch volume until received measurements replace it.
- Suppress the misleading density-trajectory prompt before fermentation begins and label every forecast quantity and vessel as projected.

Release 1.9.66

- Limit phone-width weather graphs to five time-axis ticks so labels remain visually separated.

Release 1.9.65

- Use compact time-axis labels in weather graphs so dates remain readable on phone-width screens while exact timestamps stay in the measured summary.

Release 1.9.64

- Keep graph loading and no-data placeholders hidden once a weather series has loaded, including on the mobile dark theme.

Release 1.9.63

- Make every measurable GW2000 weather card open a mobile-friendly graph backed only by stored observations.
- Add 24-hour, 48-hour, and 7-day ranges with latest reading, measured range, timestamp, units, and explicit no-data handling.

Release 1.9.62

- Reject Ecowitt weather and soil states older than ten minutes so a communication lapse cannot be recorded as a current vineyard observation.
- Add explicit soil-probe communication health to the managed Vineyard Overview dashboard.
- Notify once when soil-probe communication fails and once when it recovers, without recreating dismissed alerts while the fault remains unchanged.

Release 1.9.61

- Add the live GW2000B leaf-wetness reading, its recorder history, and its battery voltage to the managed Vineyard Overview and iPad weather dashboards.
- Add a dedicated 48-hour leaf-wetness history chart for canopy and disease-pressure review.

Release 1.9.60

- Keep GW2000 leaf-wetness auto-discovery from mistaking the companion battery-voltage entity for the wetness percentage when the sensor is received through a second gateway.

Release 1.9.59

- Normalize the local Tuya generator series from kW to W in the live Electricity chart so generator input and estate load share one readable scale.
- Finish the native Home Assistant Energy cleanup with concise device names and valid fan and pressure-pump energy meters.

Release 1.9.58

- Classify a router's per-port online-detection setting as switching telemetry so a disabled secondary-port probe cannot contradict the healthy Internet-link sensor or create a false critical router outage.

Release 1.9.57

- Stop optional or unavailable network telemetry and unused router LAN ports from appearing as critical WAN outages; show missing data as an amber coverage state while preserving real disconnects as red.
- Make the live integrity audit discover the running add-on address automatically and use the current year instead of fixed deployment values.
- Correct the estate energy source mapping so the local Tuya Bluetti entity represents the physical generator breaker, while the separately named Generator Main Breaker remains identified as the inverter circuit.
- Add a clearer live-electricity dashboard with separate solar, generator, inverter, estate-load and battery-flow readings.

Release 1.9.56

- Normalize the operational grape total to three decimal places so the API, Today screen and projection charts expose the same exact value without a binary floating-point artifact.

Release 1.9.55

- Make recorded harvest the authoritative floor for current-vintage projections, preserving actual crate counts and applying scenario ranges only to fruit that remains unpicked.
- Use the configured planning crate size for projected fruit instead of a hard-coded 15 kg divisor, while clearly separating projected crates from received crates.
- Add a live Today view of harvested kilograms and crates, remaining varietal quantity, measured cellar liters, picked-fruit wine equivalents and the working finished-wine outlook.
- Reconcile current-year scenario, variety, allocation and multi-year charts to the same recorded-plus-remaining operational basis.

Release 1.9.54

- Turn every ingested laboratory report into automatic decision support with normalized measurements, current-vintage movement, prior-vintage context, harvest implications, missing evidence and the next checks.
- Interpret APA/YAN by red- or white-fermentation context while keeping nutrient additions, harvest confirmation and cellar actions under explicit human or enologist control.
- Show the summary on the latest-report panel and inside each laboratory sample, while retaining the original report values and audit trail as authoritative evidence.

Release 1.9.52

- Recommend a supported yeast strain without waiting for repeated YAN/APA testing; one valid YAN/APA result is sufficient for the nutrition decision and rate refinement.
- Keep missing chemistry as visible yeast-plan guidance instead of blocking the strain recommendation.

Release 1.9.51

- Keep laboratory-supported pre-harvest recipe steps visible when received fruit weight or must volume is the only missing quantity input.

Release 1.9.50

- Build a visible, non-persistent pre-harvest winemaking process from the latest reviewed grape chemistry when the variety has not yet entered a physical cellar lot.
- Carry YAN/APA, potential alcohol and the complete maturity panel into the recipe plan while keeping exact quantities and addition recording pending until received fruit weight or must volume exists.

Release 1.9.49

- Retry the Vineyard North fixed-view watch five minutes after a transient RTSP/P2P failure instead of suppressing recovery for the normal one-hour capture interval.
- Add a targeted visual-watch refresh for operational recovery without rerunning the full camera-awareness pipeline.

Release 1.9.48

- Let an operator start a batch-specific laboratory request directly from the necessary-tests list; a scheduled or sampled YAN/APA test immediately opens a provisional yeast and nutrition plan, then resolves the exact dose when the result arrives.
- Restrict the working recipe to products supported by current-vintage measurements, grape fit, fermentation trajectory, exact-lot laboratory evidence, or an active laboratory plan instead of filling product categories.

Release 1.9.47

- Make BF/BT tank-name normalization idempotent so repeated imports, service restarts and power cycles never duplicate the primary- or small-tank qualifier.

Release 1.9.46

- Identify the laboratory's Grecanico BT series as the primary tank and BF series as the small tank, while retaining both original codes and keeping the physical sample series separate.

Release 1.9.45

- Normalize Italian and English laboratory matrix wording into the canonical sample type, while preserving BF/BT and process qualifiers as distinct physical sample variants.
- Keep every laboratory's original sample label unchanged in the audit field while presenting consistent variety-first English labels in selectors, comparisons and projection pipelines.

Release 1.9.44

- Restore every estate utility dashboard to the Tuya Local switch and metering entities so critical circuits no longer depend on cloud entity IDs.
- Accept a plain WhatsApp `Approve` or `Reject` when exactly one manager approval is pending; ambiguous replies remain read-only and ask for the six-digit code.
- Make WhatsApp laboratory approval ingest the complete report, retain every sample and result, and queue the harvest projection refresh instead of only changing the inbox status.
- Send bounded WhatsApp reminders for review-ready and overdue intake items to configured managers, with actionable codes, a three-item pass limit, and a twelve-hour cooldown.

Release 1.9.43

- Show Battery 3 as a prepared Modbus address 3 slot before installation, alongside the two active Felicity packs.
- Automatically include the third pack in live bank status and reserve calculations after the battery monitor activates it.
- Use the monitor's live nominal bank capacity so energy forecasts increase from 10.24 kWh to 15.36 kWh without manual reconfiguration.

Release 1.9.42

- Keep every recorded product addition in the full lot recipe after its stage has passed, positioned in process order and labeled with its actual quantity, application time, product lot and recorder alongside the current recommended steps.

Release 1.9.41

- Replace manufacturer-sized enology recipes with a short, ordered cellar plan that selects one exact product per purpose and keeps every applicable cross-manufacturer option in a step-level comparison dropdown, regardless of cellar stock.

Release 1.9.40

- Keep Admin and other read-only operational pages available during a transient Home Assistant Supervisor 502 by returning the last state snapshot, or an empty degraded snapshot before the first successful read, instead of raising a page-wide 500.

Release 1.9.39

- Move the Bluetti main-breaker and kitchen pressure-water controls to their healthy Tuya cloud entities; redundant failing Tuya Local entries are retired from Home Assistant with registry backups retained.

Release 1.9.38

- Show the crystalMUSTGRAPE alcohol-consistency calculation directly in the winemaking recipe and prediction views, including current-to-target projection, prior additions, post-addition retest guidance and a non-blocking compliance warning.

Release 1.9.37

- Follow the audited Eufy rename for the fixed vineyard analysis fallback: `camera.vineyard_north_2` is Vineyard North, while `camera.vineyard_north` is now Main Parking.

Release 1.9.36

- Keep the last accepted cistern reading when the Home Assistant state feed times out, with a specific recoverable alert instead of a failed processing job.
- Route the overview hot-water control to the working cloud Tuya entity and prepare retirement of the broken duplicate Tuya Local entry.
- Cache the large read-only Enology dashboard briefly, invalidating it immediately after every enology write.
- Add a transparent alcohol-consistency calculation for split lots using the documented crystalMUSTGRAPE conversion, verified volume, current potential alcohol, target, prior additions and an explicit compliance warning.
- Correct the live lab duplicate-code audit so blank sample codes are not reported as duplicates.
- Persist the confirmed 2026 vintage on five first-year must samples without auto-approving their scientific results.

Release 1.9.35

- Renamed the last winemaking “blocked gates” summary to plain-language missing inputs so technical measurements are not mistaken for approval steps.

Release 1.9.34

- Removed the last approval-gate wording from the Winemaking workspace and made its direct operator workflow explicit.
- Reframed the product library as lot-data-driven recommendations while retaining visible technical inputs and traceability.

Release 1.9.33

- Make authenticated enology operators authoritative for process stages, additions and active lot profiles without a second role-based approval gate.
- Add purpose-specific working-dose recommendations, whole-batch quantities, dose rationale and one-click prefilled planning across prediction and manufacturer-recipe views.
- Keep safety-critical input checks for missing volume, unsupported units, product lots and required bench trials while presenting them as actionable missing data rather than approval queues.
- Correct Nutriferm Special to its official 30–40 g/hL inoculation use, include its approximate YAN contribution, and keep the distinct stuck-fermentation starter use out of whole-tank arithmetic.
- Add recommended next actions throughout the winemaking workflow and route Brettanomyces evidence into the post-fermentation laboratory pipeline.

Release 1.9.31

- Recalibrate automatic cistern vision for the repositioned centered camera view and retire the old diagonal corner-view geometry from prompts and operator guidance.
- Use the owner-confirmed approximate 80% new-view reference as the first physical anchor for subsequent camera comparisons.

Release 1.9.29

- Start a new cistern calibration epoch after the internal camera was repositioned, preventing old-view estimates from being treated as current water evidence.
- Plot only physically calibrated, confidence-qualified water readings on a true time axis; retain legacy estimates for audit without drawing false fills or drawdowns.
- Show the last accepted reading separately from the latest pipeline attempt and expose why a camera frame was not accepted.
- Restrict the water equipment list to hydraulic measurements and controls instead of camera settings and unrelated battery-flow entities.

Release 1.9.26

- Add seven current Enartis products and convert twenty additional LAFFORT products into unit-safe, purpose-specific projection candidates using official manufacturer sheets.
- Expand recipe coverage across white and red yeasts, clarification enzymes, measured nutrition, MLF support, fining, tannins, mannoprotein programs and post-fermentation microbial control.
- Keep the new quantities enologist-controlled with exact-lot laboratory gates, stage checks, bench trials, contact-time checks and separate dose bases for fruit, must and finished wine.
- Recognize Brettanomyces laboratory results for the corresponding evidence-gated post-fermentation decision pipeline.

Release 1.9.25

- Recover older Instagram images from the current public embed document when Meta's Graph response still contains an expired CDN signature, then persist the working image in Baiamonte's local media cache.

Release 1.9.24

- Serve Instagram post thumbnails through an authenticated Baiamonte media endpoint instead of Instagram's retired permalink `/media/` shortcut.
- Persist successfully fetched Meta images locally so older posts remain visible after temporary CDN URLs expire, with a safe branded fallback when Meta cannot refresh an image.

Release 1.9.23

- Balance the desktop and tablet Admin submenu into two intentional seven-column groups, keeping communication tools together and starting security/system tools on a clean second row instead of leaving TV orphaned.
- Preserve a single evenly justified Admin row on wide desktop displays and the existing two-column mobile menu.

Release 1.9.22

- Prevent company, cellar-manager and grape-history writing from clipping on short cellar tablets by reserving taller detail cards, allowing contact lines to wrap and giving longer legal text a wider column.

Release 1.9.21

- Separate the cellar container illustration and large percentage into reserved tablet-layout columns, with additional side clearance so values such as 42.3% no longer overlap the vessel or clip against its card.

Release 1.9.20

- Render the essential cellar tank label directly in the initial page so older tablets still show level, liters, capacity, wine, vintage, stage and readings when their JavaScript enhancement or service-worker refresh fails.
- Keep the existing live charts and detailed legal label enhancement when the tablet browser is healthy, with a new display cache version for immediate rollout.

Release 1.9.19

- Keep an explicitly assigned pre-press wine lot visible on cellar labels and dashboards when its eventual liquid volume is not measured yet, instead of suppressing its lot and laboratory context.

Release 1.9.18

- Add a lightweight two-second live revision watch to every cellar tablet, refreshing the full label immediately when web, WhatsApp or manual temperature/Babo readings change while retaining a 30-second recovery refresh.
- Avoid repeated full-payload queries and screen repainting when nothing changed, with a slower hidden-tab cadence and offline-safe fallback.
- Merge trusted current SG, Brix and pH values into the matching trend point so the upper charts and lower readings agree, and link the exact 2026 Grenache lot to M-01 without misrepresenting pre-press occupancy as juice volume.

Release 1.9.17

- Tie the cellar-label service-worker cache directly to the served asset release so installed tablets cannot keep the old four-reading shell after an update.
- Refresh both physical labels to show Babo alongside temperature, density, Brix and pH, and replace the generic Mustalone tank symbol with a wide open-top food-grade fermentation-vat silhouette.

Release 1.9.16

- Preserve the retired `T-03` setup placeholder under an explicit archive identity, then assign `T-03` to the owner-confirmed physical Tank 3 without changing its lot, history or permanent digital-label URL.

Release 1.9.15

- Reconcile the large 2026 white vessel to its physical `T-03` / Tank 3 identity and standardize all three current-vintage digital-tag names without changing their stable database ids or label URLs.
- Force cellar tablets onto the current five-reading label layout so temperature, Babo, density, Brix and pH refresh together from the latest trusted values.

Release 1.9.14

- Fill safe digital-tag vintage and wine-stage defaults from the assigned vessel record when legacy lots lack a linked season or exact legal-profile phase.

Release 1.9.13

- Remove the obsolete blend-plan fallback that interrupted kiosk rendering and identify the current program as separate varietal vinification.

Release 1.9.12

- Restore live kiosk rendering on older display browsers, report bounded client render errors to the add-on log, and keep automatic release reloads working after deployment.
- Feed digital tags from the latest non-null cellar observations, including temperature, Babo, density, Brix and pH, instead of showing blank manual-profile fields.
- Standardize digital-tag titles as `code · name` and add dedicated open-top Mustalone artwork on the physical tank display.

Release 1.9.11

- Make the kiosk connectivity probe return success, publish the running release in its data payload, and reload the page automatically when deployed assets change.
- Limit camera outage alerts to the actively configured kiosk and cistern cameras so retained retired aliases remain auditable without creating false operational warnings.

Release 1.9.10

- Render newly fetched kiosk data immediately while the Today page is paused instead of buffering it indefinitely until page rotation resumes.
- Use the same preferred human harvest-plan selection on the kiosk as the operational dashboard, preventing an obsolete forecast date from replacing the owner-approved Nerello working date.

Release 1.9.9

- Accept Home Assistant's root `HEAD /` availability probe so Supervisor reports the running add-on as healthy instead of marking successful API service as an error.

Release 1.9.8

- Normalize laboratory measurements into canonical enology units, enforce plausible ranges, and prevent incompatible or ambiguous results from unlocking additive quantities or stage gates.
- Automatically map new laboratory analyte names and units with AI, retain the original reported result, reuse validated mappings on later reports, and keep uncertain mappings visible but operationally quarantined.
- Preserve and graph every numeric laboratory result, route a broader professional wine-analysis registry, show unmapped analytes explicitly, and correctly share multi-lot laboratory identity in charts.
- Align persisted and detailed cellar stages—including malo, pressing, settling and stabilization—with next-test and product-decision pipelines, remove duplicate near-dry test recommendations, and strengthen must and post-fermentation evidence gates.
- Preserve prior tank chemistry during partial web or WhatsApp updates, display the latest non-null value for each field, and anchor Babo progress to the first accepted reading instead of a later maximum.
- Enforce product protocol process stages, bulk-load lot evidence, deduplicate unchanged prediction snapshots, and render heavy laboratory and vintage charts only when their workspace is open.

Release 1.9.7

- Show every laboratory test necessary for the current wine-lot decision rather than applying an arbitrary numeric limit; continue suppressing irrelevant, premature and freshly completed tests.

Release 1.9.6

- Show manual temperature, Babo and reading time directly on Cellar and Enology tank cards, preserve Babo in web-entered history, and expose calculated Babo progress alongside exact-lot potential alcohol, YAN/APA and turbidity.
- Resolve shared must reports through the laboratory sample-to-lot link table so primary and final-press fractions receive their authoritative laboratory evidence and dependent additive predictions.
- Route NTU/turbidity, catechins, Brix and the complete recognized enology analyte set into their relevant testing stages and unit-safe current-vintage and year-over-year charts, with source-report drilldown from chart points.
- Add a dynamic premium-wine laboratory panel capped at three decision-relevant next tests per lot; stage, fermentation progress, wine color and exact-lot result freshness control what appears and when.

Release 1.9.5

- Replace the generic quantity-waiting additive cards with an enologist-controlled, lot-specific full batch recipe using recorded cellar products, current volume, exact-lot labs and fermentation stage.
- Use dated Babo readings as fermentation-progress evidence when density is unavailable, recommending first-third nutrition only in its active window and keeping NUTRIFERM NO STOP conditional on a supported sluggish or stuck trajectory.
- Correct EnartisZym RS(P) timing to post-press juice settling, retain official whole-batch dose ranges and directions, and distinguish recommended, planned, applied, upcoming, passed and not-indicated states without repetitive review-required labels.
- Add recipe and alternative-manufacturer dropdowns for direct product-plan comparison, link technical sheets inside the recipe, and refresh the active winemaking recipe automatically as new labs or tank readings arrive.

Release 1.9.4

- Show available YAN/APA values inside the selected wine-lot view instead of reducing unlinked variety-matched reports to `link_required`.
- Add audited many-to-many laboratory-sample links so one representative must sample can support multiple separated wine fractions; link the Sept. 11 Grecanico must APA result (124 mg/L) to both the primary and final-press lots per owner confirmation.
- Add exact-lot linking controls for reviewed must and wine reports while keeping nutrient dosing blocked until that explicit relationship exists.

Release 1.9.3

- Make the grape dashboard’s displayed plan date and status use the same preferred, owner-protected harvest plan as its recommendation, eliminating an obsolete minimum scheduler date from the Nerello row.

Release 1.9.2

- Corrected harvest-date fusion so sparse, low-confidence historical lab comparisons stabilize rather than overpower the weather/GDD model, and protected the owner’s September 23 Nerello Mascalese working plan.
- Added automatic canonical ingestion for complete laboratory reports received from trusted email senders and authorized WhatsApp intake, retaining the original source and audit trail while routing ambiguity to review.

- Removed Fatture in Cloud purchases and zero-value evidence from labor queues, merged duplicate 2026 harvest crews, and added the source-backed Giancarlo labor/service reconciliation without inventing rates or payment status.
- Removed the obsolete Today-page crate-production card and made the harvest list prefer the approved operational recommendation.

Release 1.9.1

- Add dedicated Mustalone artwork to cellar tank cards and tank management, using the real vessel name so M-01 renders as its squat, open-top food-grade fermenter while retaining its existing database classification.
- Preserve liquid-level and fermentation-stage animation inside the Mustalone silhouette, including the recorded 90% occupancy view before pressing.

Release 1.9.0

- Replace the active Grenache/Nerello crate-mixing workflow with independent Grecanico, Grenache and Nerello Mascalese production planning across harvest, projections, TV, imports and WhatsApp.
- Reconcile the 2026 cellar to the three real lots: primary Grecanico, Grecanico tail fraction and Grenache; merge explicit stage progress, remove two obsolete planning lots and exclude their seven pre-harvest test readings from kinetics.
- Record the 500 L Mustalone at approximately 90% occupancy (about 450 L including skins), with pressing planned for 16 September and pressed-juice quantity explicitly pending measurement.
- Split the large Winemaking workspace into Process Workflow, Testing & Charts, Additions & Recipes, and Product Catalog sub-views while keeping the selected wine lot in context.
- Preserve historical blend-era migrations as audit evidence while removing current endpoints, forms, calculators, navigation and live planning dependencies.

Release 1.8.9

- Route Manager and Reporter WhatsApp text notes and voice-note transcripts through the deterministic tank command path before freeform AI, so natural tank updates are audited, saved, and pushed into enology predictions.
- Accept natural English and Italian requests for tank lists and recent Babo/temperature history, including common transcriptions such as `T 44`, `T dash 44`, and `T trattino 44`.
- Resolve physical labels such as “tank 3” against the live tank registry while preferring exact system codes such as T-44; ambiguous labels are rejected rather than guessed.
- Keep writes safe by requiring an explicit tank reference and a labeled Babo, temperature, or volume value; casual tank discussion is never stored as a reading.

Release 1.8.8

- Import Wendy's complete 2026 Grecanico and Grenache harvest/cellar note as durable source evidence, including photographed gross weights, explicit crate tare, reconciled net weights, crew windows, transport, press, racking, yeast-preparation and sampling instructions.
- Add the dated Grecanico and Grenache Babo/temperature series and punch-down events to the exact wine lots, preserving date-only or approximate-time evidence labels instead of inventing precision.
- Record the owner-confirmed 15 September readings: small white Tank #44 at Babo 2.2 and 26°C; big white Tank #3/T-06 at Babo 12.2 and 18°C.

- Schedule separate catechin and NTU evidence requests for Tank #3 and Tank #44, retain ambiguous quantities verbatim, and do not mark unconfirmed preparation or racking instructions as completed.

Release 1.8.7

- Add 20 official-source decision products and 21 purpose-specific protocols from Lallemand Oenology, Perdomini-IOC, Oenobrands and Enartis; retain the complete synchronized LAFFORT catalog already in the system.
- Add laboratory-gated decisions for malolactic inoculation, must clarification, lees aging, traditional-method tirage and pre-bottling sensory/stability trials.
- Extend manufacturer recipe comparison to wine bacteria and stabilizers while keeping exact rate, sachet coverage, trial results, local-rule review and enologist approval as explicit blockers.
- Store official product and technical-sheet links and avoid treating prestige or marketing claims as evidence of lot suitability.
- Let Manager and Reporter WhatsApp users save Babo and temperature with a compact message such as `T-06 16.9 18.4`, bypass a stale generic form, and receive the three latest readings immediately after each save; `LAST T-06` retrieves recent history directly.

Release 1.8.6

- Add the purchased ENARTIS and ENODORO cellar products, visible inventory evidence, official technical-sheet links, verified preparation directions and product-specific laboratory/bench-trial gates.
- Add full-batch recipe views for every catalog manufacturer plus a clearly qualified best-evidence-fit comparison based on current lot evidence and recorded cellar stock, never automatic product approval.
- Recalculate yeast, enzyme, nutrient, tannin, fining and acidification quantities from the current verified liquid volume or fruit weight while blocking decisions that lack exact-lot laboratory evidence, a required bench trial, legal review or enologist approval.
- Add authenticated Manager and Reporter WhatsApp commands to list tanks, record exact-code Babo, temperature and volume readings, and retrieve the last 1–14 days of timestamped Babo and temperature history.
- Keep Babo distinct from Brix in storage and display, add Babo to current-lot and year-over-year fermentation charts, audit every WhatsApp reading and immediately refresh the enology additive, testing and manufacturer-recipe prediction pipeline.
- Record the current Tank 3, 300 L white tank and 500 L Mustalone assignments and volumes without inferring an unmeasured liquid volume for the Grenache before pressing.

Release 1.8.5

- Finish camera-name standardization across the secondary Home Assistant dashboard views, including concise mobile camera cards.

Release 1.8.4

- Restore the live Admin Security dashboard while adding camera-pipeline reconciliation to its response.

Release 1.8.3

- Reconcile the Eufy dashboard against the complete Home Assistant camera inventory so migrated or unavailable devices stay visible instead of silently disappearing.

- Standardize current Eufy operational names across Camera Center, Network, Security, TV and staff camera selectors, including Back Driveway Mid, Backyard and Fox Alley.
- Add explicit camera-configuration audit status, event-evidence coverage, security-pipeline reconciliation and numeric low-battery detection.

Release 1.8.2

- Keep overnight-readiness sensors ahead of detailed cell telemetry in the bounded live energy inventory so the gauge remains populated with both battery packs connected.

Release 1.8.1

- Add a prominent overnight-readiness gauge driven by native Home Assistant battery and solar-clock forecasting.
- Show the exact stored-energy target, forecast night demand, remaining kWh shortfall, and average net charging power needed by sunset or sunrise.
- Keep the forecast conservative with a protected 30% reserve and a 15% load uncertainty margin.

Release 1.8.0

- Replaced the unreadable raw energy entity inventory with five clear power-flow cards.
- Added total estate load calculation from live solar, generator, grid, and signed battery power when the main load meter is unavailable.
- Load values now identify whether they are measured or calculated and include a confidence label.

Release 1.7.99

- Keep bank and per-pack summary readings ahead of detailed cell rows so both batteries remain complete within the bounded live inventory.

Release 1.7.98

- Prioritize every direct Felicity BMS entity in the bounded energy inventory so Battery 1 and Battery 2 always render completely.

Release 1.7.97

- Show the connected Felicity bank as live instead of commissioning or provisioning.
- Rename the remaining checklist to system readiness and keep separate inverter/load gaps distinct from battery status.

Release 1.7.96

- Make the direct read-only Felicity RS485 bank the preferred battery source for power status and reserve learning.
- Add live bank, per-battery, balance, temperature and cell-spread status to the Solar & Power workspace.
- Remove retired Bluetti devices from the estate energy inventory.

Release 1.7.95

- Repair one-click approval for mixed grape-and-wine laboratory reports whose grape campaign is written as a split year such as `2025/26`, while preserving explicit wine vintages and the original report evidence.
- Keep approval errors visible inside the report review panel instead of showing only a transient notification.
- Add a dedicated Enology YAN/APA dashboard with current and five-vintage charts, latest sample evidence, laboratory provenance, lot-profile values, working targets and safe decision boundaries.

Release 1.7.94

- Restore Social feed thumbnails by using Instagram's stable public post-image route instead of expired cached CDN URLs.
- Reuse matching Instagram media for Facebook cross-post cards so shared posts remain visual on iPhone and desktop views.
- Make large Instagram relationship imports substantially faster with three parallel ingress-safe upload streams, automatic retry, accurate progress, and a separate processing step.
- Batch relationship database writes so final import processing does not stall after the upload reaches 100%.

Release 1.7.93

- Finish the Social-page startup repair by deferring official-document filter callbacks until their feature script is available.

Release 1.7.92

- Separate the current Instagram account totals from named-export coverage in Social auditing, recording the owner-confirmed September 6 profile totals of 313 followers and 1,055 following.
- Keep the last complete 308-follower export as the named audit basis and fix a startup reference error that could leave Social data blank after a page refresh.

Release 1.7.91

- Validate named Instagram relationship exports against the latest Meta account totals and reject date-limited archives before they create false unfollower events.
- Quarantine the incomplete September 6 import, retain it for audit, restore the August 29 valid comparison, and explain that Accounts Center must use JSON with the All time date range.

Release 1.7.90

- Make large Instagram follower exports import reliably through Home Assistant by uploading them in authenticated 512 KB pieces and assembling them safely inside the add-on.
- Show live import progress, enforce the existing 512 MB archive limit, reject interrupted or out-of-order uploads, and remove incomplete temporary files automatically.

Release 1.7.89

- Replace raw ADS-B/AIS/weather iframe 502 JSON with a branded reconnecting map state during transient traffic-service interruptions.
- Retry the affected map automatically every 15 seconds, retain a manual retry control, and leave the rest of the TV display operational while the receiver returns.

Release 1.7.88

- Restore a usable vineyard Atlas on phones by fitting all saved land into view and compacting the layer control, legend, toolbar, and attribution without removing map information.
- Treat saved parcel boundaries as mapped parcel evidence and derive map centers from their geometry, eliminating the misleading zero-locations status when authoritative outlines are present.
- Keep the official cadastral overlay optional and preserve the saved Baiamonte parcel geometry when the external Agenzia delle Entrate service is unavailable.

Release 1.7.87

- Expand the authoritative GW2000 contract to include station-native feels-like temperature, dew point, VPD, rain intensity, both wind directions, daily maximum gust, leaf wetness, soil temperature, and weather-sensor power health with explicit unit normalization.
- Persist the expanded live readings and recorder history, expose them on the application and TV weather views, and make the future GW2000 leaf-wetness channel self-discovering without inventing a reading before it is installed.
- Route appropriate expanded weather evidence into disease pressure, treatment learning, spray-window outcomes, harvest context, operational alerts, visual monitoring, assistant context, public/MCP feeds, and automated sensor-quality checks while preserving GW2000 as the authoritative observed source.
- Divide the Agronomy forecast scenario and vintage-total charts into source-backed Grecanico, Grenache and Nerello Mascalese contributions while retaining the authoritative total above every bar.
- Add exact, mobile-scrollable composition tables below both charts and label any unreconciled portion explicitly instead of implying unsupported variety precision.
- Carry recorded historical wine volume into the variety history used by liters and bottle-equivalent comparisons, while keeping the current vintage as one projected bar rather than duplicating it as actual history.

Release 1.7.86

- Move all 14 mapped Netatmo lights and outlets on the overview, iPad, and display dashboards to their local HomeKit entities, including Palmento and Fox Den, while keeping the Netatmo app out of the left sidebar.

Release 1.7.85

- Rebuilt harvest-date synthesis so the current approved grape report is the primary maturity signal, while historical weather/GDD and exact prior harvest dates provide a transparent stabilizing reference.
- Compare current chemistry with only one nearest like-for-like sample per prior vintage, preventing repeated reports from one year from dominating the result.
- Stop counting potential alcohol as independent evidence when Babo is present, restrict timing to one coherent current report date, and expose current values, historical values, deltas, matched markers and missing calibration.
- Restore variety-linked historical lab samples to the matcher and keep confidence low when a reported malic-acid result lacks comparable historical evidence.
- Fuse the laboratory and historical/weather dates once, then prohibit the narrative AI layer from applying the same laboratory evidence as a second hidden adjustment.

Release 1.7.84

- Made approved pre-harvest grape laboratory reports move provisional harvest dates through a deterministic, auditable chemistry model instead of relying only on narrative AI adjustment.
- Normalized Babo, potential alcohol, pH, total acidity and malic-acid aliases before comparing current samples with same-variety historical samples paired to exact harvest dates.
- Required at least two matching markers, exposed the selected historical comparisons, and bounded laboratory movement to seven days while preserving human approval of the actual pick.

Release 1.7.83

- Add an inline missing-attachment recovery workflow to laboratory email review: attach the original PDF or report image, retain its relationship to the email, analyze it immediately, and reopen it for whole-report approval.
- Clearly distinguish a genuinely attachment-free forwarded email from a parser failure while continuing to prohibit authoritative laboratory records based only on email text.

Release 1.7.82

- Replace sample-by-sample form entry with one report-level approval that saves every recognized sample and measured result from the retained original PDF or report image.
- Route grape laboratory evidence to Agronomy before harvest and must/wine evidence to Enology after harvest; only approved current-vintage grape tests may move harvest forecasts.
- Recalculate harvest projections immediately after a complete grape report is approved, preserve idempotent duplicate protection, and archive the redundant forwarded-email body after its attached report is accepted.
- Show a precise attachment-required message when a forward contains only its email body, and expose related Gmail items so a forwarded PDF can be opened and approved from the same review.

Release 1.7.81

- Keep every saved parcel boundary intact while making the drawn geometry reliably visible above satellite and cadastral layers with a map-initialized SVG layer, stronger high-contrast outlines, registered-parcel labels, and a clear map legend.
- Refit the Atlas only when a stale saved view does not contain the complete mapped estate, then redraw nested Leaflet layers after hidden-workspace, resize, and orientation transitions.

Release 1.7.80

- Normalize legacy cadastral sheet and parcel identifiers imported with a `.0` suffix, then populate and verify the official 9,144 m² parcel breakdown in the live atlas.
- Keep atlas reconciliation tolerant of legacy numeric formatting while retaining the official `83/76`, `83/77`, and `83/93` legal references.

Release 1.7.79

- Centralize the complete 9,144 m² authoritative vineyard register, incomplete 5,461 m² reference extract, and approximately 3,000 m² pending planting in a source-backed official-facts pipeline.
- Feed the reconciled official facts into the atlas, individual parcel records, dashboards, TV data, reference API, treatment planning, prediction context, AI context, MCP tools, and data-quality monitoring.
- Use purpose-specific area semantics: 0.9144 ha for current productive calculations, approximately 1.2144 ha for the planted treatment footprint and for projected productive calculations from 2027.

- Add SQL reporting views and parcel-level reconciliation flags so a later document update cannot silently create conflicting area values.

Release 1.7.78

- Add an administrator-only official-document registry with retained original PDFs, search, inline viewing, downloads, dated facts, and future upload intake.
- Keep the old Italian system's complete 9,144 m² record as the authoritative current vineyard surface and retain the new system's 5,461 m² output as incomplete, reference-only evidence until Italy corrects it.
- Separate the documented 0.9144 ha current productive surface from approximately 0.3000 ha of pending new planting, expected to become productive in 2027.
- Link cadastral parcels to their official evidence in the vineyard atlas and expose a transparent planted/registered/productive reconciliation panel.

Release 1.7.77

- Repair the estate power dashboards to use the current Tuya entity IDs for the camera, LTE, lighting, and outlet circuits instead of stale Tuya Local duplicates.
- Use the live Bluetti phase-power meter in place of the retired aggregate-power entity.
- Apply the corrected circuit mappings consistently across the overview, admin, wall-display, iPad, and legacy dashboard sources.

Release 1.7.76

- Correct the Admin Network JavaScript and stylesheet URLs so the dedicated module actually loads through Home Assistant ingress.

Release 1.7.75

- Load Admin Network telemetry whenever the view is restored, opened directly, or selected, instead of leaving its panels stuck at `Checking…` until a second click.
- Replace the oversized mobile clock and HomeBase alarm panel on the estate overview with compact Rome-time and alarm-mode cards while retaining all three arming controls.

Release 1.7.74

- Restore BBQ Front and Top Vineyard Path to still-image camera cards; neither dashboard now starts a background livestream, and new Eufy event thumbnails can populate the persistent snapshot cache.

Release 1.7.73

- Render BBQ Front and Top Vineyard Path as live camera cards because their shared Solar Wall Light Cam model streams normally but does not expose dashboard still snapshots.

Release 1.7.72

- Fix the saved TV camera migration startup path by importing its regular-expression dependency.

Release 1.7.71

- Replace the retired Gate Doorbell camera and companion entities throughout managed dashboards, TV rotation, event imagery, detection controls, and person-presence inputs with the active Front Gate Doorbell.

- Keep the separate Inside Front Gate camera available under its accurate name and migrate saved TV camera lists without duplicating the new entity suffix.

Release 1.7.70

- Make Solar Room HomeBase Professional the primary estate-security status across the overview, admin, wall-display, and iPad dashboards.
- Keep the legacy S380 visible as an in-service migration dependency and label its connected sensor accurately as an active P2P session rather than general HomeBase health.
- Clarify the camera-health inventory so current cameras and retained legacy unavailable entities are counted separately.

Release 1.7.69

- Corrected T-04 Grenache to its true 2025 vintage and added source-backed documentary bottling records for DDT 14/2026: 2,916 red LRB25 bottles and 1,176 white LBB25 bottles.
- Active cellar tanks now retain and display their actual vintage even when it differs from the currently selected year.
- Documentary bottling records distinguish delivery dates from bottling dates and do not invent source volume or current finished-wine inventory.

Release 1.7.68

- Added an auditable live-cellar reset operation that safely marks unassigned test tanks empty, clears stale display values, preserves historical readings, and refuses to erase a tank with a positive linked wine lot.

Release 1.7.67

- Keep PLAATO demo telemetry off manual vessels and legal labels so simulated readings cannot appear as live cellar evidence.
- Switch current-vintage per-bottle calculations to completed bottling output when recorded cellar volume reaches zero, even if empty lot rows remain.
- Stop aging, bottled and closed wines from showing false active-fermentation stall warnings while retaining their historical readings.
- Audit grape-to-wine yield, 750 ml equivalents, packaging totals, per-bottle economics, laboratory normalization and vintage evidence gates.

Release 1.7.66

- Add a reliable Clear action for every completed intake review while retaining the source, audit trail, and linked vineyard records.
- Open already-ingested laboratory alerts directly in the laboratory record instead of the generic intake form.
- Replace fragile embedded PDF frames with a server-rendered first-page preview and make original attachments open inline.
- Remove acknowledged alerts immediately without an expensive full-dashboard reload.

Release 1.7.65

- Normalize Italian lab sample types such as `UVA`, `mosto`, and `vino` into API-ready grape, must, and wine records during intake analysis.

- Fix the linked-report detail lookup so reviewed laboratory PDFs open from the lab record and downstream charts.

Release 1.7.64

- Fixed Gmail intake for Apple Mail forwards whose PDF is an inline file nested under a multipart alternative body.
- Trust the estate mailbox, David's forwarding address, and active staff email addresses recorded in the people directory while continuing to quarantine unknown senders.

Release 1.7.63

- Makes the product search global when text is entered and maps common Italian cellar terms such as `Enzime`/`Enzima` and `Lieviti`/`Lievito` to the canonical enzyme and yeast classes.

Release 1.7.62

- Fixes the root cold-start defect by including the enology process and LAFFORT catalog in the optimized Today-page deferred loader; opening Winemaking after startup now has catalog data without requiring Refresh.

Release 1.7.61

- Extends bounded background catalog recovery across the first 25 seconds of a cold Home Assistant ingress load, covering the observed period of browser connection saturation without requiring a manual refresh.

Release 1.7.60

- Schedules enology recovery before the general dashboard renderer, isolating catalog recovery from unrelated section-rendering failures.

Release 1.7.59

- Adds deferred background recovery for the large enology payload after the initial dashboard has rendered, so catalog data recovers automatically even when Home Assistant ingress drops both immediate attempts under startup load.

Release 1.7.58

- Automatically retries the enology process request after the main parallel dashboard load, preventing transient ingress failures from leaving the LAFFORT catalog in a permanent-looking pending state.
- Replaces the ambiguous static catalog placeholder with accurate loading and recoverable error messages.

Release 1.7.57

- Restores the Entrance, Vineyard and Etna camera pages on Cog/WPE television kiosks; low-power displays now use the existing sequential scheduled-image cache instead of hiding camera content.
- Keeps camera requests serialized and refresh-limited so the restored TV pages do not create parallel Eufy streams or add unnecessary load to Home Assistant.

Release 1.7.56

- Adds vintage-over-vintage fermentation overlays for specific gravity, soluble solids, temperature and pH, aligned in 12-hour buckets from each lot's first recorded fermentation observation.

- Adds vintage-over-vintage chemistry overlays with comparable variety/block filtering for pH, total acidity, Babo, potential alcohol, potassium, YAN/APA and every supported cellar analyte.
- Expands canonical cellar laboratory coverage to actual alcohol, residual sugar, volatile acidity, malic and lactic acids, free and total sulfur dioxide, turbidity and dissolved oxygen while preserving every reported unit.
- Expands the post-fermentation testing pipeline with dryness, volatile-acidity, malolactic and sulfur-dioxide evidence expected for professional enologist review.

Release 1.7.55

- Adds a source-verified additive protocol library that preserves purpose-specific dosage ranges, timing, preparation, prerequisites, contact time and incompatibilities instead of collapsing a product to one ambiguous dose.
- Adds an additive decision pipeline for each wine lot with due, forecast, blocked and applied states, projected quantity ranges, density-gate timing and explicit missing-evidence reasons.
- Adds YAN/APA, potential alcohol, turbidity, fruit condition, laccase, anthocyanin/tannin ratio, inoculation time and planned filtration evidence to the enology profile.
- Persists auditable prediction snapshots every 30 minutes while keeping product selection, exact rate, legal review and application under enologist approval.

Release 1.7.54

- Adds a source-backed enology product database that refreshes every official LAFFORT winemaking range and keeps official product, product-sheet and safety-sheet links.
- Adds product-class and text filters, a full directory, selected-lot review candidates and unit-safe volume projections when a verified technical-sheet dose is available.
- Ranks candidate yeasts, enzymes, nutrients, tannins and other cellar products from wine color, variety, target style, process stage and YAN evidence without turning suggestions into automatic instructions.
- Keeps every planned or applied product subject to the current technical sheet, applicable rules, recorded product lot and enologist approval.

Release 1.7.53

- Repairs the administrator People OpenStreetMap so its map library loads on demand instead of depending on another map having been opened first.
- Uses the Home Assistant Person entity as the authoritative current location, with its selected device tracker shown transparently and other phone trackers used only as fallback.
- Shows current positions without 24-hour trails or zone-driven zoom, and includes Carmela and Nunzio in the administrator tracking map.

Release 1.7.52

- Adds a dedicated Enology → Winemaking page with lot-scoped process control from fruit reception through must analysis, inoculation, fermentation, pressing, post-fermentation stability and aging.
- Adds enologist-controlled stage status and audit history, with completion and skipped-stage decisions restricted to authorized enology roles.
- Adds lot-volume projections for source-backed yeast and enzyme rates, while refusing nutrient and tannin quantities until YAN evidence and exact product rules are available.
- Brings must chemistry, unit-safe fermentation charts, additive approvals, product-lot traceability and density-based completion projections into the same active-lot workspace.

Release 1.7.51

- Route the Cameras breaker and power meter dashboard cards through Tuya Local.

Release 1.7.50

- Added a traceable enology process model from the September 2 agronomist meeting, including proposed yeast, enzyme, nutrient and tannin stages with mandatory enologist approval.
- Added fermentation trend forecasts from dated density evidence, stalled-fermentation review flags, YAN evidence gates and volume-based proposed-quantity disclosures.
- Added a stage-aware laboratory testing pipeline and scheduled the September 3 request for separate representative grape samples: pH, total acidity, Babo, potassium and potential alcohol calculated from Babo.
- Added an estate-calibrated Babo-to-potential-alcohol model that learns only from paired reviewed laboratory values and refuses to invent a fallback factor.
- Added enologist views for unit-safe maturity chemistry overlays, lot-by-lot fermentation kinetics and a traceable additive ledger, with canonical bilingual analyte names and calculated values labeled separately from measurements.

Release 1.7.49

- Removes the retired main daily-consumption entity that still rendered as Entity not found after the Tuya Local migration.
- Uses verified local main voltage and breaker-protection entities in its place across the estate, kiosk, display and administrator dashboards.

Release 1.7.48

- Migrates the estate distribution, kiosk, display, iPad and administrator dashboards from retired cloud Tuya entity IDs to verified Tuya Local entities.
- Adds safe local utility monitoring and confirmed hold-to-toggle controls for the legacy Bluetti breaker, hot water and kitchen pressure-water circuits.
- Leaves the offline camera circuit visibly unavailable until its device returns to the Baiamonte LAN instead of assigning an unverified entity ID.

Release 1.7.47

- Defines the red-wine planning yield explicitly as litres of estimated finished wine per kilogram of grapes, with the current factor, formula, inverse, source and 67–70% typical range exposed in projection data.
- Labels calculated wine and bottle outputs as estimates throughout Harvest, Vintage, TV and Bottling views and states that recorded cellar or bottling volume replaces them.
- Separates production yield from physical wine mass conversion; litres of wine are converted to kilograms only when a measured density is supplied.

Release 1.7.46

- Adds labeled, color-matched harvest-date lines to the multi-vintage Laboratory trend, using recorded first-pick dates and a dashed current forecast only when no actual pick exists.
- Makes source-backed points in the exploratory, trajectory and like-for-like endpoint charts open the retained original laboratory report, with saved sample detail as the fallback for older evidence.

- Preserves each point's source report metadata through Laboratory comparison and vintage-projection APIs without changing measured values or existing averages.

Release 1.7.45

- Keeps the Main Parking/Vineyard North visual-watch pipeline operational when its optional local RTSP address is offline by falling back to a genuinely fresh Home Assistant camera frame.
- Rejects cached or saved camera images as current visual evidence and reports which protected capture path supplied each successful health probe.

Release 1.7.44

- Adds a signed Twilio WhatsApp Business Calling gateway with a bilingual Tenuta Baiamonte company reception IVR, personalized Manager and Reporter menus, live estate answers, call-status auditing and a safe public-caller reception boundary.
- Keeps Meta Business messaging separate from Twilio voice calling and exposes a credential-free calling readiness panel in Admin Messaging.
- Adds protected configuration for the Twilio account, API key, TwiML Voice Application, WhatsApp sender and exact public HTTPS webhook while leaving voice disabled until every external link is complete.
- Extends the same signed, personalized IVR to Twilio PSTN telephone numbers and authenticated SIP/VoIP domains, with independent channel switches, line counts, source-aware auditing and read-only resource verification.
- Adds protected PSTN/SIP forwarding and administrator-confirmed outbound PSTN or SIP calls from a fixed verified Baiamonte caller ID; destinations and credentials are never returned to the browser.
- Reorganizes Admin → IVR into channel readiness, call routing, outbound calling, forwarding, WhatsApp follow-up and personalized contact controls without adding a polling loop.

Release 1.7.43

- Recognizes Meta WhatsApp call-permission replies as quiet system-control events, preserving the consent audit without sending them to the IVR, AI assistant or human-review queue.

Release 1.7.42

- Parses Meta WhatsApp interactive buttons and list replies as their visible IVR choices instead of sending them to unsupported-message/AI fallback handling.
- Accepts menu emoji plus label combinations such as “□ Today” and “□ Foxes this month” through deterministic local routing.

Release 1.7.41

- Identifies owner-assisted cistern reference readings distinctly throughout the dashboard and TV feed.

Release 1.7.40

- Recalibrates cistern vision around the owner-confirmed full ledge immediately below the access door and fixed corner perspective.
- Explicitly excludes damp masonry, drying bands and historical tide marks from current water-surface detection.
- Adds administrator owner-assisted level calibration plus guarded “Mark current frame as full” and “Mark current frame as empty” controls on Operations → Water.

- Publishes calibrated level evidence to Home Assistant and feeds it into shadow learning and future litre-capacity calibration.

Release 1.7.39

- Aligns the fixed cistern camera calibration with the owner-marked full reference: the top inner wall ledge appears diagonal from lower-left to upper-center because of perspective, and the model must measure physical fill height rather than dark image area.

Release 1.7.38

- Allows the first owner-landmark-calibrated cistern reading to replace a substantially different legacy estimate; the large-change safeguard now compares only like-for-like calibrated readings.

Release 1.7.37

- Recalibrates the fixed cistern view to the owner-confirmed physical references: full is immediately below the upper access door and empty is the visible floor/base; glare, shadows and wet-wall boundaries can no longer be treated as the waterline.
- Retires the misleading 88% legacy reading from operational use, keeps it as audit history, starts a clean door-calibrated v2 shadow model, and replaces the unsupported WebRTC card with Home Assistant's built-in live camera card.
- Defines every Nunzio water delivery as 5,000 L, migrates existing delivery records, shows the quantity in contractor/admin workflows, and uses confirmed two-camera plus calibrated before/after level evidence to learn the cistern's real usable capacity.
- Adds a conservative liters-available projection with uncertainty range and a durable Home Assistant liters sensor; no liter quantity is invented before a physically calibrated delivery delta exists.

Release 1.7.36

- Makes every visible WhatsApp IVR menu emoji behave exactly like its numbered option for Manager, Reporter and Guest profiles, including emoji variation-selector forms sent by iPhone and Android keyboards.
- Repairs Wendy's emoji-only fox shortcut so ■ immediately returns the friendly monthly fox update and latest confirmed image instead of entering the general intake flow.
- Keeps emoji-enhanced text responses while preserving natural spoken descriptions for voice replies.

Release 1.7.35

- Reorganizes Admin Security around a compact decision queue, daily movement ledger and reviewed known-vehicle registry, while moving camera-pipeline configuration into a clearly separated expandable control.
- Improves Email with a cached unread-first overview, accurate unread counts, a prominent new-mail action and a collapsed compose area so routine triage is faster and less cluttered.
- Verifies the live WhatsApp Cloud API, personalized IVR and voice-entry paths while preserving the current local-first routing, fox update and routine-conversation filtering.

Release 1.7.34

- Repairs managed Home Assistant dashboards after recent integration changes by removing retired entity references, replacing placeholders with live navigation, and clearly separating Solcast estimates from future Growatt actual production.

- Makes gate-camera health use the working camera entity and removes obsolete RF Bridge, lightning-distance and Uptime Kuma references so Lovelace no longer creates configuration-error cards for those retired targets.
- Reduces TV and local-service load by refreshing traffic overlays once per minute, caching short-lived responses with a bounded stale fallback, and sending only the AIS connection and local-vessel fields used by the display.

Release 1.7.33

- Adds a dedicated Admin Security workspace with a daily camera-backed vehicle entry, exit, parking and passing ledger, retained-frame review, flags, categories, staff links and conservative plate transcription.
- Adds a confirmed known-car database and statistics card; administrator tags and corrections become bounded candidates for later sightings while attendance, access decisions and payroll remain separately reviewed.
- Adds configurable entry, parking, doorbell and perimeter camera sources with direction calibration and an explicit always-analyze override for a primary battery parking camera; Eufy events and wired cameras feed the same efficient pipeline.
- Allows an optional deliberately recorded plate on a person's configured vehicle profile and keeps all security evidence administrator-only with the established encrypted retention policy.

Release 1.7.32

- Adds a guarded Admin AI switch for local camera intelligence: shadow collection and weekly validation are enabled now, while a primary-mode request remains queued until every evidence gate passes.
- Requires a 30-day forward shadow period, four weekly scorecards, at least 25 paired decisions, at least 90% agreement and no more than 5% local failures before local camera AI can become authoritative.
- Records Eufy on-device vehicle detections beside the established vision result without retaining another image, runs the due scorecard from the camera process, and preserves the established camera service as automatic fallback.

Release 1.7.31

- Adds a private, personalized contractor dashboard with Nunzio's confirmed and reported water deliveries, fixed-price jobs, payment status, estate priorities, weather, and existing time clock.
- Allows contractors to add work-plan items and separately report a water delivery from the same dashboard while preserving administrator approval and payment controls.
- Reconciles a contractor-reported delivery with camera-route and cistern-rise evidence into one canonical delivery, preventing duplicate records and duplicate payment jobs whether the report or automatic evidence arrives first.
- Adds clear emojis between every WhatsApp IVR menu number and description, including the fox update at option 12.

Release 1.7.30

- Adds an event-driven West Etna fox watch using Eufy animal or motion evidence plus bounded nighttime stills, with conservative fox-versus-dog/cat classification and silent handling of uncertain wildlife.
- Retains deduplicated encrypted fox frames, records activity and grape-risk context, exposes monthly sightings and the latest confirmed image in Camera Center, and adds review-gated fox-watch metrics to Admin AI.

- Sends Wendy a friendly fox photo only for credible sightings and only inside Meta's open conversation window, with a six-hour anti-noise cooldown.
- Adds a cute monthly Fox update to the WhatsApp Manager IVR, including the latest confirmed picture when available.
- Confirms a water delivery from two distinct delivery-camera observations plus a measured cistern rise, then creates a fixed-price water-delivery job for Nunzio in the existing labor/services queue on verification hold without executing payment.

Release 1.7.29

- Standardizes operational camera labels across the Eufy dashboard, TV payloads, Admin/People selectors, WhatsApp Manager and AI workflows without changing durable Home Assistant entity IDs.
- Establishes Main Parking, Rear Gate, Rear Gate 360, Rear Entrance Path 360 and Cistern 360 as canonical estate-facing names while retaining the original Home Assistant friendly name for diagnostics.

Release 1.7.24

- Calibrates the fixed Main Parking view for its real working distance: a unique configured color and body-style combination can match without unsupported badge-level make/model detail.
- Makes an administrator confirmation of an uncertain vehicle match promote that reviewed observation into high-confidence presence evidence for location history and timesheet reconciliation.

Release 1.7.23

- Opens encrypted retained vehicle evidence inside a protected in-app viewer so Home Assistant ingress no longer loses the administrator session when displaying a frame.
- Adds an administrator-only fresh parking-frame analysis action to each worker vehicle profile and records every manual run in the audit trail.
- Allows an explicit administrator scan outside the normal work schedule while keeping scheduled captures bounded to the configured work window.

Release 1.7.22

- Makes Admin/People and the scheduled vehicle analyzer share the same canonical worker-vehicle defaults, so Giancarlo's silver Volkswagen Golf remains an active candidate before a profile has been manually re-saved.
- Preserves saved vehicle edits and explicit tracking opt-outs while repairing partially persisted profiles for Giancarlo, Carmela, and Luca at analysis time.
- Classifies the legacy `camera.vineyard_north` entity by its current Main Parking role so sightings, location history, and review evidence use the correct estate zone.

Release 1.7.21

- Adds an encrypted, content-addressed worker-camera evidence archive that preserves the original reviewed frame—including incidentally visible faces and plates—without duplicating one image for every worker candidate.
- Keeps biometric recognition, face templates and plate OCR out of attendance and payroll decisions; retained images are administrator-only human-review evidence and vehicle observations remain advisory.
- Audits every evidence view, stores files outside MariaDB with AES-256-GCM protection, uses no browser cache, and applies bounded 90-day retention, 180 days after review, or an explicit administrator legal hold.

- Links Eufy vehicle/motion frames and scheduled parking frames to vehicle-learning observations while retaining metadata after an expired image is removed.

Release 1.7.20

- Expands worker-vehicle learning from one parking frame to a load-spread list of Eufy or wired Home Assistant cameras, with automatic estate zones and reviewed arrival/departure transitions.
- Makes vehicle evidence primary for attendance support, records the matched one of a worker's configured vehicles, and adds review-gated accuracy by camera to Admin → People and Admin → AI.
- Uses Eufy's on-device familiar-person label only as secondary corroboration when it maps unambiguously to one configured person; unlabeled person detections never create an identity.
- Retains timestamps, zones, source, confidence, model and review outcomes without retaining camera images, face templates, driver identity or license plates.
- Adds current last-seen zone, recent evidence review, multi-camera configuration, daily timesheet comparison and unknown Eufy-label auditing for future camera expansion.

Release 1.7.19

- Staggers scheduled Home Assistant and cloud-backed integration work so camera traffic, MQTT keepalives, mail, planning and external sources do not all compete on the same event-loop turn.
- Applies bounded exponential backoff after a scheduled source failure while retaining the configured normal cadence after recovery.
- Preserves persisted process timestamps across add-on restarts, preventing a deployment from creating an immediate all-source refresh burst.

Release 1.7.18

- Adds a dedicated Admin Network operations console covering Starlink/WAN, routing, managed switching, Wi-Fi, tunnels, LTE/radio, connected devices, cameras, internal services and public website publishing.
- Discovers presentation-safe Home Assistant network and camera entities automatically, while excluding credentials and private attributes from the API.
- Shows status lights, the monitored network path, real utilization and link-quality meters, searchable infrastructure, camera connectivity, endpoint freshness, recent incidents and explicit instrumentation gaps on desktop, tablets and phones.
- Reuses the shared 30-second Home Assistant snapshot so the expanded monitoring view does not multiply state-inventory requests or estimate metrics that are unavailable.

Release 1.7.17

- Restricts the Solar workspace and reserve learner to identified estate-energy equipment, preventing phones, tablets, watches, and solar-camera batteries from being mistaken for the off-grid battery bank during commissioning.

Release 1.7.16

- Added dedicated Operations Water and Solar workspaces with responsive system inventories, history, forecasts, AI/learning panels, commissioning states, and expansion space.
- Added five-minute estate energy evidence capture plus a guarded overnight-demand and battery-reserve learning foundation for the Growatt/Felicity installation.

- Kept automatic power control locked during commissioning: missing telemetry is never treated as zero and no load can be switched until its entity and authority are explicitly approved.

Release 1.7.15

- Uses active rear/front gate and doorbell motion, vehicle, or ring events with an available event image as low-cost triggers for worker-vehicle screening.
- Checks whether a car is actually visible before comparing it with configured worker vehicles; motion alone never creates worker-presence evidence.
- Records only positive worker-vehicle matches from event views, because a gate image cannot prove another worker is absent, and retains no event image or license-plate data.
- Deduplicates screened event frames by fingerprint so an unchanged active motion event does not repeatedly consume visual-analysis resources.

Release 1.7.14

- Corrects the MariaDB-compatible closing structure of the 1.7.13 worker-vehicle profile seed so migration 130 completes and the application starts normally after update.

Release 1.7.13

- Adds editable vehicle profiles and optional normal schedules to each Admin → People record, including Giancarlo's silver Volkswagen Golf, Carmela's blue Fiat Punto, and Luca's white Renault Kangoo or older red Fiat Panda.
- Uses one fresh Main Parking frame at a bounded cadence to check every configured worker vehicle together, retaining only timestamped status, confidence, review state and a frame fingerprint rather than images or license plates.
- Keeps Giancarlo's Monday–Saturday 07:00–14:00 schedule as expected-time context while treating Carmela and Luca as observation-only workers with no inferred lateness or missing hours.
- Adds durable daily first/last-seen history and a reviewable comparison with recorded timesheets without identifying the driver or altering payroll automatically.
- Feeds capped vehicle evidence into the existing timesheet-presence workflow when stronger Home Assistant person, GPS or named camera evidence is unavailable.

Release 1.7.12

- Reorganizes Admin → Social around a compact audit overview while retaining every publishing, account-history and relationship tool; detailed follower lists now live in a collapsed panel at the bottom.
- Adds automatic six-hour collection of supported Facebook and Instagram account insights, post engagement, publishing health, source freshness, 30-day movement and 90-day audience trends.
- Makes optional Meta insight collection resilient to individual retired or unavailable metrics so one rejected metric cannot stop the rest of the audit refresh.
- Extends official Instagram relationship comparisons with mutual counts, follow-back and reciprocity rates, new followers, recent unfollowers, import coverage and direct public-profile review links.
- Adds responsive social-audit layouts for phones, tablets and desktop displays while retaining the policy-safe official-export workflow for named relationships.

Release 1.7.11

- Improved the TV AIS display with consistent SVG vessel badges and country-code labels that render reliably on embedded TV browsers.
- Suppressed the duplicate triangle fallback markers whenever the live embedded AIS map is already displaying its detailed ship shapes.

Release 1.7.10

- Reconciles the PTZ dashboard with the live Eufy capability inventory, retaining Fox Den and adding the omitted South Vineyard 360 controls.
- Starts a bounded P2P session before PTZ, preset, or calibration commands and keeps P2P start/stop explicit on the dedicated control pages.
- Removes automatic live playback from camera-wall overview cards so merely opening a dashboard does not wake cameras or create background stream workers.
- Adds Driveway Entrance, Main Entrance, and Top East Vineyard so every currently enabled Eufy camera is reachable from the camera wall.

Release 1.7.9

- Keeps successfully handled WhatsApp IVR menu choices, live snapshots, settings, calculations, field-entry steps, manager controls and assistant answers out of Human Review.
- Retains every source message and a new audited archived disposition rather than deleting conversation evidence.
- Preserves genuine submissions, human handoffs, unknown senders, failed routes and intervention-required messages in the review workflow.
- Routes assigned voice notes through the same disposition and avoids launching a competing generic intake analysis for the original audio.

Release 1.7.8

- Accepts complete official Meta exports up to 512 MB, fixing rejection of media-heavy archives such as the 148 MB Tenuta Baiamonte export.
- Streams the uploaded archive to a temporary file instead of holding it in application memory, then reads only the small follower and following JSON members.
- Ignores bundled account photos and videos, retains the existing relationship-data decompression limits, and always removes the temporary archive after processing.

Release 1.7.7

- Adds a safe 10-day Instagram relationship-export cadence without scraping Meta or automating account credentials.
- Shows the current export state and next due date in Admin → Social, with direct access to the official Meta Accounts Center export workflow.
- Creates one informational reminder when the official export is due and resolves it automatically after a valid ZIP or JSON import.
- Automatically imports, compares and retains every supplied official export while the supported Meta API continues collecting aggregate audience totals every six hours.

Release 1.7.6

- Adds durable Facebook and Instagram audience snapshots to MariaDB, scheduled every six hours, with retained account metrics and verified forward-only net gain/loss events.
- Adds 30-day audience gains, losses, net change, current account totals, recent change history, and account-source status to Admin → Social.
- Adds a policy-safe Instagram relationship importer for official Meta ZIP or JSON exports, including people who do not follow back, people the estate does not follow back, and exact recent unfollowers after a second import.
- Keeps all imported relationship snapshots and source metadata for historical comparison without scraping Instagram, automating follow actions, or falsely attributing aggregate count changes to individuals.
- Extracts the administrator-only social API and audience browser UI into focused backend and frontend modules to preserve the established application module boundaries.

Release 1.7.5

- Reorganizes the second page of every tank legal label around the fermentation decisions that matter first: calculated phase, guidance, completion-review readiness, confidence, current density and target, activity and pace, temperature range, progress, stability, remaining time, and projected finish window.
- Replaces the generic density panel with a combined measured density and fermentation-activity curve, an FG target, and a clearly distinguished projected finish segment while keeping a dedicated temperature-control graph.
- Adds the newest exact-lot laboratory confirmation to the tank display with sugar, alcohol, pH, malic acid, lactic acid, source, review state, and interpretation; name-only matches are deliberately excluded from public tank labels.
- Retains estimated alcohol, peak activity, reading freshness, sensor battery and Wi-Fi, and vintage history as secondary context rather than removing existing information.
- Adds responsive tablet and phone layouts for the richer fermentation page and advances the tank-label asset version so installed displays replace their cached copy.

Release 1.7.4

- Expands each PLAATO V2 tank into a wine-specific fermentation digital twin with density, temperature, and fermentation activity aligned on one active-batch timeline while retaining the original detailed graphs and telemetry access.
- Replaces the old seven-day sensor request with the active batch window, bounded at 90 days; the full returned series drives calculations while a labeled, evenly sampled curve protects Home Assistant, phones, and tablets from oversized rendering work.
- Preserves PLAATO's activity measurement when supplied by the API and otherwise labels the rolling density-slope calculation explicitly, instead of presenting derived activity as a direct sensor value.
- Adds wine fermentation phase and pace, peak activity, apparent-stability duration, projected target time and range, automatic sensor milestones, and a guarded completion-review gate.
- Tailors process guidance to alcoholic wine fermentation, cap-management context, reference density/laboratory checks, and pressing or racking readiness; malolactic completion remains exclusively laboratory-confirmed.
- Extends the raw telemetry table with activity and source provenance, and keeps device health, batch metadata, cellar observations, vintage history, and enologist authority visible in the same process panel.

- Integrates received must and wine laboratory reports into the matching tank's sensor timeline by exact wine-lot link, exact lot/tank code, or a clearly labeled normalized wine/vintage inference; only reviewed exact matches become authoritative.
- Reconciles like-unit laboratory density and alcohol results with the nearest sensor reading and calculated estimate while retaining pH, sugar, malic acid, lactic acid, flags, source reports, review status, and decision actions in the same wine-batch view.

Release 1.7.3

- Gives the LAN TV process an independent, lightweight INGV webcam-pointer refresh at the configured Etna cadence, so the Etna photo stays current even while the slower full official-source job is queued.

Release 1.7.2

- Reloads the shared Etna status file when the scheduler writes a newer snapshot, keeping the separate LAN TV process synchronized without requiring an add-on or television restart.

Release 1.7.1

- Repairs the Etna TV camera feed so every successful INGV source check advances a bounded image token, preventing kiosk browser caches from holding an older photo when the upstream timestamp or path is unchanged.
- Preserves the last verified Etna image and token when INGV is unavailable instead of presenting a failed fetch as current data.
- Supervises the recurring integration scheduler and restarts it after an unexpected loop-level fault, preventing Etna and other scheduled sources from becoming silently stale while the web application remains online.

Release 1.7.0

- Makes the Operations Today page usable from a compact first data phase while the larger Agronomy, Enology, laboratory, history, map, and finance workspaces hydrate in the background.
- Lazy-loads the external map library only when the Estate Atlas is opened and gives versioned JavaScript and CSS immutable browser caching for faster repeat visits.
- Fixes the public-feed health checkpoint to use the database's local clock consistently, preventing a healthy publisher from being shown as overdue because of a UTC/local-time mismatch.
- Audits live integrations, scheduled processes, database and payment integrity, storage, runtime resource use, API latency, payload size, compression, JavaScript syntax, dependencies, and the complete automated test suite.
- Updates and rebuilds the owner manual for the 1.7.0 performance and reliability baseline.

Release 1.6.98

- Fixes the iPhone tank-label page flow so the header, legal card, and footer stack vertically at full width instead of collapsing into narrow side-by-side columns.
- Advances the tank-label asset version so installed iPhones immediately receive the corrected responsive stylesheet.

Release 1.6.97

- Adds a dedicated iPhone and small-phone tank-label layout with safe-area padding, readable wrapped headings, full-width legal fields, two-column compact readings, and natural vertical page scrolling.

- Makes the Tank Sensor process panel phone-native with a compact persistent header and independently scrollable KPI, chart, process-health, and vintage-history content.
- Preserves the fixed, no-scroll landscape-tablet label while using pointer and viewport detection to support both portrait and landscape phones.

Release 1.6.96

- Reorganizes the bilingual WhatsApp IVR into compact Manager, Reporter, and Guest menus aligned with the current Operations, Agronomy, Vintage, Enology, Tank Sensor, Olives, Estate Systems, Hospitality, team, finance, and record-entry data sets.
- Adds live composite summaries with role and administrator boundaries, including contact-safe public hospitality responses and administrator-only team and finance data.
- Analyzes approved incoming WhatsApp photos and up to six chronological representative video frames as review-gated visual evidence, preserving the original attachment and requiring human review when video decoding or visual certainty is insufficient.
- Bounds WhatsApp responses below the channel message ceiling while preserving complete data rows wherever possible.

Release 1.6.95

- Keeps the complete Tank Sensor health reading on the cellar legal-label screen while moving it into the primary readings strip, preventing the bottom row from being clipped on landscape tablets.
- Preserves battery, Wi-Fi, batch, and sensor status details without enabling scrolling on the main legal-label screen.

Release 1.6.94

- Turns the green tank-label status light into a Tank Sensor process control: one tap opens live monitoring for ten seconds, a second tap pins it, and the close control dismisses it.
- Adds Tank Sensor gravity and temperature graphs, activity, estimated alcohol, progress, finish outlook, device health, source-backed vintage and grape history, and the complete linked cellar process to Enology and tank panels.
- Keeps the operator interface vendor-neutral under the Tank Sensor name, enables realistic moving data with the default local demo key, and fits the legal-label main screen without scrolling or bottom clipping while retaining internal scrolling in the detailed panel.

Release 1.6.93

- Opens a complete fermentation and process panel from every Cellar or Enology tank, with PLAATO temperature and gravity graphs, raw seven-day samples, device health, batch metadata and linked cellar observations.
- Adds transparent forward calculations for fermentation phase, progress to the configured final-gravity target, recent activity, estimated alcohol, remaining time and projected finish, with visible confidence and enologist-review guardrails.
- Adds the local-only `demo` PLAATO key, enabled by default for this release, which generates realistic moving telemetry for active tanks without contacting PLAATO Cloud; entering a real API key restores mapped live-cloud operation.

Release 1.6.92

- Adds first-class, read-only PLAATO V2 Pro automatic tank monitoring through the official PLAATO API, including temperature, specific gravity, Plato, derived gravity-drop rate, batch progress, ABV, attenuation, battery, Wi-Fi, firmware, freshness and seven-day trends.
- Adds a protected PLAATO API key and explicit tank mapping in Home Assistant App Configuration, plus a PLAATO automatic mode in Enology while retaining manual and generic Home Assistant sensor modes.
- Surfaces PLAATO evidence and device health on enology tank cards and tank labels without treating batch volume as a live tank-level reading or fabricating pH data.
- Resumes scheduled integration cadence from durable history after an add-on restart, preventing a simultaneous camera, mail, weather, planning and full-refresh burst during Home Assistant recovery.

Release 1.6.86

- Uses enabled local RTSP feeds from either S380 HomeBase 3 or HomeBase Professional for Kitchen and Vineyard North camera cards, matching the current-frame path already used by Front Yard and Main Parking.
- Retains the verified cached-cover fallback for cameras without an enabled RTSP capability, avoiding P2P wake-ups and estate-wide stream bursts.

Release 1.6.85

- Replaces the stale, foggy Eufy bridge stills for Front Yard and Main Parking with current live frames while the camera page is open.
- Keeps every other camera on the existing low-load cached path so the clarity repair does not create an estate-wide stream burst.

Release 1.6.84

- Adds a low-load, hourly fixed-view Vineyard North evidence pipeline with daylight-quality screening, forward-only scene and canopy-color trends, conservative AI interpretation, and persistent review-gated alerts.
- Correlates suitable visual observations with current weather, recent vineyard work, scouting and treatments without turning camera imagery into disease diagnoses or automatic treatment recommendations.
- Adds the latest fixed-view image, quality, visibility, activity, change, and inspection context to the authenticated Today dashboard and LAN TV, plus learning coverage in Admin AI.

Release 1.6.83

- Uses the RTSP demuxer's supported socket timeout, verifies Vineyard North frame delivery, and distinguishes a credential-safe authentication rejection from transport or frame timeouts.

Release 1.6.82

- Negotiates bounded TCP and UDP RTSP transport for the Eufy fixed-view sources, retaining the low-load local capture path while supporting camera servers that reject interleaved TCP media.

Release 1.6.81

- Adds an administrator-only, credential-safe health probe for both protected RTSP sources so camera reachability and current-frame capture can be verified independently from AI analysis.

- Bounds RTSP acquisition and reports only sanitized failures, preventing one camera connection from consuming the full scheduled-process window or exposing a protected stream address.

Release 1.6.80

- Uses the cistern camera's protected, always-on local RTSP stream for illuminated AI readings, falls back to the new Eufy bridge's generated still, preserves the permanent stream, and wakes P2P only when both current-frame sources are unavailable.
- Adds a protected Vineyard North RTSP source slot for future fixed-view visual recognition without exposing credentials or allowing distant camera observations to become automatic disease diagnoses.
- Records whether each accepted cistern reading came from the generated bridge image or live camera so future learning and camera diagnostics remain auditable.

Release 1.6.79

- Repairs every native Home Assistant camera card to use the supported `camera_image` field, eliminating the camera-page configuration errors while retaining cached, on-demand imagery.
- Removes four retired Eufy camera aliases, replaces the retired IP cistern camera with the current Eufy cistern entity, and keeps the complete current camera set visible across overview, wall and health views.
- Updates the default TV camera rotation from the retired driveway alias to the current Front Yard camera.
- Migrates the cistern capture and AI pipeline from the retired IP entity to `camera.cisterna` while preserving custom camera selections, historical estimates, conservative alerts and the side-by-side shadow-learning release gate.
- Uses the Eufy bridge's stable device relationship to illuminate the internal cistern, waits for exposure, wakes its on-demand stream before capture, restores the previous light/stream state, and reports the actual recorded level range so repeated readings cannot masquerade as learned accuracy.

Release 1.6.78

- Camera dashboards now render the integration's current app-style camera frame instead of forcing the lower-resolution Eufy event-thumbnail entity. This improves clarity and freshness while retaining cached, non-live cards so phones, tablets, TVs, battery cameras, and Home Assistant remain efficient.
- Camera refresh controls now target the current bridge snapshot index rather than a single legacy camera entity.

Release 1.6.77

- Aligns Admin Control with the camera scheduler's current `camera-awareness` event name so a successful Eufy run immediately clears the red process card instead of leaving a stale legacy 502 visible.
- Retains `camera-snapshot-cache` as a recovery-only alias for historical failures while ensuring it cannot override current camera health.

Release 1.6.76

- Updates Camera Center for Baiamonte eufy Bridge 1.6.15 and the 9.6.4 Home Assistant integration while preserving stable device-key joins and existing entity IDs.
- Adds the bridge's recognized-person, unrecognized-person, dog, dog-behavior, sound and crying classifications to estate awareness without exposing identities, raw recognition payloads or unrestricted audio.

- Carries privacy-safe snapshot source and bridge schema, authentication and freshness diagnostics into the camera API so administrators can verify that the new bridge contract is active.
- Keeps all camera and PTZ controls capability-gated; fixed cameras cannot receive movement commands and sleeping battery cameras remain healthy rather than falsely offline.

Release 1.6.75

- Switched the Home Assistant weather radar to the keyless OpenStreetMap basemap, removing the CARTO `API KEY REQUIRED` watermark while retaining RainViewer animation, controls and attribution.
- Home Assistant state reads now retain the last complete snapshot across brief Supervisor/Core 502 hand-offs, preventing the Eufy awareness scheduler from staying failed after a transient Home Assistant reload.
- Prevents WhatsApp video, audio, HEIC/SVG and unknown binary attachments from being sent to the OpenAI Responses API as unsupported document `file_data`.
- Preserves unsupported media in the intake record, supplies safe attachment metadata to the analysis, forbids inference about unseen content and requires human review.
- Retains direct visual analysis for supported images and document analysis for supported PDF, office, spreadsheet, text and structured-data files.
- Adds MIME and extension regression coverage, including a mislabeled video fallback, so one unsupported attachment cannot create a persistent system error.

Release 1.6.74

- Rebuilds Camera Navigation as a phone-readable responsive section instead of squeezing five controls into one fixed row.
- Gives Refresh and HomeBase half-width vertical tiles and retains PTZ, Health and Wall as evenly balanced navigation controls.
- Shortens the mobile labels to `Refresh`, `HomeBase`, `PTZ`, `Health` and `Wall` while preserving every action and destination.

Release 1.6.73

- Makes Eufy live-view shutdown backward-compatible with browser and Home Assistant Companion pages left open during an upgrade: legacy stop requests are acknowledged without forwarding a duplicate command to the bridge.
- Adds an end-to-end regression proving the live stream generator remains the only Eufy cleanup owner for both current and cached clients.

Release 1.6.72

- Completes the post-install Eufy audit across the bridge, Home Assistant integration, Camera Center, cached imagery, events, alerts, PTZ capability gating and system load.
- Removes the duplicate live-stream stop request when an operator closes an on-demand camera view; the server stream generator remains the single cleanup owner, preventing harmless but noisy `LivestreamNotRunningError` entries.
- Adds regression coverage that preserves one-stop live-stream cleanup while retaining server-side protection for interrupted browser and Companion sessions.

Release 1.6.71

- Adds a durable Eufy estate-awareness pipeline that records on-device motion, person, vehicle, pet and doorbell events plus genuine camera outages and low-battery transitions without storing raw recognition data.
- Distinguishes healthy sleeping battery cameras from unavailable cameras, delays outage alerts for fifteen minutes, and correlates multiple failures in one estate area with shared power, network and station guidance.
- Joins cameras to renamed battery, Wi-Fi, image, detection and control entities through the integration's privacy-safe device key instead of fragile entity-ID assumptions.
- Restores cached Eufy event evidence after restarts, repairs Mega catalog-health reporting, and adds a scrollable event review timeline with acknowledgement history.
- Adds low-load on-demand live camera viewing that starts only when requested and stops when the view closes, while retaining slow cached refreshes on TV and overview pages.
- Adds administrator-only PTZ preset save/delete and verified doorbell canned-response controls; alarms, locks, microphones and unrestricted audio remain excluded from ordinary operations.
- Feeds camera health into the existing alert, Today, TV and WhatsApp flows, and adds recent authorized camera activity to the Manager IVR response.

Release 1.6.70

- Adds an Operations Camera Center with a complete Eufy estate overview, area and attention filters, online, activity, battery and PTZ summaries, cached event imagery, and a privacy-safe on-device finding.
- Adds capability-gated PTZ, 360 movement, presets, calibration, snapshot refresh, camera, light, motion detection, motion tracking, night-vision and recording controls; unsupported commands are rejected server-side and alarms, locks, microphones and speakers remain excluded.
- Keeps camera load bounded by using Eufy's cached event evidence in the camera grid and requesting a current snapshot only when one camera is opened.
- Enriches the TV camera walls with PTZ, battery and active Eufy detection details without increasing their existing fifteen-minute image refresh cadence.
- Integrates the Eufy 9.2.0 camera capability contract so controls come from advertised model support rather than names or assumptions.

Release 1.6.69

- Extends the complete treatment workflow to olives with olive fruit fly and olive peacock spot pressure screening, olive-specific growth stages, current and historical simulation, and crop-filtered pressure history.
- Adds olive trap, fruit-damage and leaf-symptom observations plus automatic and explicit paired pre/post-treatment scouting so verified outcomes feed treatment effectiveness learning.
- Separates olive and vineyard weather precedents, effectiveness, product duration, spray-window profiles and FRAC rotation sequences so learned grape behavior cannot drive an olive recommendation.
- Keeps weather-only olive signals at monitoring status and requires matching field evidence, a current reviewed Italian olive use, verified formulation, inventory, PHI, sprayer, area and Agronomist approval before presenting a product program.
- Uses the olive harvest forecast for PHI screening, requires the true olive treatment area instead of inheriting vineyard hectares, and calculates carrier batches from the selected sprayer rather than assuming two fills.
- Adds regression coverage for olive pressure, field evidence, prediction isolation, scouting vocabulary, dashboard wiring and simulator availability.

Release 1.6.68

- Restores normal mobile workspace filtering so the expanded navigation shows only the pages belonging to the selected Operations, Agronomy, Enology, Hospitality, Register, or Admin workspace.
- Keeps all workspace choices visible and retains the persistent Menu/Close behavior, allowing a workspace and one of its pages to be selected from a single opening.
- Adds regression coverage for every workspace-specific mobile page row and prevents the generic open-menu layout from overriding those filters.

Release 1.6.67

- Keeps the mobile two-level navigation open while a user selects a workspace and then a page, allowing both choices from one menu opening.
- Closes the mobile navigation only through its Menu/Close control and preserves the current page label after closing.
- Confirms the embedded application and Home Assistant Companion header surfaces both use the exact `#303030` charcoal.
- Adds regression coverage for the persistent mobile navigation state.

Release 1.6.66

- Reduces Entrance and Vineyard television camera reloads from every page rotation to independent 15-minute refresh windows while preserving immediate first-page loading and safe handoff after slow cameras.
- Reloads the Etna camera only when its official source update marker changes instead of attaching a new timestamp during every display-data update.
- Matches the application header to Home Assistant Companion's `#303030` charcoal so the embedded mobile interface reads as one continuous application surface.
- Adds regression coverage for camera refresh cadence, camera-page handoff, stable Etna imagery, and the shared Home Assistant header color.

Release 1.6.65

- Prevents scheduled television data updates from rebuilding the visible Today page and producing a refresh flash when the display rotation returns there.
- Continues fetching fresh estate data on schedule, applies a pending update as soon as the display advances, and keeps initial loading and manual refresh immediate.
- Adds regression coverage for stable Today rendering and the revised live-refresh contract.

Release 1.6.64

- Consolidates overlapping browser status timers into one visibility-aware 30-second poll while retaining minute-level release checks and live Admin monitoring.
- Reuses one Home Assistant state inventory for 30 seconds across operations, worker and display consumers instead of repeating the same Supervisor request for nearby page loads.
- Reuses the complete television aggregation for 90 seconds so staggered displays share expensive database, weather and planning work while still rebuilding before the next normal 120-second refresh.
- Suspends nonessential admin uptime and TV overflow animation work while the page is hidden, reducing browser and embedded-display load without removing information or controls.

- Adds dedicated efficiency regression coverage and retains the bounded database pool, serialized integration scheduler, camera cache and failure backoff.

Release 1.6.63

- Expands the large-screen Today weather card with GW2000 condition and update time, gust, pressure, solar radiation, UV, soil moisture, dew point, VPD, and the observed 24-hour temperature range.
- Adds a compact three-day forecast with useful recent-weather fallbacks when Home Assistant forecast data is unavailable.
- Adds live planning, hospitality, task, and treatment-weather context to the Today work and decision cards so large televisions remain informative without removing existing operational data.
- Rebalances the Today grid for wide and narrower television aspect ratios and adds compatibility regression coverage for the enriched display.

Release 1.6.62

- Locks the Baiamonte header and section navigation into a fixed application shell while allowing the operational canvas beneath them to scroll independently on touch and desktop devices.
- Reworks iPhone navigation into a compact closed state with the current page, a complete three-column workspace selector, and readable two-column page buttons when opened.
- Removes clipped desktop-style mobile tabs, preserves every workspace and page, and adds proportional sizing for narrower iPhones without changing worker, tank, or television layouts.
- Keeps the frontend core within its modularization size budget and adds regression coverage for the fixed shell, touch scrolling, responsive workspace navigation, and mobile menu state.

Release 1.6.61

- Restores native vertical touch scrolling across Android, iOS, coarse-touch browsers, the worker portal, and Home Assistant embedded WebViews after the Android viewport optimization.
- Keeps page height and overflow owned by the browser while retaining dynamic viewport sizing, safe-area spacing, responsive layouts, and touch-sized controls.
- Replaces the header's opaque logo tile with the transparent Baiamonte artwork and matches the app header to the surrounding charcoal Home Assistant and browser chrome.
- Adds regression coverage for touch panning, worker-page scrolling, embedded WebViews, transparent header branding, and the shared app-chrome color.

Release 1.6.60

- Optimizes the main dashboard, worker portal, cellar tablets, and tank displays for Android browsers, Fully Kiosk, and the Home Assistant Companion app without removing operational data.
- Adds keyboard-aware dynamic viewport sizing, safe-area handling, coarse-touch targets, portrait and landscape tablet layouts, native maskable Android icons, and cache-busted tank-display assets.
- Replaces the Today card's decorative rings with a lightweight live weather illustration that follows Baiamonte conditions and Europe/Rome day or night, including sun or moon, vineyard ridges, vine rows, clouds, rain, fog, snow, and storms.
- Preserves readable high-contrast text, reduced-motion support, and the existing live weather evidence while avoiding a large background-image download on field devices.

Release 1.6.59

- Repairs the projections and alert-settings routes after modularization by restoring their explicit dependencies and adding executable regression coverage.
- Improves responsive dashboard hierarchy and navigation without removing visible data, including restoring vintage detail that had been hidden on narrow screens.
- Optimizes iPhone, iPad, Safari, and Apple home-screen behavior with safe-area support, accessible touch targets, stable dynamic-height dialogs, and a native 180-pixel touch icon.
- Preserves full-height television layouts and verifies the dashboard at phone, tablet, desktop, and television viewport sizes without horizontal overflow.

Release 1.6.58

- Extracts the AI administration, durable-learning rebuild, assistant question, and review-gated suggestion routes into a focused intelligence router without changing their public contracts.
- Preserves administrator and write authorization, request normalization, provider checks, cost and profile validation, suggestion intake, and human-review safeguards.
- Adds executable intelligence route, authorization, normalization, and validation contracts while reducing direct route ownership in `app/main.py` from 75 to 68.

Release 1.6.57

- Strengthens the modular backend with executable FastAPI route and authorization contracts for worker, payroll, cellar, dashboard, intake, attachment, and public boundaries.
- Centralizes attachment sanitizing, hashing, persistence, and rollback cleanup so failed database writes cannot leave orphaned worker or operational files.
- Makes payroll presence transport-neutral and translates domain validation errors only at the HTTP boundary.
- Moves worker review calculation, approval locking, payment-state initialization, and auditing into the payroll service while preserving the existing API contract.
- Adds architecture guards against route duplication, missing authorization, transport coupling in services, storage duplication, and domain routes moving back into the composition root.

Release 1.6.56

- Completes the next behavior-preserving backend modularization sequence by extracting the worker portal, payroll administration, cellar and tank operations, dashboard and historical aggregation, alerts and intake, attachments, public feeds, and static-page routes from the application composition root.
- Preserves every pre-extraction route contract, authorization boundary, response path, and operational workflow with no duplicate registered routes.
- Separates payroll presence and summary assembly into focused services so worker entry and administrator review share authoritative calculations without importing the application root.
- Reduces `app/main.py` by more than 2,100 lines and documents the remaining intelligence-service extraction boundary.
- Expands assembled-source and architecture regression coverage across the extracted domains while retaining the full application test suite.

Release 1.6.55

- Completes the behavior-preserving communications modularization by extracting the official Meta WhatsApp aggregate, outbound operations, sender selection, groups, contact book, and assistant administration from the application composition root.
- Moves the signed Meta webhook and stateful personalized WhatsApp assistant into dedicated modules while preserving verification, signature checks, sender isolation, quarantine, media intake, voice routing, manager confirmations, auditing, and human-review guardrails.
- Centralizes normalized Meta contact and assistant state so inbound, outbound, people, and IVR workflows share one durable source without importing the application root.
- Preserves all 38 communications routes and their authorization boundaries with no duplicate routes, and reduces `app/main.py` by more than 1,000 lines.
- Adds focused Meta route, webhook-ownership, and architecture boundary tests while retaining the complete Gmail and linked-system WhatsApp test coverage.

Release 1.6.54

- Extracts the complete linked-system WhatsApp control plane into a dedicated communications router and shared account-status service.
- Preserves all account, contact, catalog, history, membership, backup, sending, auditing, scope enforcement, deduplication, and authenticated inbound-bridge contracts.
- Keeps the official Meta WhatsApp sender and the cached communications aggregate intact while reducing the application composition root by another 365 lines.
- Adds route-uniqueness, admin-protection, private-token, account-scope, and send-guard boundary tests.

Release 1.6.53

- Continues the behavior-preserving backend modularization by moving all eight Gmail mailbox routes into a dedicated communications router.
- Preserves existing paths, read/write permissions, cached mailbox behavior, message actions and auditing, downloads, attachments, send limits, and error contracts.
- Adds route-boundary and behavior tests and records the safe communications sequence: system WhatsApp control, Meta outbound operations, intake/webhooks, then assistant orchestration.

Release 1.6.52

- Begins the behavior-preserving backend modularization by extracting Admin Control process health, runtime, connections, storage, setup warnings, and recovery status from the application composition root.
- Keeps the existing protected Admin Control aggregate contract intact and adds a protected runtime sub-endpoint for independently loadable operational status.
- Adds pure process-health classification and route-boundary contract tests, plus a documented sequence for extracting people, labor, communications, and intelligence safely.

Release 1.6.51

- Adds dedicated Agronomy Admin and Enology Admin pages and moves each area's configuration into one authoritative location while preserving all operational tools and evidence.
- Reorders Agronomy and Enology navigation around daily workflow, with clear links from operational pages to the corresponding administrative settings.

- Uses damage-adjusted production and blend-yield projections for current-vintage bottle equivalents and profit/loss per bottle until authoritative completed bottling data is available.
- Labels projected and actual bottle bases explicitly and switches automatically to completed-run or historical bottled totals when those records become authoritative.

Release 1.6.50

- Expands every television page's primary content grid to consume the full space between the fixed header and navigation across standard 1080p and shorter 1366-by-768 display viewports.
- Preserves every existing television card while eliminating unused lower-screen bands on Vintage, Work Plan, cameras, traffic, Enology, Weather, Etna, and Communications pages.
- Adds live estate context to the Vintage and Work Plan pages with current weather, next harvest, disease pressure, treatment outlook, cistern and solar resources, and estate readiness.
- Lets the Vintage history chart grow with the available panel height while retaining the full production outlook table.

Release 1.6.49

- Refreshes the complete visible Facebook and Instagram post sets before Meta's temporary media links age out, replacing expired links instead of retaining them indefinitely.
- Keeps stale social media available during a failed Meta refresh without falsely marking the failed attempt as a fresh cache.
- Shows actual per-person IVR usage on Admin -> People cards and adds a saved-form count only when a structured submission was completed.
- Moves the Register's Manual charge form below the product catalog so normal product selection remains primary.
- Constrains Register Admin fields to their responsive cards so long values and placeholders no longer overflow into neighboring panels.

Release 1.6.48

- Adds a durable local cistern level model that runs in shadow beside the existing Camera AI estimate without taking control prematurely.
- Backfills every eligible historical cistern observation with strict chronological walk-forward predictions that never use future readings.
- Creates each new live prediction before the next Camera AI result exists, then scores it only when that later result is accepted.
- Reports all-data, historical-backfill and new-live mean error, agreement within five percentage points, maximum error and last-value baseline evidence.
- Keeps Camera AI authoritative until the shadow model has sufficient historical cases, prospective live cases, distinct levels, observed changes and live changes, as well as the required error and agreement scores.
- Treats low-confidence evidence, duplicates and large refill or regime-change jumps as explicit data-quality evidence rather than silently improving the score.
- Shows Camera AI and learned-shadow levels side by side in Today, the cistern detail view, WhatsApp and Admin -> AI.
- Backfills automatically after migration on the first startup and continues learning after every newly accepted camera result.

Release 1.6.47

- Sends both readable text and matching audio for Manager choices 1 through 7, 10 and 11: Today, Weather, Work, Disease, Harvest, Cellar/Labs, Cistern, Power and Etna/traffic.
- Keeps Cameras and Team Presence in their appropriate existing formats instead of adding unnecessary duplicate audio.
- Personalizes these high-value live summaries with the person's saved first name, language and individual voice.
- Adds natural topic-specific introductions while preserving the deterministic live evidence and human-readable date, time and weather formatting.
- Retains text delivery if voice synthesis is temporarily unavailable.

Release 1.6.46

- Makes the Admin WhatsApp address book authoritative for assistant access, so a saved Reception, Reporter or Manager no longer remains blocked by the legacy environment allowlist.
- Keeps unknown and AI-disabled contacts quarantined while allowing explicitly assigned people such as Wendy to use their saved role immediately.
- Recognizes the approved invitation replies `START` and `INIZIA` as deterministic local menu commands instead of sending them through the AI fallback.
- Adds a per-person keep-alive/reopen action to Admin -> People, automatically using the existing approved no-variable Baiamonte invitation template.
- Shows whether the 24-hour conversation is open and explains that the person must reply to renew it; an outbound template alone is never presented as extending Meta's conversation window.

Release 1.6.45

- Recognizes + as the global WhatsApp menu command even when no guided form is active, preventing it from falling through to the general assistant.
- Moves every numbered Manager and Reporter information choice onto deterministic live database and Home Assistant snapshots, so ordinary IVR navigation continues working when the AI service is unavailable.
- Adds direct Today/urgent-alerts, Cellar/Labs and Cameras responses and routes Reception weather and harvest choices through verified public snapshots.
- Makes the * Back, + Menu and = Cancel controls consistent across Manager, Reporter, guided-form, invalid-choice and fallback messages.
- Routes common spoken topics such as weather, work, disease, harvest, cellar, cameras and power locally, with natural high-quality summaries; AI remains available for genuinely open-ended interpretation instead of being required for basic facts.
- Adds a database-backed public estate-and-wines response and deterministic media-submission guidance for Reception.
- Adds conservative durable IVR learning: each worker can explicitly choose SAME to reuse the last successfully saved location for that form, while route coverage and AI fallthrough are measured for the Admin monitor without retaining message content.
- Adds a private IVR profile to every Admin -> People card with linked access, language and reply configuration, 30-day routing and completion statistics, and automatic per-person workflow learning from successfully saved forms.

- Lets administrators set the repeated-history threshold for learned locations; the worker must still explicitly confirm SAME, and disabled or retired locations are never silently selected.
- Personalizes each linked person's IVR independently: the menu highlights that person's most-used permitted local choices, while saved-location learning remains isolated by sender and form type.
- Consolidates all person-specific IVR controls in Admin -> People -> Person, including number, access, language, reply medium, individual voice, personalized shortcuts, automatic learning, learned-location confidence and optional AI fallback.
- Defaults replies to the incoming medium so text receives text and voice receives voice; sending both formats requires an explicit per-person selection.
- Normalizes legacy contacts that previously received both formats to Match mode until an administrator or the user explicitly chooses Both again.

Release 1.6.44

- Formats WhatsApp dates, times, forecasts and operational responses in natural language instead of exposing database timestamps, machine condition codes or internal timezone names.
- Repairs guided-form Back navigation so the first question returns to record selection and later questions move back exactly one step without retaining the replaced answer.
- Adds a first-class Entire estate choice for scouting, treatment reports and completed vineyard work; whole-estate scouting is stored with authoritative estate scope rather than a misleading block assignment.
- Shortens vineyard choices to field-friendly variety, vine-age and block-code labels and explicitly supports spoken or typed numbered answers.
- Adds fast field controls: * Back, + Menu and = Cancel, while retaining the existing word commands for text and voice use.

Release 1.6.43

- Makes the Admin Ministry-catalog process use its recorded integration identity so successful manual and scheduled runs finish as healthy instead of remaining at "Not run yet."
- Runs power continuity monitoring independently from slow integration work so a long Ministry import cannot create a false heartbeat gap.
- Suppresses add-on, Core and host restart gaps as power-outage alerts, clears unconfirmed prior conditions at startup, and retains the restart gap only in the audit trail.
- Changes recovery wording to state that utility loss is unconfirmed unless direct power evidence is available.

Release 1.6.42

- Uses small transactional Ministry-catalog write batches so the full official JSON audit record imports within the live Home Assistant MariaDB per-query timeout.
- Reports zero rather than an unknown count before the first successful official-catalog synchronization.

Release 1.6.41

- Adds the official Italian Ministry plant-protection product catalog as a weekly, manually refreshable regulatory source.
- Overlays exact registration and reviewable name matches on Baiamonte products; trusted revoked, suspended or expired products are blocked from projected treatments.

- Allows an Agronomist to add a product outside the estate's historical use pattern only after recording its current crop, target, dose limits, PHI, REI and authorized label source.
- Marks products without completed Baiamonte applications as low-confidence first-use candidates and requires Agronomist approval plus paired pre/post-treatment scouting.
- Adds an Agronomy treatment panel for official catalog search, local overlay review, product adoption and crop-use authorization without treating national registration as a label direction.

Release 1.6.40

- Expands the owner system manual with the complete authoritative release history from 1.6.0 through 1.6.40.
- Generates the release appendix directly from the project changelog so maintenance fixes and smaller workflow changes are retained in the durable operator record.
- Rebuilds the downloadable PDF and all 36 in-window preview pages with synchronized release metadata and navigation.

Release 1.6.39

- Updates the full owner manual through the current Agronomy, Enology, Laboratory, Admin AI, Gmail-cache and guided WhatsApp workflows.
- Documents the weather-first disease model, Agronomist review, paired scouting, learning gates, one-pass two-batch treatment process and young-vine TerraPlus boundary.
- Rebuilds the downloadable PDF and every in-window manual preview page with current release metadata.

Release 1.6.38

- Starts the dashboard only after every Agronomy and Enology feature renderer is registered, eliminating intermittent cold-load failures.
- Preserves the active Agronomy or Enology workspace immediately while changing vintages instead of temporarily resetting navigation to Operations.
- Normalizes authoritative laboratory aliases and vintage suffixes during source auditing and corrects four source-confirmed historical sample types.
- Constrains laboratory endpoint and AI-assisted projections to physically possible nonnegative values while retaining the measured adjustment as evidence.
- Guards the embedded weather-map observer against an unavailable document root during iframe startup.

Release 1.6.37

- Corrects the worker portal task queue to use the current planned/in-progress statuses and normal priority, restoring assigned work visibility.
- Makes guided WhatsApp issue priorities compatible with the Operations schema and adds the missing lot/tank reference to cellar-operation submissions.
- Expires abandoned WhatsApp forms and calculators after 24 hours, and makes MENU reliably close either workflow before returning to the main profile menu.
- Updates FastAPI, Starlette, the multipart parser and MCP runtime to secure compatible releases after dependency auditing; the resulting environment reports no known vulnerabilities.

Release 1.6.36

- Replaces the partial WhatsApp intake menu with complete bilingual guided workflows for Agronomy/Field, Operations and Enology/Cellar records.
- Adds saveable field scouting, phenology, planned treatment reports, completed work, labor hours, issues/tasks, equipment service, fruit maturity, fermentation checks and cellar operations.
- Makes WhatsApp voice notes first-class form answers; complicated reports can be spoken once, reviewed as a transcript summary and explicitly saved into the Operations review queue.
- Adds consistent BACK, CANCEL, MENU, RECORD/REGISTRA and final SAVE recovery paths so active sessions do not strand workers or silently create records.
- Keeps treatment field reports planned and unapproved until the Agronomist completes the legal, safety, weather and product review.
- Adds Admin WhatsApp IVR health, active/stalled session counts, 24-hour completions, voice readiness, workflow coverage and worker command guidance.

Release 1.6.35

- Adds durable disease-onset forecasts that project when weather-driven pressure may cross the actionable threshold, with walk-forward validation and explicit confidence.
- Learns treatment effectiveness, observed duration and retreatment cadence from paired pre/post scouting by disease, mixture, dose, block and weather without treating reconstructed pressure as proof of causation.
- Adds chronological active-ingredient and FRAC-group rotation review, blocking repeated groups for Agronomist review when the current product records support the comparison.
- Adds young-vine TerraPlus learning from mapped applications, growth scouting, soil and tissue evidence while retaining the documented-need and Agronomist-approval gates.
- Adds adaptive sensor, duplicate-observation and unreliable-label detection to operational data quality and Admin → AI.
- Adds bounded block-specific disease calibration from parcel/map context, canopy/training system, elevation/aspect, variety and localized scouting.
- Adds outcome-conditioned spray-window learning from wind, temperature, carrier coverage and rain in the first 48 hours after application; legal label and Agronomist limits remain authoritative.
- Rebuilds all connected learning manifests after treatment, scouting and Agronomist review, exposes them in Admin → AI, and provides an administrator rebuild endpoint.

Release 1.6.34

- Adds a durable disease-pressure calibration model trained only from explicit Agronomist decisions and comparable field scouting while retaining the weather/phenology rules score as an auditable baseline.
- Applies bounded, small-sample-shrunk disease adjustments to current pressure, Treatments, alerts, WhatsApp and AI consumers; sparse feedback can never move a score by more than 20 points.
- Adds corrected risk scores to the Agronomist review workflow and automatically rebuilds disease learning after reviews, scouting, startup and scheduled pressure refreshes.
- Adds leave-one-case-out calibration error, training-label provenance, season coverage and validation gates to Admin → AI; validation requires at least eight labels across two seasons and performance no worse than the rules baseline.

Release 1.6.33

- Adds an audited Agronomist review workflow to current disease-pressure assessments; rejected findings stop driving treatment predictions, while materially changed scores return to pending review.
- Adds treatment-linked before/after scouting with strict block, date-window and disease-marker checks, plus conservative automatic pairing only when one treatment match is unambiguous.
- Rebuilds treatment outcome learning and Admin → AI evidence metrics immediately after paired scouting, while retaining weather as context rather than proof of effectiveness.
- Adds a Treatments follow-up panel for missing baselines, due or overdue post-treatment checks, completed pairs and observed outcomes.
- Caches Gmail folders and message summaries in MariaDB during scheduled intake so Email opens from local data; explicit receive and refresh controls still perform live Gmail checks.

Release 1.6.32

- Adds a dedicated **Admin** → **AI** console containing OpenAI service and credit status, effort and speed controls, budget and projected cost, feature usage, model pricing, and durable-learning health.
- Monitors Laboratory, Agronomist treatment, Harvest, and disease-pressure intelligence with explicit model versions, evidence coverage, data freshness, validation methods, accuracy or error metrics, readiness thresholds, and review-gate warnings.
- Distinguishes measured held-out accuracy from deterministic rules: disease pressure reports Agronomist review coverage without claiming outcome-trained accuracy, and an unavailable model cannot hide the other learning processes.
- Keeps Operations Control focused on system operations and limits the Audit Trail viewport to the latest six visible activities while retaining the latest 100 entries in a scrollable list.

Release 1.6.31

- Keeps the durable Laboratory model in an explicit learning-active, low-accuracy state when historical walk-forward direction accuracy is below 60%, even when case and vintage coverage thresholds are met.
- Runs both Laboratory and Treatment durable-learning backfills during application startup so newly installed migrations and historical evidence are reflected immediately in their current pipelines.

Release 1.6.30

- Adds durable vineyard-treatment learning with pre-treatment weather, leaf wetness, solar, phenology, scouting, objectives and treatment cadence; 14-day post-treatment evidence windows are cut off before the next application.
- Separates weather-reconstructed pressure from field-observed effectiveness, versions the treatment model and feature schema, calibrates similarity from the 2023–2026 GW2000 history, and keeps the model provisional until adequate seasons and comparable scouting outcomes exist.
- Normalizes historical laboratory identities while retaining every original report label; Nerello, Narello and Nerello/Narello Macalase resolve to the canonical Nerello Mascalese series, alongside documented Grecanico and Grenache aliases.
- Adds durable laboratory cutoff cases, leakage-safe historical walk-forward outcomes, versioned model manifests, per-analyte-and-unit error metrics, and automatic rebuilding at startup and after every new, corrected or reviewed laboratory result.
- Exposes transparent Laboratory and Treatment learning status in their current pipelines and APIs without converting predictions into automatic agronomy, cellar, harvest or enology approvals.

Release 1.6.29

- Passes the latest GW2000 disease-pressure snapshot into current treatment simulations so weather-pattern similarity is calculated rather than left unavailable.
- Learns the authoritative product/date/water program from every structured completed treatment, including safety-restricted Treatment 5, while continuing to prohibit historical safety evidence from authorizing a current prescription.

Release 1.6.28

- Narrows TERRAPLUS SOLUB NPK 8-7-6 to the small, young vines only and labels its procurement card accordingly.
- Removes TERRAPLUS from general mature-vine, olive and whole-estate nutrition baselines; it remains unavailable unless mapped young vines have a documented need and an Agronomist-directed root-zone or fertigation plan.

Release 1.6.27

- Corrects TERRAPLUS SOLUB NPK 8-7-6 from broadcast land/soil fertilizer to vineyard root nutrition for fertigation or localized root-zone application.
- Separates land/soil fertilizer procurement from vine root-nutrition procurement so TERRAPLUS appears with vine nutrition without being mislabeled as a foliar canopy spray.

Release 1.6.26

- Rebuilds vineyard treatment simulation around the Agronomist's complete Treatments 2–5 rather than selecting isolated products, while retaining current label, necessity, exact-mixture, weather, PHI/REI, PPE, inventory and Agronomist approval gates.
- Calculates every weather-matched historical program for the current Baiamonte process: one vineyard pass using 400 L total carrier prepared as two identical 200 L batches.
- Makes GW2000 weather the explicit treatment rationale by comparing temperature, humidity, 72-hour and seven-day rain, soil moisture and other available markers with the weather preceding completed treatments.
- Adds continuous, leakage-safe learning: each confirmed completed vineyard treatment stores only the seven days of weather available before application, reconstructed disease pressure, the full recipe and a stable program signature.
- Uses the learned weather match in both live next-treatment review predictions and the simulator, with transparent similarity, model-version, training-case and leave-one-treatment-out validation evidence.
- Keeps restricted historical treatments as behavioral learning evidence only; they never become current reusable prescriptions without fresh safety and authorization review.

Release 1.6.25

- Rebuilds the Agronomy treatment simulator around the Baiamonte field process: one complete vineyard pass using 400 L total carrier prepared as two identical 200 L batches.
- Leads with a readable full-treatment recipe showing each necessary product, the amount in each 200 L batch, the whole-treatment amount, inventory readiness, and the current evidence supporting its inclusion.
- Prevents inventory and prior use alone from promoting support, adjuvant, or nutrition products; those products require documented current stress or another explicit evidence-supported need.
- Screens independently significant same-date disease pressure for additional necessary control while retaining exact-mixture compatibility, label, weather, safety, and Agronomist approval gates.

- Adds Laboratory variety-standard cards that show recorded approved analyte markers by variety, sample type, stage, and unit, while clearly leaving missing standards unconfigured.

Release 1.6.24

- Removes the red Live cellar records readiness banner from the Cellar page; real tank alerts remain visible, while the informational Demo-mode banner is retained only when Demo mode is active.
- Preserves user-selected values systemwide when live refreshes rebuild a select box's options, including the Laboratory Sample and Measurement selectors, without restoring values that are no longer valid choices.

Release 1.6.23

- Preserves the former shared Live/Sandbox choice on the first upgrade to independent US and Italian PayPal selectors, even when Home Assistant pre-populates the new fields with schema defaults.
- Records the one-time migration in persistent add-on data so later operator changes to either account remain independent and are never overwritten on restart.

Release 1.6.22

- Adds independent Live/Sandbox selectors for the US and Italian PayPal Business accounts in protected Home Assistant configuration.
- Uses the selected environment for each account's OAuth token, order creation, capture, checkout label, and status light, preventing one account's mode from changing the other.
- Keeps the former shared PayPal environment as a migration fallback for existing installations.
- Makes guest-inquiry deletion persistent by retaining an invisible tombstone linked to the Gmail intake item, so dashboard refreshes and later Gmail checks cannot recreate a deleted inquiry.
- Keeps Labels & dedicated displays selected after saving a tablet assignment instead of returning to the Cellar overview.
- Adds live Online/Offline heartbeat lights to each manually registered cellar tablet, including Tablet 1 and Tablet 2.
- Keeps Laboratory sample and measurement selectors useful when the selected vintage has no numeric report by clearly showing the latest available vintage, while refusing to imply that historical evidence is a current-year measurement or projection.

Release 1.6.21

- Exports the protected US and Italian PayPal client IDs, client secrets, and environment from Home Assistant app options into the Register API process.
- Restores the already-configured US PayPal Business account in checkout while preserving the existing EUR-to-US-account fallback when no Italian account is configured.
- Preserves the explicitly selected PayPal environment from protected Home Assistant options; the Baiamonte installation is configured for live PayPal after deployment.
- Adds persistent US and Italian PayPal Business status lights to the sale page, showing Active/Offline and LIVE/SANDBOX before checkout.
- Compacts the oversized Register header pill to a short FIC, PayPal-account, and environment summary; the detailed sync timestamp remains in Integration status.
- Moves Register category buttons to a dedicated wrapping row so Wine, Oil, Hospitality, and later categories are never clipped by the search field or Manual item button.

Release 1.6.20

- Routes clear guest tasting, visit, tour, dinner, and reservation inquiries into Hospitality even when the Gmail label or configured subject phrase is missing; the messages remain review-only until explicitly converted.
- Rechecks previously downloaded Gmail message bodies so the existing “Inquiry about Classic Tasting at the Estate” request can enter Guest inquiries.
- Restores saved Atlas parcel and block shapes with an SVG overlay that remains correctly sized when Agronomy is opened after page startup.
- Separates saved boundary counts from parcel center locations so Atlas no longer reports point-only records as verified boundaries.
- Keeps the active workspace, page, and subpage selected when changing years instead of returning the navigation to Operations.
- Suppresses cellar monitor alerts for tanks whose recorded volume is zero; readings remain visible, and guard checks resume automatically when a positive volume is recorded.
- Makes every Finance summary metric clickable and keyboard accessible, opening the selected year’s supporting invoices, monthly result calculation, cash accounts, or individual approved labor and payment records.

Release 1.6.19

- Added transparent AI-assisted Laboratory projections: exact prior-vintage evidence is preferred, with a low-confidence 14-day current measured trajectory only when at least two dated readings exist; every refresh and new lab arrival recalculates the outlook.
- Added newest-report AI-assisted finding cards to Laboratory and Vineyard fertilization, with source/review limitations and no automatic cellar, harvest, product, rate, or application decision.
- Restricted Fertilizer procurement to explicitly classified land/soil products; foliar vine-treatment fertilizers remain in Treatments and Crop nutrition.
- Added a dedicated Email tab beside Messaging in Administration, containing mailbox folders, reading/actions, attachments, received/sent history, connection controls, intake refresh, and compose.
- Splits the oversized Laboratory series menu into Sample and Measurement selectors.
- Keeps samples with comparable prior-vintage evidence first while retaining clearly labeled current-only samples.
- Limits the measurement menu to the selected sample, sample type, and stage so unlike records cannot be chosen accidentally.
- Makes “Mark Enologist approved” save immediately and clears the sample’s source-review flag so approved reports stop appearing red.

Release 1.6.18

- Opens Laboratory on a series that actually has comparable prior-vintage evidence instead of defaulting to a current-only measurement.
- Groups the selector into measurements with comparable history and measurements that have only a current reading, so unavailable projections are explicit.
- Replaces the misleading mixed-sample annual-average chart with like-for-like vintage endpoints for the selected wine, sample type, stage, analyte, and unit.
- Replaces the process-wide mixed table with the final measured endpoint for that same sample definition in each available vintage.

- Refreshes report history, decision status, trends, comparisons, and projections immediately after a laboratory result is corrected, then reopens the detail with the saved value.

Release 1.6.17

- Consolidates safe laboratory label variants such as Grecanico 25/2025, Bianco–Grecanico, Granache/Grenache, and Nerello/Nerello Mascalese for vintage comparison.
- Keeps unrelated wines, sample types, process stages, analytes, and units separate so projections do not combine unlike measurements.
- Adds sample type and process stage to the Laboratory selector, removing visually duplicated choices and making grape, must, and wine series explicit.
- Adds Giancarlo's source-backed December 2024–November 2025 monthly labor to reconciliation: 988 known hours plus one 15-day attendance record whose hours remain explicitly unknown.
- Applies the owner-confirmed €10/hour rate to known historical hours, records all prior labor as paid with ledger entries for the 11 known-value months, and preserves November 2025 as paid with its unknown hours and amount explicit.
- Prevents mirrored source rows from double-counting labor totals and excludes paid historical attendance with an unknown amount from the zero-value invoice warning.

Release 1.6.16

- Keeps the active Enology, Laboratory, Bottling, Fertilization, or Nutrition page selected while changing vintages.
- Orders vintage-specific refreshes after the shared year state changes and ignores stale responses from rapid year switching.
- Repairs the Laboratory outlook endpoint under Home Assistant ingress so recorded vintage measurements and projections load instead of an empty fallback.
- Reloads analyte comparisons on every vintage change and exposes the existing source-backed 2023–2025 bottle-equivalent backfill in Bottling.

Release 1.6.15

- Defers direct-route activation until the signed-in workspace permissions are available, restoring Laboratory and Messaging links.
- Safely binds the later-loaded laboratory outlook renderer so opening Labs cannot interrupt the page.
- Preserves the Tank Details workspace-save repair and verified Etna status from the previous patches.

Release 1.6.14

- Completes the workspace startup repair by safely binding the later-loaded cellar-history renderer.
- Restores the remaining Agronomy, Etna, Laboratory, and Messaging initialization path exposed during live 1.6.13 verification.

Release 1.6.13

- Repairs Admin Messaging when Review has already loaded the shared communications data.
- Rebuilds the Messaging panel placement and restores its Contacts, Meta Business, and System-account tab state whenever the page opens.

- Prevents the later-loaded analytics bundle from stopping dashboard startup before Agronomy/Etna, Laboratory navigation, and Messaging are initialized.
- Keeps Tank Details and other Agronomy saves in the current workspace and page instead of returning to Today.
- Keeps direct navigation and cached communications responsive without requiring a full dashboard reload.
- Adds regression coverage for Messaging page activation and stale-panel recovery.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.12

- Isolates Messaging, Social, and TV browser startup so an error in one page cannot prevent the other pages from loading.
- Adds an independent page-loader fallback for direct navigation and restored sessions.
- Keeps hospitality Gmail inquiry refreshes active while restoring access to the shared Messaging interface.
- Adds regression coverage for page startup isolation and hospitality inquiry synchronization.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.11

- Aligns the Admin Docs card with the installed owner’s manual release 1.6.4 instead of the obsolete 1.5.0 label.
- Updates regression coverage so the displayed manual release cannot silently drift again.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.10

- Routes Gmail messages tagged with the configured Hospitality label into Guest Inquiries, including nested labels and messages that were downloaded before the label was applied.
- Retains subject-based hospitality routing as a fallback and exposes configurable inbound labels and subjects in Hospitality Admin.
- Adds a manual Gmail refresh action and regression coverage for label normalization, re-routing, and source metadata.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.9

- Records harvest picks from multiple vineyard blocks while preserving every legal parcel in the grape-to-tank traceability chain.
- Treats the entered crate weight as net kilograms per crate, calculates the field total, and supports a later authoritative winery weight without erasing the original field measurement.
- Keeps the complete harvest form visible on phones and separates harvest routes from the main application module for easier maintenance.
- Restores laboratory outlooks when switching years by using each report's authoritative vintage and keeps current-vintage projections separate from historical comparisons.
- Resolves Today greetings through the linked Home Assistant person or worker identity while excluding tablet, kiosk, iPad, and service accounts.

- Adds product images to register items and a fast editable manual-charge item while preserving local ledger and inventory behavior.
- Adds regression coverage for harvest weighing and multiple blocks, parcel traceability, laboratory year selection, register behavior, and greeting privacy.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.7

- Prevents overlapping System Documentation refreshes from leaving the control stuck on “Loading…”.
- Adds a Rome-time greeting on Today and includes a first name only when Home Assistant supplies a natural full human name.
- Shows continuously updated application uptime in Admin Control.
- Adds live Up/Down heartbeat badges for active tank-label tablets based on their latest check-in.
- Adds regression coverage for greeting privacy, documentation refresh recovery, uptime, and label-device status.
- Leaves the full-version manual unchanged because this is an operational reliability patch release.

Release 1.6.6

- Keeps every TV card and both sides of split layouts visible on narrower or overscanned displays.
- Replaces unsupported relational CSS selectors with explicit compatibility classes for older Samsung and embedded TV browsers.
- Loads vessel rendering before the main TV bundle and reduces continuous chart and overflow work for smoother long-running displays.
- Adds regression coverage for TV layout preservation, legacy-browser CSS, helper order, and refresh throttling.
- Leaves the full-version manual unchanged because this is a TV reliability patch release.

Release 1.6.5

- Reuses healthy MariaDB connections through a bounded, thread-safe pool instead of reconnecting for every small dashboard query.
- Compresses large API and TV-display responses for substantially lower tablet and kiosk network transfer.
- Preserves the existing short-lived TV payload cache while making cold refreshes and concurrent displays more efficient.
- Leaves the full-version manual unchanged because this is a performance patch release.

Release 1.6.4

- Adds a complete hospitality partner directory with referral attribution, configurable commission rules, due and approval states, partial payments, payment history, and auditable corrections.
- Connects partner liabilities to the Finance payable total and provides a dedicated partner payment queue and annual summary.
- Expands the System Manual to 24 portrait pages with intelligence-pipeline descriptions, decision trees, evidence gates, partner workflows, and current operational guidance.
- Rebuilds the downloadable PDF and in-window page previews for the updated manual.

Release 1.6.3

- Adds Home Assistant-managed network receipt printing through a configured `notify`, `script`, or `shell_command` service.
- Lets administrators select browser printing or the Home Assistant IP-printer path and optionally provide a target printer entity.
- Preserves the existing browser receipt as an automatic fallback when the Home Assistant printer service is unavailable.
- Keeps receipt print counts and audit history consistent for both printer paths.
- Leaves the full-version manual unchanged for this patch release.

Release 1.6.2

- Adds an explicit touch-friendly cash-received confirmation with amount due, cash tendered, and change retained in the audit trail.
- Allows administrators to correct payment metadata or void a local payment while preserving its audit history and automatically restoring unposted inventory.
- Protects captured online PayPal payments from local deletion and directs refunds through PayPal before ledger reconciliation.
- Replaces register payment browser prompts with purpose-built tablet dialogs, clarifies the operator-confirmed PayPal phone workflow, and keeps online PayPal API verification separate.
- Refreshes the ECB EUR/USD reference rate once daily while keeping the checkout rate editable and preserving the applied rate on every sale.
- Compacts register item cards and administration controls, and standardizes register buttons for clearer touch targets and states.
- Leaves the full-version manual unchanged for this patch release.

Release 1.6.1

- Corrects Giancarlo Pafumi's paid €20 gas reimbursement from the erroneous future date 27 August 2026 to the source-supported date 27 July 2026.
- Repairs the archived July timesheet source alongside the labor ledger and records both before/after values in the audit log without changing the amount or paid status.
- Corrects the archived source title from Mattia to Giancarlo and leaves the full-version manual unchanged.

Release 1.6.0

- Adds a tablet-oriented Register workspace with Sale, Inventory, Ledger, and Admin pages plus dedicated Register and Cashier access.
- Mirrors sellable Fatture in Cloud products and prices read-only, includes local Hospitality packages, and supports authorized manual items, editable prices, and discounts.
- Keeps EUR as the authoritative reporting and VAT base while allowing EUR or USD collection at the saved checkout conversion rate.
- Supports both Italian and US PayPal Business credentials, shows the active account at checkout, and preserves the selected account on each transaction.
- Adds an English/Italian checkout switch that controls PayPal locale and the printed receipt language.

- Supports hosted PayPal/card checkout and operator-confirmed PayPal Tap to Pay on an NFC phone without storing raw card data or claiming browser-side POS verification.
- Preserves EUR base values, collected currency and amount, conversion rate, PayPal account, language, and payment reference in receipts and monthly CSV exports.
- Keeps register sales posting to Fatture in Cloud disabled until a separately approved reconciliation release.
- Updates and visually verifies the portrait, scrollable System Manual for the full 1.6 release.

OPERATING REFERENCE

33. Glossary

Actual: A completed, confirmed event or measurement. **Evidence:** The source message, document, sensor, note, or record supporting a value. **GDD:** Growing degree days, a temperature accumulation measure. **Ingress:** Home Assistant's authenticated route into an add-on interface. **Intelligence proposal:** A traceable calculated result that remains separate from an approved fact.

LocalTuya: The Home Assistant integration used for local control of supported Tuya devices without the cloud as the normal command path.

Purchased fruit: Grapes received from an external source and included in production/cellar totals while excluded from estate parcel yield and estate-harvest completion.

Report timestamp: Home Assistant's indication that an integration reported an entity, even when its numeric value did not change.

Pipeline: The ordered evidence, calculation, review, approval, and refresh path for a domain decision.

Provisional: A system result temporarily available for planning while its required human review is pending. **Hospitality Manager:** The role responsible for guest inquiries, packages, private experiences, guest readiness, deposits, and confirmations.

Cashier / Register user: A limited role that can operate estate sales without receiving finance, vineyard-write, payroll, or administration access.

EUR base: The authoritative euro value used for VAT and management reporting even when the customer pays in USD.

Tender amount: The amount actually collected in the selected payment currency. **MCP:** Model Context Protocol, the controlled interface used by Codex and approved automation. **NDVI / NDRE:** Satellite vegetation indices used as trend evidence. **Planned:** Intended but not completed or approved. **Review item:** A visible uncertainty that must not be guessed. **Vintage:** The harvest year to which grapes or wine belong, even when later work occurs in another calendar year. **Observed L/kg:** Current measured cellar liquid divided by grapes received for that variety; it is operational evidence, not automatically a training value. **Predictive L/kg:** The weighted reconciled prior-vintage finished-wine yield for a variety, or a clearly labeled configured/fallback value. **YoY:** Year-over-year comparison.

OPERATING REFERENCE

34. Operating principle

The system is designed to be useful without pretending to know more than the evidence supports. Confirmed facts remain authoritative, forecasts remain provisional, unknowns remain visible, and sensitive actions remain under human control.